The Inner Beauty of Firms

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August 7, 2026

Abstract

Using millions of task assignments from salon management software, I find significant establishment-level dispersion in labor productivity and internal task specialization and a strong association between the two that is largely unexplained by establishment size. The 25% most specialized salon-quarters are on average 60% more productive than the bottom 25%. To rationalize these facts, I identify and estimate a model in which firms characterized by different organization costs hire and assign tasks to workers with multidimensional skills in order to improve quality in an oligopolistic product market. I show that internal reorganization changes the partial and industry equilibrium effects of economic shocks. Neglecting task reassignment within the firm can therefore cause a researcher to misstate or reverse the aggregate productivity and specialization impacts of economic shocks.

JEL Codes: J2, L2

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I am grateful for helpful suggestions from Tara Sinclair, Maurizio Mazzocco, Simon Board, Daniel Haanwinckel, Moritz Meyer-ter-Vehn, Alex Bloedel, Robert McDonough, Ichiro Obara, Jay Lu, Felipe Goncalves, Till von Wachter, Coen Teulings, Tyler Ransom, Rodrigo Pinto, Jesse Shapiro, John Kennan, Stepan Aleksenko, Marco Duarte, Andrii Babii, Jonathan Kowarski, Victoria Barone, and Pauline Mourot. I thank Malayvardhan Prajapati and Will Firmin for excellent research assistance. This work received support from the Institute for Humane Studies under grant number IHS017120.

"Of all the things I've done, the most vital is coordinating those who work with me and aiming their efforts at a certain goal." — **Walt Disney**

1 Introduction

There are large differences in productivity across similar firms (Syverson 2004). Also, documented differences in management practices (Bloom and Van Reenen 2007) suggest that some firms may be unable to assign the right task to the right worker. Three natural questions emerge from these observations. First, are there large differences in internal task specialization across similar firms? Second, if such differences exist, to what extent are they related to individual firm productivity? Finally, how is aggregate productivity determined in equilibrium when firms differ in their ability to assign tasks?

These questions are critical for understanding the productivity implications of many economic shocks, including the diffusion of management practices, taxation, immigration, and increased market concentration. However, answering these questions poses a dual challenge. Empirically, one must look inside the black box of the firm and observe how workers are assigned to tasks. Further, this must be done not just for one firm but for many comparable and competing firms. Theoretically, it is necessary to develop a model where organizationally unique firms choose their task assignments in product and labor market equilibrium.

In this paper, I overcome the empirical challenge using novel data from a management software company. I document a robust association between task specialization within the establishment and productivity, one not accounted for by other observables such as establishment size or location. I overcome the theoretical challenge by proposing a model where task specialization within the firm is heterogeneous and endogenous. Tractability, identification and estimation are achieved by modeling firm-specific organization costs as proportional to the task specialization index (the s-index), which measures how far a firm's chosen task assignment is from a generalist benchmark where tasks are assigned randomly to workers. I use the model to study the market-level task-specialization and labor productivity effects of several counterfactual economic shocks. I show that allowing firms to internally reorganize in response to a shock qualitatively changes the productivity and task specialization implications, both directly because economic shocks change the incentive to task specialize and indirectly because reorganization shifts equilibrium wages and prices.

In the first part of the paper, I use the revenue information in the data to confirm past work. Labor productivity is highly persistent within establishments and highly dispersed across estab-

lishments: The salon-quarter at the 75th percentile is 1.90 times as productive as the salon-quarter at the 25th percentile. I then use the millions of task assignments and a new measure of internal task specialization (the s-index) to show that task specialization within the establishment exhibits similar persistence and dispersion: The salon-quarter at the 75th percentile is 13 times as internally specialized as the salon-quarter at the 25th percentile. There is also a robust positive correlation between the s-index and labor productivity: The top 25 percent of salon-quarters in terms of specialization are on average 60 percent more productive than the bottom 25 percent, generating on average $0.83 more revenue per minute. Stated differently, a one standard deviation increase in task specialization is associated with a 0.28 standard deviation increase in labor productivity, and this effect remains essentially the same after controlling for establishment size, county and quarter fixed effects.

Importantly, 80% of the dispersion in labor productivity and 75% of the dispersion of task specialization remain after accounting for the task mix, as well as county, time and establishment-size fixed effects. The coefficient from a regression of labor productivity on task specialization is similarly stable even after adding county, time and establishment-size fixed effects. Establishments of similar sizes vary in task specialization, and among similar-size establishments task specialization is correlated with labor productivity. That internal specialization patterns are not driven primarily by establishment size is difficult to rationalize by models where organizations pay the fixed costs of specialization only when they expect to spread these costs over a large amount of output. Further, by decomposing the link between productivity and task specialization, I find that rather than serving a greater number of customers, task-specialized salons generate more revenue per customer and have higher customer return rates, indicating that productivity gains manifest via quality differences. Finally, the s-index is positively correlated with other potentially productive management practices including teamwork (assigning multiple workers to the same customer on a single date) and early adoption of management software features.

In light of these facts, I design a model which generates dispersion in establishment productivity both due to traditional demand and Hicks-neutral cost shocks and due to differences in internal task specialization across establishments. This is accomplished via establishment-specific organization costs based on the Kullback–Leibler divergence that are identifiable from the data, theoretically and computationally tractable, yet rich enough to allow for heterogeneous patterns of behavior across establishments operating in the same labor and product markets. The model allows me to understand via counterfactual exercises how reorganization of work inside the establishment mediates the productivity impacts of policy.

In the model, workers have multidimensional skills, and wages are worker-specific. Strategic firms engage in Bertrand-style price competition while also choosing the skill composition of their workforce and the task assignment of each hired worker. Better assignment of workers to tasks increases product quality but requires the firm to bear a greater organization cost. Importantly, organization costs are firm-specific and impact marginal rather than fixed costs. I do not impose conditions on the magnitude or distribution of organization costs across establishments, but rather allow them to be identified in the data.

Heterogeneous organization costs capture non-transferable differences in management practices that change the productivity of labor across establishments. Organization costs make more intricate assignments of workers costly, and I interpret them as a reduced-form representation of many frictions within establishments, including coordination costs, incomplete contracts, and lumpy demand, among many others. Many of these frictions are exactly what the management software that generated the data used in this paper was designed to overcome. Among hair salons in the United States, contracting frictions are particularly important, as there is wide variation in whether salons have traditional employees, independent contractors (booth or chair renters), or both (Lee 2007).

Consistent with the stylized facts, the model generates a positive relationship between firm performance in the product market and task specialization within the firm. Profit-maximizing firm- and worker-specific task assignments take a logit-like form. For any fixed wages, firm strategies exist and are essentially unique. Tractability is maintained even in equilibrium because the pricing and task assignment decisions can be separated, and organization costs are proportional to the mutual information, making the task assignment decision equivalent to a well-studied problem in behavioral economics (i.e., rational inattention) and computer science (i.e., rate-distortion). I also show that organization costs based on the mutual information can be microfounded by costly communication between a manager and workers within the firm.

I prove that each firm's organization cost parameter, worker skills and worker wages are identified from data which contain task assignments, firm prices and firm market shares. The identification proof is constructive and motivates an estimation procedure which does not require solving the model. In the first step of the procedure, workers with the same skill set are identified by comparing the task assignments of pairs of coworkers across firms, and firm organization costs are obtained by comparing how firms utilize pairs of workers with the same skills. In the second step, wages, skills and other parameters are obtained by solving a linear system of moment conditions. I implement a version of this procedure and estimate the model for salons in New York County,

NY, Cook County, IL, and Los Angeles County, CA, for 12 quarters between 2018 and 2021.

Finally, I study how four counterfactual economic shocks change task specialization and labor productivity: low-wage immigration, a sales tax increase, the diffusion of management practices, and increased product market concentration. The model allows shocks to be decomposed into a reallocation equilibrium which focuses only on how the shock redistributes labor across establishments, and a reorganization equilibrium which captures both reallocation and internal task reassignment. I show that internal reorganization changes the partial and industry equilibrium implications of economic shocks. Neglecting reorganization when it occurs can therefore cause a researcher to misstate and in some cases reverse the aggregate productivity and specialization impacts of economic shocks. In the most salient example, when reorganization is shut down, increasing the sales tax on services in New York County by 4 percentage points reduces aggregate task specialization by 3.0% and reduces aggregate labor productivity by 1.2%. When reorganization is allowed to operate, the sales tax increase reduces aggregate task specialization by 6.3% and increases aggregate labor productivity by 0.7%. The reversal occurs because a sales tax discourages task specialization within the firm. This has a direct negative partial equilibrium effect on firm-level productivity, but also a positive industry-equilibrium selection effect because it advantages productive firms. The positive industry equilibrium effect dominates the negative direct partial equilibrium effect.

The counterfactual exercises reveal two general lessons. First, some shocks, like sales tax increases and the diffusion of management practices, directly change the incentive for all firms to engage in costly specialization. Neglecting reorganization in these situations misses the creation or destruction of productive specialized jobs. In New York County, if management practices diffused such that each salon's organization cost parameter becomes the organization cost parameter of the salon one rank above it, absent reorganization, task specialization increases by 0.9% and productivity increases by 0.3%. Once reorganization is incorporated, task specialization increases by 2.6% and productivity increases by 2.0%. Second, task reassignment within the firm plays a key role in determining which firms are favored in industry equilibrium by a shock. In Los Angeles County, a 5% increase in total labor from low-wage immigration results in a 5.9% increase in aggregate task specialization and a 1.7% decrease in native productivity when firms are not allowed to reorganize. With reorganization, the same shock reduces aggregate task specialization by 0.2% and reduces native productivity by 0.9%. The discrepancy occurs because natives with the immigrant skill set tend to be employed at specialized but not particularly productive firms in the initial equilibrium, and when reorganization is neglected these firms are assumed to gain

a competitive advantage from labor cost reductions. With reorganization, some firms specialize while others generalize in order to share in the labor cost reduction.

Researchers may be tempted to use observable features of firms in an initial period to try and to predict the aggregate impact of an economic shock. As an example, a researcher might consider using the share of workers at a firm that share the skills of immigrants as a proxy for which firms will benefit from immigration. When internal reorganization is believed to be an important margin of adjustment, such exposure- or shift-share-based logic can be misleading because it assumes task assignments remain fixed within the firm. Unfortunately, the bias this induces in an outcome of interest, like aggregate labor productivity, can be either positive or negative, depending on whether the exposure variable from the initial equilibrium is positively or negatively correlated with baseline firm productivity. To complicate matters further, the sales tax increase in New York County shows that because firms differ in their ability to reorganize, reorganization itself can alter the relative exposure of firms to an economic shock and therefore change the aggregate productivity impact.

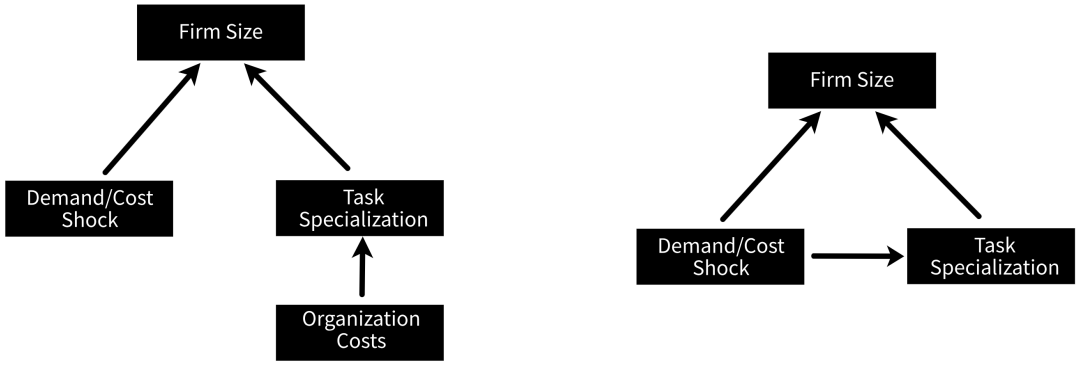
Determinants of Firm Productivity. Past studies have shown that productivity differences across firms are large (Syverson 2004) and can be linked to management practices (Bloom and Van Reenen 2007) and establishment managers (Metcalfe, Sollaci, and Syverson 2023). Much existing work focuses on manufacturing firms. This paper finds similar revenue productivity dispersion in service firms, and shows that task specialization accounts for some of this dispersion. Further, firms that engage in task specialization also use management software in more sophisticated ways. The differences in specialization, and, in particular, the large number of generalized salons in the U.S., are similar to the patterns Bassi et al. (2023) observe in Ugandan manufacturing firms. The link between productivity and task specialization is consistent with the link between productivity and coordination established by Kuhn et al. (2023).[1]

Task-Based Perspective of Labor Markets. I model labor as being divisible into tasks which can be assigned to workers with different skills, a tradition that dates back to at least Sattinger (1975) but has seen growing use since Autor, Levy, and Murnane (2003). I incorporate features present in different parts of the literature, including multidimensional worker types (Lindenlaub 2017, Ocampo 2022), firms with multiple worker types (Freund 2022, Haanwinckel 2023), coordination costs (Garicano 2000, Adenbaum 2022, Bassi et al. 2023), and firm-specific task demands (Lazear 2009).

1. This paper is also relevant to the literature on the role of firms in wage inequality (Card, Heining, and Kline 2013, Alvarez et al. 2018, Song et al. 2019). My paper positions task assignment as a mechanism through which this effect could operate.

The main innovation relative to prior work is that firms are described primarily by differences in their cost of coordinating labor, similar in spirit to Becker and Murphy (1992). These organization costs are not fixed costs per occupation, as in Adenbaum (2022). As a result, specialization is not directly caused by demand or cost shocks. Rather, it is determined separately and interacts with demand and cost shocks to determine firm size, forming the causal relationships depicted in Panel (a) of Figure 1. In other models, firms pay the costs of specialization because they expect to be large, and specialization further amplifies their size in equilibrium, leading to the causal relationships depicted in Panel (b) of Figure 1. The model can accommodate firms that are large due to specialization and firms that are large for other reasons, and will thus generate differences in task specialization among firms of the same size. The idea that coordination costs dampen or even overwhelm the benefits of specialization is consistent with recent empirical evidence that generalization is positively correlated with labor productivity (Fang et al. 2023, Liu 2025).

Figure 1: Causal Relationships



(a) This Paper **(b)** Others

The heterogeneous organization costs proposed in this paper can be micro-founded by costly communication, as in Garicano (2000). However, they also are a reduced-form way of capturing the many reasons identified in organizational economics for why firms differ in their ability to organize workers. These include monitoring (Alchian and Demsetz 1972, Baker and Hubbard 2003), relational contracts (Baker, Gibbons, and Murphy 2002), knowledge (Garicano and Wu 2012), coordination (Dessein and Santos 2006), trust (Meier, Stephenson, and Perkowski 2019), culture (Martinez et al. 2015) and choice of manager (Muñoz and Otero 2025). This paper contributes to this literature by embedding organizational heterogeneity in a model where firms interact in product and labor market equilibrium. I show how such organizational heterogeneity shapes the productivity implications of economic shocks. A secondary methodological contribution is that

my model can be identified and estimated without classic linked employer-employee data. In much of the literature, the distribution of worker skills is identified using detailed information on workers, typically wages, occupations, and education. In this paper, worker skills are recovered from task assignment patterns. Such data are becoming increasingly available across a wide range of industries, including pharmaceutical research and restaurants.[2]

Talent Allocation within the Firm. A recent body of work has shown that managers play an important role in determining worker productivity within the firm (Haegele 2022, Coraggio et al. 2023, Minni 2023). Consistent with my findings, these effects operate through the assignment of workers to jobs within the firm. While my paper studies the beauty industry, Minni (2023) studies a consumer goods multinational, Haegele (2022) studies a large European manufacturer, and Coraggio et al. (2023) study Swedish registry data. The diversity of these contexts strengthens the external validity of each paper's individual findings. This paper also complements this work by providing a way to integrate management heterogeneity into market equilibrium.[3]

2 Data

This section describes the salon management software data I use in this paper.

2.1 Context and Institutional Details

The data set was obtained from a data-sharing agreement I negotiated with a salon management software company. The software facilitates the operation of a beauty business, including scheduling, pricing, payments, inventory, staffing, business reporting, client profiling and marketing. As of July 19, 2022, a monthly subscription has a base price of $175. Although the company also markets its software to spas, tanning salons and massage parlors, hair salons and barbers make up the majority of its clients. For this reason, I analyze only hair salons and barbershops.

The software is sold to beauty businesses throughout the United States, but customer concentration is highest in Los Angeles (where the company was founded) and New York City. Although individual firms buy the software, the data document task assignments at the establishment level. Further, 85% of firms in the sample are single-establishment. I therefore use the establishment as the unit of analysis and refer to it as a salon or a firm in the model section. The data document the

2. Some examples include TruLab (trulab.com), which is used by vaccine and biotech companies to manage laboratories, and 7shifts (www.7shifts.com), which is used by restaurants.

3. Within-firm allocation is especially important in light of evidence of productivity dispersion among workers in the same occupation (Chan Jr and Chen 2022).

internal organization of salons that are geographically close and therefore likely to be direct competitors in labor and product markets. For example, I observe 10 salons in the Lower Manhattan ZIP code 10013, which is a 0.55-square-mile area.

The data document which stylist is assigned to each task and client, and record the duration of the appointment, the price paid, and a custom text description of each task. If more than one employee is assigned to a single client, this is recorded as multiple entries describing what each employee contributed. Although the data are de-identified, firm-specific IDs allow me to track employees and clients over time within establishments that share an owner, but not across establishments with different owners.

A sample from the data is provided in Table 1, with IDs replaced with pseudonyms and variable values slightly changed for confidentiality. This sample shows the different ways two salons coordinate employees to meet customer demand. Blake requested a cut, highlights and a treatment at salon 1A. The salon had a single employee, Rosy, perform all three services. Grace requested a cut and a single process (color) at salon 2A. Unlike salon 1A, salon 2A split these tasks between two employees, Tyler and Ben. These salons are in the same ZIP code. Throughout, I measure labor in units of time using the appointment duration variable recorded in the data.

Table 1: Salon Activity Data Sample

Business	Establishment/Salon	App.	Cust.	Service	Staff	Time Stamp	Price	Duration (minutes)
1	1A	123	Blake	Advanced Haircut	Rosy	3/14/2021 16:15	102	70
1	1A	123	Blake	Full Head Highlights	Rosy	3/14/2021 16:15	234	123
1	1A	123	Blake	Treatment Add On	Rosy	3/14/2021 16:15	39	67
2	2A	456	Grace	Women's Haircut	Tyler	2/11/2021 11:00	225	48
2	2A	456	Grace	Single Process	Ben	2/11/2021 11:00	200	81

Note: This table is a snapshot displaying two actual appointments at salons in the same ZIP code from the data used for the estimation. Customer, establishment, and business IDs are replaced by pseudonyms.

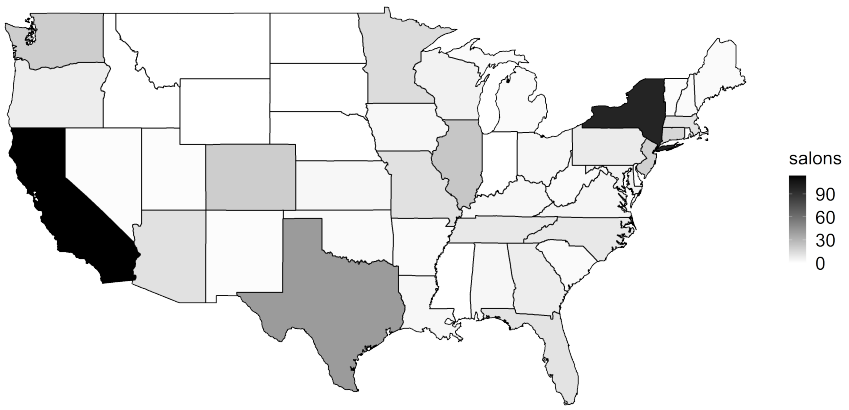
While the data are rich in terms of task content and worker assignments, information about workers is sparse. Worker demographics are not available. The software can track some compensation information (tips, commissions, employment relationships, and similar information), but these additional functions are not used consistently by client salons, as my discussions with the company and analysis of internal data revealed. Additionally, because the software is built for salons to manage their operations, it is not possible to track workers when they change firms, which prevents the use of standard techniques from labor economics to decompose firm and worker effects. Although workers can be observed at the same firm over time, many salons adopted the software relatively recently and as a result the time dimension is relatively short. For this reason, the analysis in this paper is mainly conducted at the establishment level. A licensed cosmetologist classified 20,560 of the service descriptions into six task categories. Descriptions labeled as

more than one category are split into component tasks (Appendix Section A.4), and I merge the sparse Extensions category into Blowdry/Style/Treatment, leaving the five task categories used throughout the paper. I label each task category based on the main type of task it represents, with the understanding that there is heterogeneity within a category. An additional 226 descriptions arose after the initial classification and were classified based on their similarity to the original 20,560 with a procedure described in Appendix Section A.4.

2.2 Descriptive Statistics

The data used in this section and Section 3 include all observed salon-quarters in which revenue per customer is positive. I exclude 2021 Q3, because I observe only part of the quarter. Establishments typically first appear in the data when they adopt the management software, but in some cases it appears that legacy data were imported. These legacy data are excluded. Additional details about the construction of the sample are provided in Appendix Section A.4. The data used to establish the stylized facts contain information on 507 hair salon establishments (representing 333 unique firms), 10,283 hair stylists, 1,995,928 customers and 11.2 million services performed. Figure 2 illustrates the distribution of salon establishments by state. Although the software is used by salons across the country, users are concentrated in New York and California.

Figure 2: Hair Salons by State



Note: The distribution of unique salon establishments by state. Includes all hair salons observed prior to 2021 Q3.

I aggregate the data to the salon-quarter level for analysis. Descriptive statistics at this level are provided in Table 2. The salons in the sample have an average quarterly revenue of $213,221 and an average of 13 employees. Johnson and Lipsitz (2022) study a sample of salon owners and report an average annual (not quarterly) revenue of $233,000 and an average of seven stylists. Given this, the sample in this paper should be viewed as a positively selected sample of salons, and

the heterogeneity in productivity and specialization observed in this sample likely understates the heterogeneity among hair salons nationwide. Such positive selection is reasonable given that salons must pay a subscription fee to access the software.

Table 2: Salon-Quarter Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Revenue	4,495	213,220.80	246,691.10	5.00	2,559,703.00
Employees	4,495	13.43	10.80	1	92
Customers	4,495	1,165.24	1,102.84	1	16,768
Share Haircut/Shave	4,495	0.39	0.23	0.00	1.00
Share Color/Highlight/Wash	4,495	0.40	0.20	0.00	1.00
Share Blowdry/Style/Treatment/Extensions	4,495	0.09	0.12	0.00	1.00
Share Administrative	4,495	0.05	0.11	0.00	1.00
Share Nail/Spa/Eye/Misc.	4,495	0.06	0.16	0.00	1.00

Note: Summary statistics for all salon-quarters used for the stylized facts.

In terms of task composition, salons typically spend most of their time on the Haircut/Shave and Color/Highlight/Wash tasks, but there are large differences across salon-quarters. Even though the relative intensity of tasks varies at different salons, most salons offer at least four of the five task categories in a given quarter. Throughout the paper, I refer to the task mix of a salon as the fraction of total time spent on each of the five tasks. I define a salon's price to be its total quarterly revenue divided by its total number of customers. The average price across all salon-quarters in the sample is \$199.[4]

3 Stylized Facts

This section presents four stylized facts about the relationship between task assignment within the establishment and productivity. Taken together, the data are consistent with heterogeneity in internal task specialization across establishments that is positively correlated with labor productivity, service quality, and proxies for management sophistication. These facts motivate modeling task specialization as chosen by strategic but organizationally unique establishments.

These facts require defining two concepts used throughout the paper. To begin, denote workers by the index m, establishments by the index j, and tasks by the index k.

Definition 1 *An establishment's observed organization, denoted by B_j, is a matrix where element $B_j(m,k)$ is the fraction of labor assigned to worker m and task k.*

4. I analyze establishments as offering a representative differentiated product (service) rather than multiple products (services).

Given establishment j's observed organization, I define an establishment's *generalist benchmark* as $G(B_j)(m,k) = \left(\sum_{m'} B_j(m',k)\right)\left(\sum_{k'} B_j(m,k')\right)$. This is the counterfactual task assignment that would be observed if the establishment randomly assigned workers to tasks, holding fixed the establishment-level marginal distribution of time across tasks and workers. In Figure 3 I provide an example of an observed organization and the corresponding generalist benchmark.

Figure 3: An Organization and the Corresponding Generalist Benchmark

Task Assignment (B_j)

Employee	Tasks 1	2	3	
A	1/2	0	0	1/2
B	0	1/4	0	1/4
C	0	0	1/4	1/4
Tot.	1/2	1/4	1/4	1

Generalist Benchmark ($G(B_j)$)

Employee	Tasks 1	2	3	
A	1/4	1/8	1/8	1/2
B	1/8	1/16	1/16	1/4
C	1/8	1/16	1/16	1/4
Tot.	1/2	1/4	1/4	1

Note: The left panel depicts a task-specialized salon, while the right panel depicts the corresponding generalist task assignment. Column sums represent the task mix, and row sums represent the fraction of work performed by each employee.

To study task specialization, I need to first define a notion of within-establishment specialization. I measure how far the observed task assignment is from the corresponding generalist task assignment, using the Kullback–Leibler divergence (denoted D_{KL}) as the notion of distance. In Appendix Section A.1, I show that this measure of task specialization can be micro-founded as the minimum amount of communication that must occur within the establishment for it to implement a given task assignment (Shannon 1948).

Definition 2 *The task specialization index* ***(s-index)*** *of establishment j with organization B_j is*

$$I(B_j) := D_{KL}(B_j||G(B_j)) = \sum_{m,k} B_j(m,k) \log\left(\frac{B_j(m,k)}{G(B_j)(m,k)}\right)$$

where use of the natural logarithm means that the unit of measurement is the nat; one nat is approximately 1.44 *bits.*

When an establishment randomly assigns workers to tasks, the s-index takes on its minimum value of 0. Importantly, because the s-index is defined relative to an establishment's generalist benchmark, its maximum value varies based on the distribution of time across tasks at the establishment level. For an establishment with five tasks and an even distribution of time across them, the maximum is approximately 1.61. For an establishment with all time spent on a single task, the maximum is 0 as specialization is not possible. The stylized facts are presented with the raw s-index, but the documented heterogeneity persists after normalizing the maximum possible

s-index to be 1 for each establishment. For reference, in the data, the maximum observed s-index is 1.02 while the minimum is 0.[5]

With this measure in hand, I present four stylized facts about productivity and internal organization. Throughout the rest of the paper, task assignments, and in particular the s-index, are assumed to be measured without error.[6]

Fact 1 *There is large dispersion in labor productivity and internal task specialization not accounted for by establishment size or other observables. Both labor productivity and internal task specialization are highly persistent within establishments.*

I measure labor productivity as total revenue divided by the total duration of all services. Total duration serves as a proxy for utilized labor, but a notable limitation is that wages are not observed, so it is not possible to adjust for labor quality directly. Concerns about labor quality are partially alleviated by the fact that most of the workers are likely in a single occupation (cosmetology) and work in the same high-end segment of the hair salon market.

Table 3 documents the result. In line with past work in the literature, I find large differences in productivity across otherwise similar salons. Remarkably, the ratio of labor productivity at the 75th percentile to that at the 25th percentile is 1.90, similar to productivity dispersion found by Syverson (2004) among manufacturing firms. There are also large differences in internal task specialization, measured by the s-index, among the salons in the sample. The ratio of the s-index between the 75th and 25th percentiles is almost 13-to-1. The distribution of the s-index, depicted in Figure 4 Panel A, roughly follows a power law, with a large number of generalized salon-quarters and a long right tail of specialized salon-quarters.

Table 3: Dispersion of Labor Productivity and Task Specialization

Statistic	N	Mean	Min	Pctl(25)	Median	Pctl(75)	Max
Labor Productivity	4,495	1.61	0.003	1.02	1.35	1.94	15.04
S-index	4,495	0.22	0.00	0.03	0.11	0.41	1.02

Note: The table describes the distribution of labor productivity and task specialization across salon-quarters. Labor productivity is measured as total revenue divided by total utilized labor, expressed in units of dollars per minute.

Most of the variation in both productivity and the s-index is not accounted for by establishment size, location, time, or task composition. The standard deviation in the productivity measure residualized for the task mix, establishment-size fixed effects, county fixed effects and quarter

5. In the model, I will allow each establishment to have a unique coefficient which converts the s-index into dollars.
6. Appendix Section A.20 provides evidence that measurement error is small.

fixed effects is 79.5% of the raw standard deviation. The standard deviation of the s-index residualized for the task mix, establishment-size fixed effects, county fixed effects and quarter fixed effects is 75.4% of the raw standard deviation.

Similar to past work (De Loecker and Syverson 2021), my measure of labor productivity is highly persistent within establishments. A regression of labor productivity on its one-quarter lag yields a coefficient of 0.848 (s.e. of 0.027) without establishment fixed effects and 0.666 (s.e. of 0.098) with establishment fixed effects. Task specialization shows similar persistence: a regression of the s-index on its one-quarter lag yields a coefficient of 0.954 (s.e. 0.006) without establishment fixed effects and 0.562 (s.e. 0.039) with establishment fixed effects.

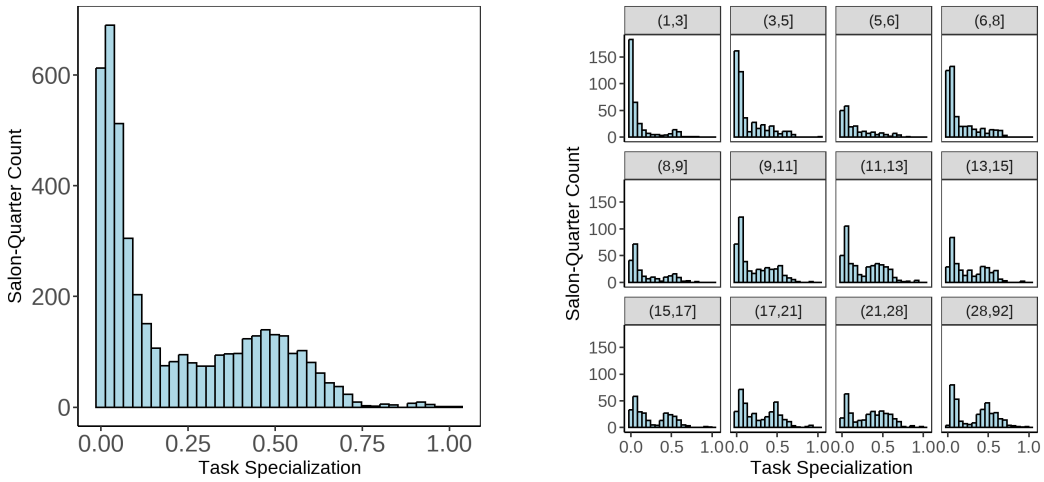
Internal task specialization varies even among establishments of the same size. I demonstrate this variation in Figure 4 Panel B, which displays histograms of the s-index among salon-quarters binned by employee count (12 quantile bins), excluding single-employee salon-quarters (for which the s-index is mechanically zero). There is significant dispersion across all size groups, which is difficult to reconcile with traditional models, where specialization differences typically arise only because firms anticipate spreading the fixed costs of specialization across a larger amount of output. I take this as evidence of organization costs that enter as marginal costs, and represent this force as a primitive feature of establishments in the model. However, it is important to acknowledge that the distribution of the s-index becomes more spread out as one moves from smaller to larger establishments in Panel B, suggesting that traditional scale-driven specialization is also an important force. Combining both motives for specialization is left for future work.

Fact 2 *Task-specialized salons are more productive than generalized salons.*

Table 4 documents this result via regressions of productivity on the s-index, while Figure 5 shows a binned scatter plot. There is a robust positive correlation between labor productivity and the s-index. A one standard deviation increase in the s-index is on average associated with a 0.28 standard deviation increase in labor productivity. To relate this finding back to the productivity dispersion documented earlier, the top 25% of salon-quarters by specialization on average generate $0.83 or 60% more revenue per minute than the bottom 25%.

Table 4 demonstrates that the correlation persists both in magnitude and statistical significance even after sequentially adding establishment-size fixed effects (indicator variables for each number of employees observed in the data spanning from 1 to 92), county fixed effects, quarter fixed effects and eventually ZIP code fixed effects. Notably, the coefficient on the s-index remains similar with and without fixed effects for establishment size. The coefficient drops by about half when ZIP code fixed effects are added, but remains economically and statistically significant. This

Figure 4: Histogram of the Task Specialization Index (S-Index)



(a) All Salon-Quarters **(b)** By Number of Employees

Note: The figure displays the distribution of the s-index unconditionally and among salon-quarters with similar numbers of employees. The distribution roughly follows a power law, with a large number of generalized salons and a long right tail of specialized salons. The upper bound of the s-index depends on the specific generalist benchmark, but for most salons it is around 1.

provides reassurance that differences in the customer base are not the sole driver of the association between specialization and the revenue-based productivity measure. Because the data are sparse in most ZIP codes, ZIP code fixed effects are similar to establishment fixed effects. Given the persistence in the s-index, it is therefore not surprising that ZIP code fixed effects reduce the coefficient on the s-index.

In the most aggressive specification, I control for establishment-size-by-ZIP-code fixed effects (2,197 indicators). The coefficient falls from 0.28 to 0.20 and remains statistically significant at the 0.05 level, even as R^2 rises from 0.08 to 0.79. This indicates that even if one controls for location-specific size effects that account for the majority of the variation in labor productivity, more than 70 percent of the association between the s-index and labor productivity remains. This finding suggests that while demand-side differences and economies of scale may play a role, they are not the main drivers of the productivity-specialization association. In Panel B of Figure 5, I make this argument graphically by showing the positive correlation between productivity and the s-index among establishments of similar sizes. Finally, I compute the uncorrelated variance share of labor productivity and the s-index conditional on establishment-size fixed effects to be 54% (Gibbons, Overman, and Pelkonen 2012). This can be interpreted as saying that conditional on establishment-size fixed effects, the s-index accounts for 54% of the jointly explained variance in labor productivity. As illustrated in Figure 1, there will be a causal link between establishment

Table 4: Regressions of Productivity on Task Specialization

Dependent Variable:	Revenue per Minute (standardized)						
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
S-Index	0.2764***	0.2663***	0.2864***	0.2997***	0.2954***	0.1625**	0.2023*
	(0.0487)	(0.0465)	(0.0460)	(0.0446)	(0.0439)	(0.0588)	(0.0867)
Color Task Mix		0.3838*	0.3041	0.4302*	0.4458*	0.3959	0.3689
		(0.1707)	(0.1593)	(0.1983)	(0.1987)	(0.2624)	(0.4703)
Blowdry Task Mix		0.5711	0.4336	0.2999	0.3388	0.4113	0.7876
		(0.3176)	(0.3134)	(0.3269)	(0.3322)	(0.4532)	(1.040)
Admin. Task Mix		-1.197***	-1.188***	-0.9629***	-0.9277***	-0.6381*	-0.2099
		(0.2439)	(0.2345)	(0.2325)	(0.2368)	(0.2929)	(0.4061)
Nail Task Mix		-0.2140	-0.2549	-0.0930	-0.0887	0.1028	-0.1649
		(0.2237)	(0.2161)	(0.2013)	(0.2022)	(0.3065)	(0.4102)
Fixed-effects							
Firm Size			Yes	Yes	Yes	Yes	
County				Yes	Yes		
Quarter-Year					Yes	Yes	Yes
Zip						Yes	
Zip-Firm Size							Yes
Fit statistics							
Observations	4,495	4,495	4,495	4,495	4,495	4,495	4,495
R^2	0.07642	0.11433	0.16708	0.40552	0.41751	0.55909	0.79269

Clustered (Establishment) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: Each column represents a regression of the labor productivity measure on the s-index. Task mix controls are the shares of total labor assigned to each task category, with all coefficients relative to the omitted haircut/shave category.

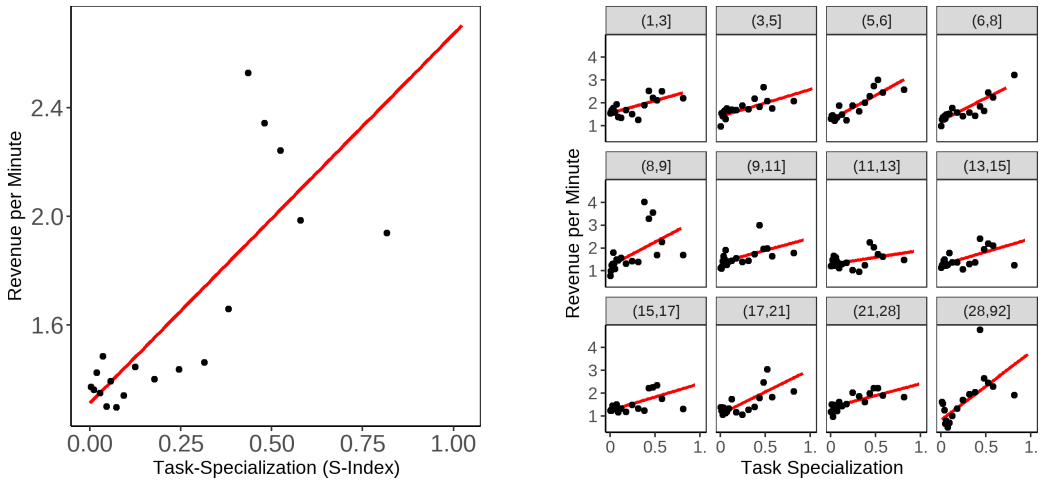
size and specialization even in models where specialization impacts only marginal costs.

Fact 3 *Task-specialized salons earn more revenue per customer and have a higher customer return rate compared with generalized salons.*

In Table 5, I study the components of the productivity-specialization association by regressing the number of customers, the revenue per customer and the future return rate of current customers on the s-index. I interpret revenue per customer as a proxy for the price of the average bundle of services purchased at the salon. I interpret the future return rate as reflecting perceived service quality.

After accounting for the number of employees, I find no statistically significant association between the number of customers and task specialization. This finding suggests that while specialized salons may be able to serve more customers because they have more employees, they are not serving more customers, holding fixed the number of employees. However, there are positive and statistically significant associations between the s-index and both revenue per customer and the customer return rate. This finding is inconsistent with task specialization reducing marginal

Figure 5: Binned Scatterplot of Productivity and Task Specialization



(a) All Salon-Quarters **(b)** By Number of Employees

Note: Salon-quarters are grouped into 19 bins defined by five-percentile increments of the s-index distribution (Panel A); Panel B applies the same bin edges within each employee-count group. Because there is a large mass of salon-quarters with an s-index of 0, the first bin includes the first 10 percent of observations. The average of the revenue-based labor productivity variable is displayed on the y-axis for each bin. The red line is from a linear regression of revenue-based labor productivity on the s-index.

costs but is consistent with specialization improving service quality.[7]

I now show evidence that the s-index is related to the management practices of a salon.

Fact 4 *Task-specialized salons engage in more teamwork and are earlier adopters of software features than generalized salons.*

This stylized fact provides suggestive evidence that the patterns of task specialization observed are related to deliberate management choices of salons. The s-index captures differences in how particular workers are assigned to tasks in a quarter. These differences include instances of teamwork, where multiple workers combine their skills to serve a single customer on a single day. But they also include cases where two workers perform different types of tasks on different days or with different customers. Although both are examples of specialization, teamwork involves more coordination and interaction, and therefore more deliberate management of workers. To measure teamwork, I divide the number of customer visits in which more than one worker is assigned by the number of customer visits in which more than one service is performed. Table 6 shows that a one standard deviation increase in the s-index is associated with a 0.65 standard deviation increase in teamwork controlling for task category composition, ZIP code fixed effects

7. I show in Appendix Section A.19 that under multinomial logit demand and marginal cost reductions, prices and specialization are negatively related.

Table 5: Decomposing the Productivity-Specialization Association

Dependent Variables:	Customer Count	Rev. per Customer	Customer Return Rate
Model:	(1)	(2)	(3)
Variables			
S-Index	-0.0178	0.1147*	0.0963**
	(0.0324)	(0.0576)	(0.0292)
Color Task Mix	-1.308***	1.906***	0.2399
	(0.2335)	(0.2705)	(0.1822)
Blowdry Task Mix	-1.146***	1.571*	-0.7907**
	(0.2086)	(0.6970)	(0.2920)
Admin. Task Mix	-0.2390	0.6471*	-0.3058
	(0.4444)	(0.2991)	(0.2384)
Nail Task Mix	0.3166	0.2260	-0.2454
	(0.3132)	(0.2419)	(0.1943)
Fixed-effects			
Quarter-Year	Yes	Yes	Yes
Zip	Yes	Yes	Yes
Firm Size	Yes	Yes	Yes
Fit statistics			
Observations	4,495	4,495	4,495
R^2	0.77588	0.70679	0.73735

Clustered (Establishment) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: The covariate of interest and each dependent variable are standardized by their respective standard deviations in the analysis sample. The return rate of customers is measured as the fraction of customers observed returning in a future quarter. Task mix controls are the share of total labor assigned to tasks of that category, with all coefficients relative to the left-out haircut category.

and quarter fixed effects.

Another way to understand whether the s-index is capturing underlying differences in management is to see whether it is correlated with how establishments engage with the software. I compute the number of days a salon takes to begin using a feature relative to the first salon observed using the feature. The four features I consider are general adoption of the software, tipping, prebooking and requesting staff. I then regress these measures of time to adoption on each salon's average s-index across time. As shown in columns 4–7 in Table 6, on average, task-specialized salons adopted all features earlier, suggesting task-specialized salons are more sophisticated users of the software. Salons with a high s-index also enter more unique text descriptions of services into the software (column 2 of Table 6) and offer a greater number of unique discounts on physical products (column 3 of Table 6).[8] Note that, where possible, additional controls are included, but because many of the regressions are effectively a single cross section, it is not possible to include

8. Because the data on physical products are limited, most of the paper focuses on the sale of services.

time-varying controls or fixed effects.

Table 6: Regressions of Teamwork and Management Software Engagement on the S-Index

Dependent Variables:	Teamwork	Service Descriptions	Product Discounts	Software Adopted	Tip Feature	Prebook Feature	Request Feature
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
S-Index	0.6536***	0.1026*	0.0893*	-0.2315***	-0.2948***	-0.2853***	-0.0844*
	(0.0489)	(0.0510)	(0.0430)	(0.0480)	(0.0555)	(0.0480)	(0.0394)
Color Task Mix	-1.107***	1.172***	0.3160				
	(0.2200)	(0.1860)	(0.2060)				
Blowdry Task Mix	-0.1427	0.1363	-0.6581				
	(0.4676)	(0.2457)	(0.3584)				
Admin. Task Mix	-0.8222	0.4277	-0.1542				
	(0.4624)	(0.3405)	(0.2472)				
Nail Task Mix	-0.7350**	0.5132*	0.3628*				
	(0.2743)	(0.2500)	(0.1576)				
Fixed-effects							
Zip	Yes	Yes	Yes				
Quarter-Year	Yes	Yes	Yes				
Fit statistics							
Observations	4,450	4,495	3,108	507	429	488	493
R^2	0.78982	0.75504	0.78943	0.05357	0.08209	0.08331	0.00724

Clustered (Establishment) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: Each column represents a regression of a variable related to management practices on the s-index. Columns 4–7 average the s-index across time for each salon and regress the average on adoption times among those salons observed adopting at some point. Product discount information is not available for all salons. The s-index and each of the dependent variables are standardized by dividing by the respective sample standard deviation.

4 Theory

In this section, I design a tractable model which generates dispersion in firm productivity both due to traditional demand shocks and Hicks-neutral cost shocks, and due to differences in internal task specialization across firms. This is accomplished via firm-specific organization costs based on the Kullback–Leibler divergence that are identifiable from the data, theoretically and computationally tractable, yet rich enough to allow for heterogeneous patterns of behavior across establishments operating in the same labor and product market. The model allows me to understand via counterfactual exercises how reorganization of work inside the firm mediates the productivity impacts of policy. Because worker and firm heterogeneity are crucial ingredients in the model, they are discussed in detail in Sections A.17 and 4.3, respectively.

4.1 Model

For ease of exposition, the model is specified for a single market and a single period, with the additional subscripts kept implicit. There are three important groups of objects in the economy: firms, indexed by $j = 1, ..., J$; *individual* workers, indexed by $m = 1, ..., M$; and task types, indexed

by $k = 1, ..., K$.

Firms and Tasks. The J firms differ in their organization cost parameter $\gamma_j \in \mathbb{R}_+$, discussed below. Each firm produces a single differentiated product, also indexed by j. Producing a single unit of good j requires $a_j \in \mathbb{R}_+$ units of labor. The fraction of total labor that must be assigned to each type of task is called the task mix ($\alpha_j \in \mathbb{R}_+^K$). Besides incurring an organization cost, a firm producing good j incurs a per unit cost $\alpha_j \cdot c + \omega_j$, where ω_j captures Hicks-neutral productivity differences across firms, and $\alpha_j \cdot c$ captures variable material costs.

Workers. The M workers are each described by an inelastic labor supply $l_m \in \mathbb{R}_+$, a skill set $\theta_m \in \mathbb{R}^K$, and a skill level $\bar{\theta}_m \in \mathbb{R}$. For expositional purposes, I define Θ as a matrix where rows are skill sets and columns are tasks. Worker m performs task k with quality $\bar{\theta}_m + \theta_m(k)$ per unit of labor. The skill set represents a worker's portfolio of skills, and determines both their absolute and their comparative advantage. There are $N \in \mathbb{N}_+$ distinct skill sets, indexed by i, with total labor L_i. Worker characteristics are common knowledge to all actors in the model.

Firm Strategies. Each firm simultaneously chooses the price of its product $p_j \in \mathbb{R}_+$, a relative labor demand for each worker $E_j \in \mathbb{R}_+^M$, and a task assignment for each worker $b_j \in \mathbb{R}_+^{M \times K}$. The sum of a firm's relative labor demand across all workers is 1: $\sum_m^M E_j(m) = 1$. The sum of a firm-worker task assignment across all task types is 1: $\sum_k^K b_j(m,k) = 1$.

Note that in equilibrium, the task assignments, labor demands and prices of all firms jointly imply a product-market demand for each firm. This product-market demand, combined with the relative labor demand E_j for each worker, implies a labor demand for each worker at each firm. The sum of the labor demand for each worker across all firms is the total labor demand for that worker. Although it is not technically necessary to have relative labor demand and task assignments be separate actions, this improves expositional clarity.

Throughout the paper I also refer to a firm's organization, which is the task assignment weighted by the relative labor demand: $B_j(m,k) := b_j(m,k) \cdot E_j(m)$. Organizations are sufficient for task assignments and relative labor demand. However, it clarifies exposition to work with organizations in some instances and task assignments and labor demands in others.

Organization Costs. Firm j's organization cost per unit of labor from worker m is proportional to the Kullback–Leibler divergence between the task assignment of that worker $\{b_j(m,k)\}_{k=1}^K$ and the task mix $\{\alpha_j(k)\}_{k=1}^K$. The total organization cost of firm j per unit of output is then given by the sum of the organization costs over all workers, multiplied by the required labor and the firm-specific organization cost parameter: $a_j \gamma_j \sum_{m=1}^M E_j(m) D_{KL}(b_j(m,k) || \alpha_j)$. I assume that the firm bears no organization cost for any workers not hired, that is, $E_j(m) D_{KL}(b_j(m,k) || \alpha_j) = 0$ if

$E_j(m) = 0$.

This functional form for organization costs has two important properties. First, it is proportional to the s-index introduced earlier, meaning it is a firm-level measure of task specialization describing how far a chosen task assignment is from a generalist benchmark where workers are randomly assigned to tasks within the firm. Second, it can be microfounded as the minimum amount of information, measured in nats per unit of labor, that the firm must communicate to each worker about that worker's task assignment.[9] Both results are shown formally in Appendix Section A.1.

Labor Market. The labor market is competitive, with worker-specific wages $w \in \mathbb{R}_+^M$ per unit of labor. The labor market for worker m clears if total labor demanded from worker m across all firms is equal to worker m's labor supply (l_m).

Consumers and Demand. There is a mass of consumers interested in purchasing at most one of the J products. Consumers observe firm task assignments and prices prior to purchase. Consumer z's utility for good j is represented by the logit utility function $u_{z,j} = \xi_j + \nu_j - \rho p_j + \epsilon_{z,j}$, where ξ_j is the average quality of the tasks performed to produce product j under the firm's task-assignment strategy, multiplied by required labor: $\xi_j := a_j \sum_{i,k} \theta_i(k) B_j(i,k)$, ν_j captures the components of quality that cannot be controlled by the firm, $\epsilon_{z,j}$ captures idiosyncratic consumer preferences over products, and ρ captures consumer price sensitivity. I assume $\epsilon_{z,j}$ is distributed as i.i.d. type-I extreme-value random variables across consumers and products.[10] The outside option for consumers is assigned index $j = 0$, and its utility is normalized to $u_{z,0} = \epsilon_{z,0}$.

Equilibrium. The equilibrium concept I use consists of two conditions. First, firm labor demand, task assignments and pricing strategies must form best responses to all other firm strategies at wage vector w. Second, firm labor demands at wage vector w must clear the labor market for each of the M workers. Note that for any fixed wage vector w, the model is a well-defined game, and the first part of the equilibrium definition amounts to a Nash equilibrium. I refer to this game as the fixed-wage subgame.

4.2 Understanding Worker Skill Heterogeneity

The model decomposes worker skills into a vertical dimension, skill level, and a horizontal dimension, skill set. I show that the skill set is the important dimension of worker skill which determines

9. Throughout, the natural logarithm is used in the Kullback–Leibler divergence so that the s-index is measured in nats. One nat is equal to around 1.44 bits.

10. Most theoretical results extend to other demand systems, including nested logit and mixed logit with a non-random price coefficient.

task assignments. In this sense, the model is primarily capturing horizontal specialization, or task assignments driven by comparative advantage. In the model exposition, the skill set is left unrestricted, and Appendix Section A.17 shows that skill sets can be redefined to capture skill level differences and cases where consumers value tasks differently.

I show that restrictions on skill sets are necessary to guarantee unique task-assignment strategies and to identify the model from task assignment data alone. The restrictions amount to requiring that skill sets are linearly independent after multiplication and exponentiation and that skill sets are linearly independent after adjusting for wages. The first condition can be understood as requiring the existence of comparative advantage, while the second can be understood as requiring that wages not be such that skills are too similar. One skill set structure that typically satisfies both is single-task specialists, where each skill set holds an absolute advantage in one task, and all non-specialists perform that task with the same lower proficiency.[11] But the restrictions allow for many different patterns, including superstar generalists (one skill set holding an absolute advantage in all tasks) and multi-task specialists.

A related question is whether the space of worker skills can be well-represented by worker types that can be monotonically ranked across all tasks. This is an additional restriction on top of the skill set restriction imposed in the main estimation of the model. It is relevant because it is close in spirit to the skill parameterizations in Haanwinckel (2023) and Teulings (1995). I estimate the model with this restriction and provide comparisons of the skill parameters in Appendix Section A.18. If the restricted estimates are far from those in the main text, this indicates that monotonicity meaningfully alters the economic content of the model.

4.3 Firm Heterogeneity

Firms differ in their required labor, task mix, Hicks-neutral productivity, exogenous product quality and organization cost. In Section 4.4, I show that the task mix and organization costs are what determine firm task assignments for any fixed wages. As a result, the other dimensions of firm heterogeneity will impact task assignments (both own and competitors') only indirectly by influencing market shares and therefore the wages of different skill sets. The other dimensions are therefore included only to allow for traditional types of productivity heterogeneity, and to allow the model to be brought to the data.

I interpret heterogeneous organization costs as a reduced-form representation of the many

11. This is satisfied when the skill sets can be stacked into a symmetric positive definite matrix; see Appendix Section A.10

organizational frictions within a firm that make perfect specialization difficult. The organization costs can therefore be viewed as a convenient way to incorporate firm-specific specialization costs in equilibrium. In Appendix Section A.1, I provide a microfoundation for the Kullback–Leibler functional form. Under that microfoundation, the firm-specific coefficient γ_j represents both the ability of the owner to communicate assignments effectively and the opportunity cost of the owner's time.

Although the relative amount of each task performed is allowed to vary across firms via the task mix α_j, an implicit assumption of the model is that this variation is exogenous. Specifically, firms view the task mix α_j as a constraint that they must satisfy when assigning workers. This amounts to assuming a Leontief task-based production function, which has been commonly used in similar models, notably Adenbaum (2022) and Haanwinckel (2023). This does restrict equilibrium behavior, and as a result I view allowing the firm to substitute across tasks as an important next step. In the context of hair salons, where tasks are often sold directly to consumers, a natural approach would be to allow firms to influence the task mix by setting service-specific prices.

4.4 Theoretical Results

This section has three goals. First, I characterize the essential heterogeneity across firms and workers that drives equilibrium behavior. Propositions 1 and 2 show that the key attribute of workers is their skill set, and it is essentially without loss to consider each firm as assigning tasks to a representative worker of each skill set. Proposition 3 shows that the important attributes of firms are their organization cost coefficient and their task mix. Second, I show in Proposition 4 and Corollary 1 that the model generates patterns consistent with the stylized facts. Third, I characterize the equilibrium properties needed to identify and estimate the model. Specifically, Theorem 1 characterizes the logit-like form of task assignments for all workers and firms, while Proposition 5 proves that, under mild conditions, fixed-wage subgames feature a unique equilibrium.

In the model, firms demand labor from specific workers and assign specific workers to tasks. This allows the model to be directly mapped to the empirical application, where rich information on how specific workers use their time is available, but direct information about worker skills is not. The cost of this realism is that whenever there are many workers in a labor market (which is almost always the case), task-assignment strategies are high-dimensional objects. The firm must choose a portfolio of workers and assign tasks given this portfolio. The first step in the theoretical analysis is to understand how firms make this choice. The key insight is that while firms can in principle make task assignments depend on all three dimensions of worker heterogeneity (i.e.,

labor supply, skill level and skill set), in equilibrium, task assignments at a given firm are the same for all workers with the same skill set.

Proposition 1 *In any equilibrium task-assignment strategy, all workers at the same firm with the same skill set are assigned the same distribution of time across tasks.*

The proof is provided in Appendix Section A.5; however, the intuition is straightforward. The firm can engage in a very complicated division of labor by specifying different task assignments along all the dimensions of worker heterogeneity. However, among workers with the same skill set, tailoring task assignments, holding labor demand fixed, has no benefit but is costly because it requires more organization. This has implications for wages.

Proposition 2 *In any equilibrium, all workers with the same skill set and skill level are paid the same wage per unit of labor, and wages are such that firms are indifferent between all workers with the same skill set but different skill level.*

The proof is provided in Appendix Section A.6. Workers with the same skill set but different skill level are ranked by absolute advantage. The proposition establishes that workers extract the value of their absolute advantage via wages in a way that perfectly offsets the productivity improvements. Specifically, for any two workers m, m' with the same skill set but different skill levels, wages are such that $w_m - w_{m'} = \rho^{-1}(\bar{\theta}_m - \bar{\theta}_{m'})$. As a result, firms are indifferent between all workers with the same skill set even when they differ in skill level. Firms that happen to employ workers with a high skill level will produce higher-quality products but will pay out the revenue from quality improvements to workers. In this way, the model can accommodate vertical skill differences without loss of tractability. This is helpful in settings where there are wage differentials among workers with the same skill set.[12]

It should be noted that Proposition 2 requires the maintained assumption that tasks enter consumer utility with equal weight. When this is not the case, a vertical specialization force will arise. Specifically, when a firm is faced with two workers with the same skill set but different skill levels, it will assign more of the important (high utility weight) task to the worker with the higher skill level. How much a firm differentiates its task assignments across skill levels will depend on the firm's organization cost parameter, and as a consequence, firms will value skill level differences differently and wages of two workers m, m' with the same skill set but different skill levels will not satisfy $w_m - w_{m'} = \rho^{-1}(\bar{\theta}_m - \bar{\theta}_{m'})$. Appendix Section A.17 shows that skill level heterogeneity and task utility weights can be represented by redefined skill sets, and in this way vertical

12. With wage data, the distribution of skill levels within skill sets could be estimated.

specialization can be incorporated. However, there are important limits: the theory and the data generally do not allow more worker skill sets than tasks. Thus, if a researcher has already specified the maximum number of skill sets ($N = K$), Proposition 2 implies that unmodeled skill level heterogeneity is relatively benign under homogeneous utility weights. But if utility weights are not homogeneous, skill level heterogeneity is not benign and the skill sets cannot be reformulated to address it.

As these propositions make clear, there are many equilibria, but the actors in the model (i.e., consumers, firms, workers) are indifferent between them.[13] Further, all equilibria imply the same task assignments and the same labor productivity. For these reasons, and with some abuse of notation, I recast all firm strategies in terms of worker skill sets (rather than worker identities), which are indexed by i. Thus, $E_j(i)$ is the relative labor demand for skill set i at firm j, and $\{b_j(i,k)\}_{k=1}^{K}$ is the task assignment of skill set i at firm j. I analyze firms as assigning tasks to a representative worker of each skill set, with the understanding that these tasks will be executed in equilibrium by potentially multiple actual workers with different labor supplies and skill levels. I recast the wage vector to be length N, with one wage for each skill set, with the understanding that this wage reflects both the wage premium of that skill set and the wage premium due to the average vertical skill level of workers of that skill set. Given the type-1 extreme-value distribution of consumer taste shocks (McFadden 1973), I can write firm j's profit-maximization problem as

$$\max_{p_j, b_j, E_j} \frac{exp(\xi(b_j, E_j) - \rho p_j)}{1 + \sum_{j'} exp(\xi(b_{j'}, E_{j'}) - \rho p_{j'})} \left(p_j - a_j \gamma_j \sum_i E_j(i) D_{KL}(b_j(i,\cdot) || \alpha_j) - a_j \sum_i w_i E_j(i) - \alpha_j c - \omega_j \right) \tag{1}$$

s.t. $\sum_i E_j(i) b_j(i,k) = \alpha_j(k)\ \forall k$.

Though pricing and internal organization may appear intertwined, the problems naturally separate in the following way.

Proposition 3 *A relative labor demand and task assignment are profit-maximizing if and only if they solve*

$$\min_{b_j, E_j} \gamma_j \sum_i E_j(i) D_{KL}(b_j(i,k) || \alpha_j) + \sum_i E_j(i) w_i - \rho^{-1} \sum_i E_j(i) \sum_k \theta_i(k) b_j(i,k) \tag{2}$$

s.t. $\sum_i E_j(i) b_j(i,k) = \alpha_j(k) \forall k$.

The proof is provided in Appendix Section A.7. The result implies that while task assignments impact prices, prices do not impact task assignments. Further, task assignments are impacted by

13. The difference across these equilibria is which individual worker is employed at which individual firm.

competitors' choices only via wages. Prices, however, depend on own task assignments, competitor task assignments and competitor prices. This separation makes the model computationally tractable, and it also allows the researcher to shut down internal organization and have a benchmark for understanding how firms would behave if they could not reorganize. I call this benchmark the reallocation equilibrium, and use it extensively in Section 7. Also, I call the minimized objective from Equation 2 a firm's endogenous quality-adjusted cost. I call $\omega_j - \rho^{-1}\nu_j$ a firm's exogenous quality-adjusted cost. Results from the literature on multinomial logit demand make clear that the sum of a firm's exogenous and endogenous quality-adjusted cost is sufficient for a firm's behavior in the pricing game; even further, it is "a measure of a firm's ability to provide utility to consumers" (Nocke and Schutz 2018). With these results in hand, I can now show directly how organization costs drive observed task specialization.

Proposition 4 *The s-index required to implement a profit-maximizing task assignment is equal to the* ***observed*** *s-index (I_j) and is strictly decreasing in γ_j for all values of firm-level heterogeneity ($a_j, \alpha_j, \nu_j, \omega_j$) until it reaches 0.*

The first part of the proof follows from the fact that all workers with the same skill set have the same task assignment, and the second part follows from applying the envelope theorem to Equation 2.[14] The proposition has important empirical content. Specifically, the observed s-index (I_j) can be inverted to recover unobserved organization costs (γ_j) despite significant firm-level heterogeneity. This is similar in spirit to how Garicano and Hubbard (2016) invert the span of control to recover a manager's skill. This property plays an important role in the identification proof and estimation strategy.

Proposition 4 is true even though firms feature several other dimensions of heterogeneity. This is because, once wages are fixed, demand shocks, cost shocks, and required labor turn out to have no impact on specialization within the firm. The task mix delineates the exact proportion of each task the firm requires to operate, and as a result it directly impacts specialization. However, because the task-mix is observed and the s-index continues to be strictly decreasing in γ_j for different values of the task-mix, γ_j can still be recovered via inversion. In this way, the model, combined with data on task assignments within the firm, allows organization costs to be identified even within firms of the same size. A firm may be large because it has positive demand shocks or negative cost shocks. Or it may be large because it has low organization costs. Both large and small firms can be more or less specialized.

14. The formal proof is in Appendix Section A.8.

A corollary of Proposition 4 is that the model replicates the positive correlation between the s-index and firm productivity documented in Section 3.

Corollary 1 *All else constant, firms with a lower organization cost parameter (γ_j) have a higher s-index, a larger market share, a higher profit, and a higher productivity in the sense of a lower quality-adjusted cost.*

The proof is provided in Appendix Section A.8. Recall that γ_j represents the management technology, relationships, knowledge and practices specific to the firm which determine the cost of communicating task assignments to workers. Corollary 1 implies that more organizationally efficient firms are larger and more profitable, and can produce better-quality goods at a lower cost. This is in line with the findings of Kuhn et al. (2023), who use surveys and administrative data to show that more coordinated or specialized firms are more profitable. In the model, firms can be large or small for reasons unrelated to organization costs. However, there is no complementarity between expected firm size and task specialization.[15]

Theorem 1 *Profit-maximizing task assignments for any worker with skill set i at firm j can be expressed as*

$$b_j(i,k) = \alpha_j(k)\frac{exp[\gamma_j^{-1}(\rho^{-1}\theta_i(k) - w(i))]}{\sum_{i'} E_j(i')exp[\gamma_j^{-1}(\rho^{-1}\theta_{i'}(k) - w(i'))]}, \tag{3}$$

*and they satisfy the following properties: (**Relative Law of Demand**) as $w(i)$ increases, skill set i's share of labor at firm j ($E_j(i)$) decreases, (**Incomplete Specialization**) all workers employed by firm j ($E_j(i) > 0$) spend a strictly positive amount of time on all tasks performed at the firm ($\{k|\alpha_j(k) > 0\}$), and (**Maximum Coworker Diversity**) either the number of skill sets employed at a firm is less than or equal to the number of tasks, or there exists another profit-maximizing task assignment strategy where this is true.*

The proof, provided in Appendix Section A.9, involves manipulating the first-order conditions of the firm and relying on the equivalence between Equation 2 and a class of problems that are well-studied in computer science and information economics. Equation 3 reveals that the profit-maximizing task assignment balances the tasks that need to be done (α_j), the firm's organization costs (γ_j), wages (w_i) and skill sets (θ_i). The result regarding incomplete specialization is obtained because of the functional form of the organization cost, specifically, the Kullback–Leibler divergence. Because many of the workers in my context are in the same occupation (cosmetology) and perfect specialization at the firm level is uncommon, it is reasonable to assume incomplete specialization here. However, in other contexts, such as manufacturing settings with assembly lines, it is

15. This is a major difference between this paper and Adenbaum (2022), where firms specialize more precisely because they expect to spread fixed costs over more output.

not. In these cases, alternative cost functions can be used which allow for complete specialization of some workers. Because the relative labor demand of each skill set ($E_j(i)$) is endogenous, the expression given for optimal jobs is not a closed-form solution. Despite this, I will show that the expression will enable identification and greatly simplify computation of equilibria.

Proposition 5 *If* $\det(exp[c\Theta]) \neq 0$ *for all* $c > 0$*, then there exists a unique Nash equilibrium in prices* (p_j)*, task assignments* (b_j) *and relative labor demands* (E_j) *for every fixed-wage subgame.*[16]

The determinant condition $\det(exp[c\Theta]) \neq 0$ requires that, for any level of organization costs, there exists sufficient variation in skill sets.[17] This is most clearly seen when there are two worker skill sets and two tasks, in which case the condition reduces to $\theta_1(1) - \theta_1(2) \neq \theta_2(1) - \theta_2(2)$, which can be interpreted as requiring that there exist comparative advantage across workers. The proof is provided in Appendix Section A.10. I utilize a uniqueness result in the rational inattention literature (Matějka and McKay 2015) and Nash equilibrium uniqueness of Bertrand pricing games with multinomial logit demand (Caplin and Nalebuff 1991). The proposition has two immediate implications. First, when strategies are formulated in terms of worker skill sets and there is sufficient variation in comparative advantage, each fixed-wage subgames has a unique equilibrium. Second, because of this uniqueness for any fixed wage, equilibria of the full model are pinned down by their wage vectors. Put another way, if wages are known, all firm strategies are uniquely determined. This property justifies treating wages as parameters to be estimated in the empirical application. However, this property does not establish equilibrium uniqueness of the full model, as more than one wage vector may clear the market.

4.5 A Simple Example

To illustrate the task-assignment decision at the heart of the model, consider the simple case with three worker skill sets, wages fixed at $w = (20, 15, 21)$, $\rho = 1$, and skill sets given explicitly below.[18] Proposition 3 implies that firms consider quality-adjusted wages (or wage-adjusted skills), which can be stacked into a matrix:

$$\begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 23 & 19 & 15 \\ 15 & 15 & 15 \\ 15 & 19 & 26 \end{bmatrix} \quad \Longrightarrow \quad \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} - \rho w = \begin{bmatrix} 3 & -1 & -5 \\ 0 & 0 & 0 \\ -6 & -2 & 5 \end{bmatrix}$$

16. Another sufficient condition is that the skill set vectors can be arranged into a symmetric positive definite matrix Θ.
17. The notation $exp[x]$ represents the element-wise exponential.
18. These parameter values are based on an example in Csaba (2021).

Consider an economy populated by two firms $j = 1, 2$ which both have a uniform task mix: $\alpha_1 = \alpha_2 = (1/3, 1/3, 1/3)$. Suppose firm 1 has no organization cost: $\gamma_1 = 0$. Firm 1 will simply choose the best worker for each task at prevailing wages, which corresponds to choosing, in each column, the row containing the highest number. In this case, it will assign all of task 1 to skill set 1, all of task 2 to skill set 2, and all of task 3 to skill set 3, resulting in a relative labor demand of $E_1 = (1/3, 1/3, 1/3)$. Suppose firm 2 has a sufficiently high organization cost ($\gamma_2 \gg 0$). It can be shown that firm 2 will choose a generalist structure, and assign all three tasks to skill set 2, generating a relative labor demand of $E_2 = (0, 1, 0)$.

This two-firm example illustrates that under the model, because firms with different organization costs assign tasks differently, they often have different labor demand functions. In the specific environment sketched in the last paragraph, firms with low organization costs like firm 1 will tend to employ a balanced workforce because they can exploit the specific comparative advantage of each worker type. Firms with high organization costs like firm 2 cannot exploit these comparative advantages. They resort to using only the generalist worker skill set. When the market is populated by a larger share of high-organization-cost firms, labor market equilibrium will require a high wage for generalist workers. When the market is populated by low-organization-cost firms, wages will tend to be more equal across worker skill sets.

Another point this illustrates is that within the model, each establishment is solving a miniature matching problem, in the spirit of Lindenlaub (2017). Absent organization costs, task assignment is a simple linear program because workers are perfect substitutes at each task. Therefore, optimal task assignments absent organization costs will involve assigning all of a task to the worker most skilled at that task. The establishment absent organization costs then operates as a frictionless team, with each worker's horizontal skills perfectly complementing the skills of all coworkers. When organization costs are added, the establishment must compromise on this ideal allocation, and begin assigning workers to similar batches of tasks. The establishment loses out on the horizontal complementarities. Once organization costs become large enough, task assignments become the same across all coworkers and in many situations the establishment therefore only hires one type of worker.

5 Empirical Application

This section shows that all parameters of the model are identified, with the exception of the organization cost parameters of zero-s-index firms, for which only lower bounds are identified. The identification proof is sketched, along with an outline of the estimation procedure.

5.1 Identification

For identification, I assume that the econometrician observes the following data. For workers, the econometrician observes only the task assignment distribution of each worker ($\{b_j(m,\cdot)\}_{m=1}^{M}$).[19] For each firm, the econometrician observes required labor, task mix, price and market share $\{(a_j, \alpha_j, p_j, s_j)\}_{j=1}^{J}$. I consider wages, skill set parameters Θ, and the type of each firm (in terms of γ_j) and worker (in terms of θ_m) as parameters to be estimated, not data. I collect all parameters of the model into three groups. First, there are market parameters, denoted by Ω: wages, material costs, consumer price sensitivity and skill set vectors. Second, there is the skill set group membership of each worker (i.e., which of the N skill sets they possess). Third, there are the organization cost parameters of all firms ($\{\gamma_j\}_{j=1}^{J}$).

I make three identifying assumptions. Denote e as an N-length vector of ones. First, idiosyncratic product quality (ν_j) and cost shocks (ω_j) are mean zero and independent of organization cost and task mixes. Second, I make a standard full rank assumption on several moments which are interactions of model objects.[20] Third, I assume every skill set is employed within a single connected set of firms, where two firms are considered connected if they both perform all K tasks and they share at least two skill sets. If there exists a firm which employs all skill sets and performs all tasks, this assumption is satisfied. I invoke this simpler but stronger sufficient condition during estimation, but not for identification.[21]

Theorem 2 *If the set of wage-adjusted skill vectors $\{\theta_i - \rho w_i e\}_{i=1}^{N}$ is linearly independent,[22] the market parameters (Ω) and the amount of labor of each skill set are identified. The organization cost parameters (γ_j) and the skill sets of all workers ($\{\theta_m\}_{m=1}^{M}$) at firms with a strictly positive s-index ($I_j > 0$) are identified. Lower bounds on the organization cost parameters of firms with an s-index of 0 are identified.*

Theorem 2 is proven formally in Appendix Section A.11. The key challenge is identifying worker types across firms that have unobserved organization cost differences. This challenge is overcome with a connected-set argument similar in spirit to those used in AKM-style decompositions (Abowd, Kramarz, and Margolis 1999), except that it relies on firms sharing pairs of similar workers in a single period rather than firms sharing the same worker in different periods.

19. Importantly, the econometrician cannot access demographic or wage data about individual workers.

20. These are the rank conditions: $rank\{E[\left(\{B_j(i,k)\}_{i,k}^{N,K} \quad a_j\frac{\gamma_j}{\gamma_1}I_j\right)\left(\{B_j(i,k)\}_{i,k}^{N,K} \quad a_j\frac{\gamma_j}{\gamma_1}I_j\right)']\} = N \times K + 1$ and $rank\left\{E\left[\left(a_j\{\frac{E_j(i)}{\sum_{i'}E_j(i')}\}_i^N \quad \frac{\gamma_j}{\gamma_1}I_j \quad \alpha_j'\right)\left(a_j\{\frac{E_j(i)}{\sum_{i'}E_j(i')}\}_i^N \quad a_j\frac{\gamma_j}{\gamma_1}I_j \quad \alpha_j'\right)'\right]\right\} = N + K + 1$

21. I must also rule out correlated mixed hiring strategies across skill levels within a worker skill set.

22. The requirement on wages is relatively weak given maintained assumptions, because if the set of skill vectors is linearly independent, wage-adjusted skills will be linearly independent except for a measure-zero set of wages.

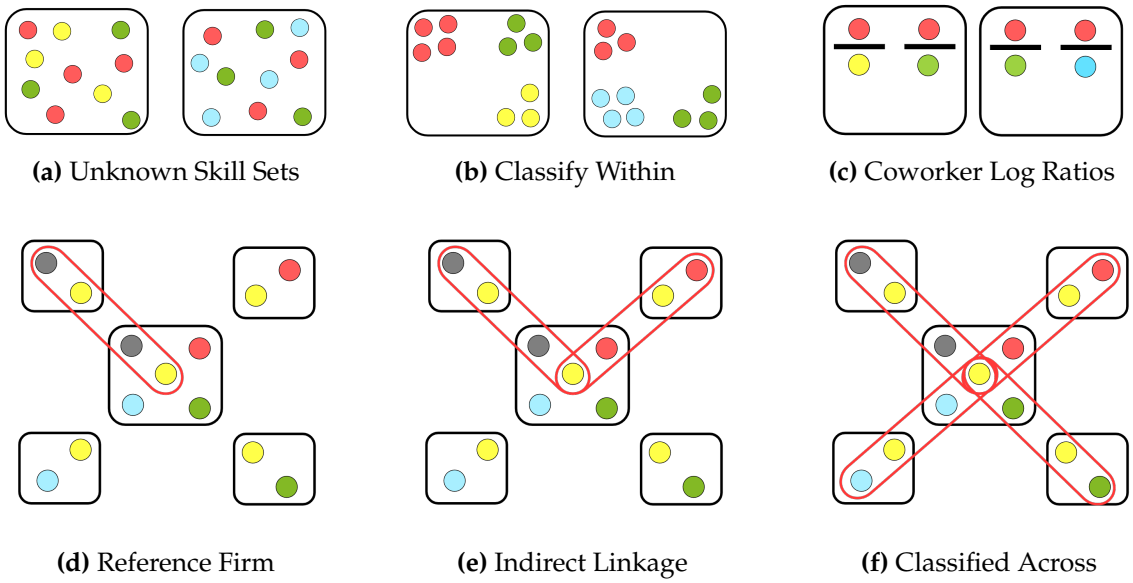
I now briefly sketch the proof. First, observe that within a firm, all workers with the same skill set share the same task assignment. This implies that within a firm, workers can be grouped together based on the similarity of their task assignments, as in Panel B of Figure 6. The next step is to classify workers into skill sets across firms, but task assignments across firms are not directly comparable due to firm-specific organization costs and task mixes. As shown in Panel C, the solution is to divide one worker's distribution of time across tasks by that of a coworker known to have a different skill set, take the logarithm, and normalize by the norm to obtain the vector:

$$\frac{\log\left(\frac{b_j(t_l,k)}{b_j(t_{l'},k)}\right)}{\left|\left\{\log\left(\frac{b_j(t_l,k')}{b_j(t_{l'},k')}\right)\right\}_{k=1}^{K}\right|} = \frac{[\theta_{t_l}(k)-\rho w(t_l)]-[\theta_{t_{l'}}(k)-\rho w(t_{l'})]}{\left(\sum_{k'=1}^{K}\left([\theta_{t_l}(k')-\rho w(t_l)]-[\theta_{t_{l'}}(k')-\rho w(t_{l'})]\right)^2\right)^{1/2}}$$

which does not depend on any firm-specific heterogeneity. The full proof shows that when worker types are sufficiently differentiated, these ratios of task assignments will match across firms if and only if the skill sets of the compared workers match. Comparison with coworkers therefore allows two pairs of coworkers to be matched across firms, as in Panel D of Figure 6. Repeating this process allows a common classification of workers into skill sets among all firms in a connected set, as illustrated in Panels E and F. The diagram provides an example where there exists a firm which employs all skill sets. This is a sufficient but not necessary condition for a connected set, and it is the one I invoke during estimation.

Once workers are grouped into skill sets, the same comparison of coworkers identifies the organization cost parameters of all firms in the connected set relative to a single reference firm. With these in hand, all other parameters are identified from two linear systems of moment conditions, one of which involves prices and the other of which involves market shares. The systems have exactly the same number of equations as parameters to be identified; thus the full rank assumption on the moments guarantees identification. The size of the correlation between prices net of markups and the s-index adjusted for a firm's relative organization cost is informative about the magnitude of organization costs. Traditional Hicks-neutral productivity shocks impact prices but are uncorrelated with the s-index, while traditional demand shocks impact prices only through markups.

Figure 6: Identification of Worker Skill Sets



(a) Unknown Skill Sets **(b)** Classify Within **(c)** Coworker Log Ratios

(d) Reference Firm **(e)** Indirect Linkage **(f)** Classified Across

Note: A graphical representation of how worker skill sets are identified via task assignments. Balls represent workers, and colors indicate their skill sets. Boxes are firms.

5.2 Estimation

It is necessary to have enough penetration of the software in a market to effectively estimate the model. Therefore, the model is estimated for the three counties with the most active software users (New York County, NY; Cook County, IL; and Los Angeles County, CA) and 12 quarters (2018Q1 through 2021Q2 excluding 2020Q2 and 2020Q3, which were impacted by the COVID-19 pandemic). This subset represents 987 of the full sample of 4,495 salon-quarters. The summary statistics in Table 7 reveal that this subset is positively selected on average relative to the rest of the data, with slightly more employees, revenue and task specialization.[23]

Model estimation requires taking a stand on the size of the potential market. I assume that the potential market is the resident population for each county-year obtained from the U.S. Census Bureau (U.S. Census Bureau 2010, U.S. Census Bureau 2024). I divide the number of unique customers at a salon-quarter by the number representing the potential market to obtain market shares. I assume the fraction of consumers choosing the outside option to be the fraction of consumers from the associated county-quarter who do not have a salon transaction in the Consumer Expenditure Survey (U.S. Bureau of Labor Statistics 2020, U.S. Bureau of Labor Statistics 2021).

I make several assumptions that allow me to exploit the panel nature of the data to improve

23. The effective number of salon-quarters that enter the GMM objectives is 472 since the market-level parameters can only use salon-quarters where all tasks are performed and at least two worker skill sets are employed.

Table 7: Estimation Sample Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Revenue	987	311,896.50	363,332.30	81.00	2,559,703.00
Employees	987	16.67	13.49	1	76
Customers	987	1,297.73	1,188.31	2	7,420
S-Index	987	0.24	0.22	0.00	0.92
Share Haircut/Shave	987	0.38	0.24	0.00	1.00
Share Color/Highlight/Wash	987	0.42	0.21	0.00	0.93
Share Blowdry/Style/Treatment/Extensions	987	0.11	0.14	0.00	1.00
Share Administrative	987	0.05	0.15	0.00	1.00
Share Nail/Spa/Eye/Misc.	987	0.03	0.10	0.00	0.79

Note: Summary statistics for the subset of data used to estimate the structural model.

power. I assume the 25 skill set parameters and price sensitivity are fixed across time within a county. I assume that the wages of each worker skill set evolve in parallel across time in each county. This has several benefits. First, it mechanically reduces the number of wage parameters that need to be estimated. Second, when this assumption is true, the model implies that wage time trends do not directly affect task assignments. Therefore I can group workers into skill sets across time using task assignments.[24]

I also include county-quarter fixed effects in both the marginal cost and quality equations, and a time-varying coefficient on required labor that allows the average wage of cosmetologists to vary flexibly across time and counties. Moreover, I restrict wages to be at least the minimum wage for small employers in that county and quarter. I allow firms' organization costs (γ_j), task mix (α_j) and required labor ($\bar{a}_j$) to vary across quarters. Finally, I assume that the material cost parameters vary across quarters but not across counties.

The estimation procedure, detailed in Appendix Section A.12, follows the spirit of the identification proof. I classify workers into skill sets within a firm using similarity of task assignments. I classify worker skill sets across firms by finding, for each firm, the labeling of workers that minimizes the distance between all pairs of coworkers and their counterparts at a reference firm that employs all worker skill sets. I apply Lidstone smoothing to task assignments within a salon so that all workers in a salon perform a positive amount of each task performed at the salon.[25] I instrument for prices in the demand equation using a variable constructed from the producer price index of hair dye interacted with the share of labor assigned to the hair dye task. First-stage estimates, standard tests of instrument strength, and alternative instrument choices are provided in

24. This is because the identification argument used to estimate the model differences out any shared wage trend.
25. Even with data generated by the model, there is always a positive probability a worker never performs a task.

Appendix Section A.2. With the estimated skill matrix in hand, I then estimate wages in each market to match the average share of labor demanded from each worker type, inverting the s-index to recover each firm's organization cost parameter for each wage guess. To simplify and speed up inversion, I use a globally convergent contraction mapping described in Appendix Section A.14. The other parameters in the pricing equation (material costs, wage level, etc.) are estimated as described in the proof, with the exception that wage levels are constrained so that implied wages of all workers are at least the minimum wage in that county-quarter.

The most computationally difficult part of estimation is finding wages that fit the observed worker skill set shares due to complicated reorganization responses within establishments. I use a series of optimization algorithms, including multiple rounds of particle swarm and Nelder-Mead optimization. Standard errors are obtained by a sandwich-style estimate of the variance-covariance matrix clustered at the establishment level, in the spirit of Murphy and Topel 2002. The procedure is such that the standard errors of demand-side parameters (skills and price sensitivities) match two-stage least squares clustered standard errors, and the standard errors from later estimation steps account for the uncertainty inherited from earlier steps. More details are provided in Appendix Section A.13.

5.3 Task Types, Worker Skill Sets and the Reference Firm

The empirical sections of this paper assume five task types, but the theoretical section of the paper is agnostic about the number of task types. The model is also identified for any number of task types. In practice, the main cost of adding a task type is that it effectively increases measurement error in the task assignment. I continue to assume five task types when estimating the model, with the understanding that it is straightforward to estimate the model with more or fewer task types.

For any fixed number of task types K, there are several important considerations when choosing the number of worker skill sets N. First, Theorem 1 proves that regardless of N, every firm can always maximize profit using at most K worker skill sets. This presents practical challenges in terms of designing an estimation routine for $N > K$, as it implies that there will never exist a firm that employs all worker skill sets. Second, when $N > K$, skill sets will typically violate the linear independence conditions required for identification via Theorem 2 and for essential equilibrium uniqueness via Proposition 5. Intuitively, firms can construct replicas of workers using convex combinations of other workers, leading to multiple optimal task assignments and causing identification using task assignments alone to fail.

When there are 5 task types, the largest number of skill sets for which I am sure the model can

be identified and estimated using task assignment data is 5. As a consequence, I set the number of skill sets to be equal to the number of task types: $N = K = 5$. To invoke Theorem 2 for identification, I assume the wage-adjusted skill vectors $\{\theta_i - \rho w_i e\}_{i=1}^N$ are linearly independent. To invoke Proposition 5 for uniqueness of firm strategies for fixed wages, I assume that there is sufficient variation in comparative advantage across all possible types of firms, formally: $\det(exp[c\Theta]) \neq 0$ for all $c > 0$. Although the assumptions needed for identification and uniqueness of firm strategies are closely related, they are distinct. In particular, the identification condition requires that wage differences do not exactly offset the comparative-advantage differences revealed by task assignments.

The model is identified whenever there is a connected set of salons employing pairs of overlapping worker skill sets. However, to estimate the model I assume there exists a reference salon which employs all N worker skill sets. The existence of a reference salon makes estimation feasible. To understand why, note that without a reference firm, any algorithm that simply begins grouping workers based on pairwise comparisons of coworkers will often run into internal contradictions or cycles. Avoiding cycles turns out to be computationally intractable. The existence of a reference firm anchors all firms in a way that avoids such cycles.

6 Model Estimates

This section reports the results of estimating the model, provides tests of model fit and validation, and relates these to establishment behavior.

6.1 Parameter Estimates

I begin by reporting the price sensitivity parameters (ρ) in Table 8, and the wages and skill set parameters (w, θ) in Tables 9, 10 and 11.

Table 8: Price Sensitivity Parameters

County	Price Sensitivity
Cook	0.049
	(.022)
Los Angeles	0.027
	(.011)
New York	0.034
	(.018)

Note: The price sensitivity parameters for the three counties analyzed in this paper.

Table 9: Worker Wages and Skill Sets in Cook County Salons

Worker Skill Set	Wage	Administrative	Blowdry/Style/Etc.	Color/Highlight/Wash	Haircut/Shave	Nail/Misc.
1	-	3.110	-1.207	3.735	-0.954	8.801
1	-	(2.104)	(6.084)	(2.696)	(1.448)	(17.498)
2	549.363	4.884	17.576	17.121	5.600	107.373
2	(576.730)	(6.270)	(41.844)	(10.478)	(6.239)	(59.868)
3	279.886	3.395	56.465	-4.796	-4.202	-6.337
3	(736.192)	(9.228)	(59.356)	(2.766)	(19.406)	(2.961)
4	-12.208	7.403	17.136	0.764	-3.557	-33.417
4	(26.161)	(3.688)	(10.110)	(1.459)	(1.879)	(17.700)
5	8.623	-12.788	18.361	-4.198	5.443	-14.223
5	(34.343)	(8.368)	(9.124)	(5.061)	(5.098)	(3.050)

Table 10: Worker Wages and Skill Sets in Los Angeles County Salons

Worker Skill Set	Wage	Administrative	Blowdry/Style/Etc.	Color/Highlight/Wash	Haircut/Shave	Nail/Misc.
1	-	1.392	-0.339	0.310	-2.910	-13.394
1	-	(1.115)	(1.593)	(1.415)	(1.762)	(5.920)
2	41.422	-0.916	-3.300	4.535	-8.136	-16.793
2	(90.430)	(1.244)	(7.155)	(2.771)	(3.351)	(4.672)
3	1.682	-0.745	2.685	2.235	-8.635	0.157
3	(86.564)	(2.336)	(4.266)	(2.440)	(3.951)	(1.775)
4	-0.320	0.804	0.933	1.466	-3.403	-8.262
4	(102.545)	(1.110)	(2.940)	(1.298)	(1.187)	(8.584)
5	379.591	2.234	-0.512	-3.488	7.791	-2.964
5	(470.687)	(7.479)	(1.840)	(6.698)	(22.713)	(1.117)

The worker skill sets are numbered arbitrarily. It is exactly the skill parameters in these tables which make each skill set distinct. For example, Skill Set 3 in New York County is a haircut/shave and administrative specialist because it has the highest values in the columns corresponding to those two tasks. Wages are relative to Skill Set 1. The skill set estimates imply that, at a typical salon,[26] reassigning 1% of the time spent on color tasks from the least-skilled to the most-skilled worker increases consumer willingness to pay by \$5.63 in Cook County, \$1.93 in New York County, and \$3.73 in Los Angeles County. Equivalently, reassigning 1% of the time spent on color tasks from the lowest to the highest skill worker (in that task) increases typical salon market share relative to baseline by 27.67% in Cook County, 6.56% in New York County, and 10.22% in Los Angeles County. All of the skill set parameters are presented in these intuitive units in Appendix Table A9.

Estimated wages should be interpreted as the average wage of all workers with that skill set. This interpretation follows from the assumption of random sorting by skill level combined with the no arbitrage condition across skill levels within a skill set that is derived in Proposition 2. Specifically, for any two workers m, m' with the same skill set but different skill levels, wages are such that $w_m - w_{m'} = \rho^{-1}(\bar{\theta}_m - \bar{\theta}_{m'})$. When $\hat{\rho} = 0.027$ as in Los Angeles County, this implies

26. I define a typical salon for this exercise as one with average market share and average required labor

Table 11: Worker Wages and Skill Sets in New York County Salons

Worker Skill Set	Wage	Administrative	Blowdry/Style/Etc.	Color/Highlight/Wash	Haircut/Shave	Nail/Misc.
1	-	-15.677	-3.587	6.952	-3.758	-6.729
1	-	(12.692)	(8.391)	(5.474)	(2.546)	(11.538)
2	-30.813	-2.106	3.281	1.051	-1.112	-4.056
2	(79.805)	(3.067)	(2.626)	(2.589)	(2.779)	(6.090)
3	62.801	73.288	-13.700	5.605	1.917	2.874
3	(71.847)	(47.493)	(8.711)	(3.101)	(4.411)	(15.447)
4	-35.319	-83.620	-1.187	4.575	-3.329	4.897
4	(57.367)	(67.373)	(7.328)	(3.595)	(2.592)	(6.449)
5	12.115	12.331	-2.765	7.414	-5.678	-9.890
5	(67.801)	(9.079)	(1.000)	(5.023)	(2.777)	(11.736)

that each unit of skill level difference commands an equilibrium $37 wage differential. Thus, a specific skill set may have a high estimated wage either because that skill set is scarce or because workers of that skill set have a high average skill level. Task assignment data alone do not typically distinguish between the two cases.

The material-cost, demand-level, cost-level, and wage-level parameters across all market-quarters are presented in Appendix Table A11. Proposition 2 also implies that variation in the wage level parameter across time within a county is suggestive evidence of variation in skill level in a labor market. One organization cost parameter (γ_j) is recovered for each salon-quarter.[27] Per-unit organization costs, which are $\hat{\gamma}_j$ multiplied by required labor multiplied by the observed s-index, have a 25th and 75th percentile of 11.43 and 52.78 dollars, respectively. These represent a modest share of prices: at the median salon-quarter, organization costs are 12.32% of the observed price.

6.2 Model Fit and Validation

I assess model fit by comparing data- and model-generated moments used in estimation in Table 12. I group the moments by estimating equation and by the associated instrument, and I compute the average of the moments as well as the r-squared between the data moments and the model-generated moments. Because the estimation procedure can be thought of as a series of just-identified almost linear GMM estimators, it is unsurprising that model fit is close to perfect. Because the log market share moments are matched via two-stage least squares (a just-identified linear system), the model matches the moments exactly. The labor demand moments from the data are matched exactly by the model. The price moments have the worst model fit, because that portion of the estimation routine imposes the minimum wage in each county and quarter as a constraint.

27. Note that 43 salon-quarters have an s-index of 0 and as a result only a lower bound on their organization cost parameter is identified.

Table 12: Model Fit

Equation	Instrument	Count	Avg. Model	Avg. Data	R2
Log Market Share	County-Dye Instrument	3	-135.03	-135.03	1.000
Log Market Share	County-Quarter	33	-0.22	-0.22	1.000
Log Market Share	County-Task Assignments	75	-0.26	-0.26	1.000
Labor Demand	County-Skill Set	12	0.07	0.07	1.000
Price	County-Quarter	36	4.15	4.45	0.977
Price	Quarter-Task Mix	48	2.16	2.29	0.971
Price	County-Quarter-Labor	36	10.48	10.23	0.999

Note: This table summarizes how well the model matches the moments used in estimation.

To validate the model, I assess how well it can match the observed task content of jobs at the worker level. Although firm-level relative labor demands and task specialization were used in estimation, the task content of individual worker jobs was not. Despite this, I show in Table 13 that the model can match many, but not all, of the patterns observed in the task assignment data. These include the sign and magnitude of the correlations between different task dimensions, which represent the way in which tasks are packaged together. I provide several additional illustrations of the model's ability to predict job task content in Appendix Figures A5 and A4. Heterogeneity in specialization patterns across salons, combined with workers that differ flexibly in five dimensions of skill, allows the model to reproduce the tri-modal marginal distributions of time spent on the haircut/shave task and the color/wash task.

Table 13: Validating the Model Using Job Task Content

	Task	Variance	Cor. Task 1	Cor. Task 2	Cor. Task 3	Cor. Task 4	Cor. Task 5
Model	1	0.101	1.000	-0.768	-0.166	-0.194	-0.171
Data	1	0.105	1.000	-0.760	-0.229	-0.265	-0.153
Model	2	0.105		1.000	-0.254	-0.176	-0.177
Data	2	0.100		1.000	-0.113	-0.159	-0.247
Model	3	0.019			1.000	0.014	-0.037
Data	3	0.014			1.000	0.028	-0.089
Model	4	0.016				1.000	-0.078
Data	4	0.018				1.000	-0.018
Model	5	0.016					1.000
Data	5	0.020					1.000

Note: This table summarizes how well the model matches moments not used in estimation. Variances and correlations are weighted by the labor time associated with each job.

Labor Productivity Measure. For the counterfactual exercises and the coworker productivity spillovers documented Appendix Table A6, I define labor productivity of workers with skill set i (A_i) in the following way. Recall the utility delivered by firm j to consumer z, given a fixed task assignment, relative labor demand and price, is $u_{z,j} = a_j \sum_{i,k} E_j(i) b_j(i,k) \theta_i(k) + \nu_j - \rho p_j + \epsilon_{z,j}$.

Given that $\nu_j + \epsilon_{z,j}$ is exogenous quality, $a_j \sum_{i,k} E_j(i) b_j(i,k) \theta_i(k)$ is the only part of consumer utility that is endogenous, depending on both the sorting of workers to establishments and their assignments. As a result, I define labor productivity of workers with skill set i to be the average of endogenous quality across all establishments weighted by the amount of labor employed by that salon in equilibrium.[28] To ensure that this is nonnegative, I make all skills relative to the lowest-skill worker in that task, yielding the final productivity measure

$$A_i = \frac{\sum_j M \cdot s_j \cdot a_j \sum_k E_j(i) b_j(i,k) [\theta_i(k) - \min_{i'} \theta_{i'}(k)]}{\sum_j M \cdot s_j \cdot a_j}.$$

Partial Equilibrium Heterogeneous Firm Behavior. Firms respond heterogeneously in partial equilibrium to the same wage shock, both in terms of which workers they hire and how they assign tasks. I document heterogeneous labor-labor substitution patterns in Appendix Table A5, and show how pairs of workers can be complements at one firm and substitutes at another in the same labor market. I document how heterogeneous task assignment responses translate into coworker productivity spillovers in Appendix Table A6.

7 Counterfactuals

This section shows that neglecting internal reorganization when it occurs can cause a researcher to misstate, and in one case reverse the sign of, the productivity impacts of economic shocks. There are two general reasons this occurs. First, some shocks change the incentive for all firms to engage in costly specialization, and neglecting reorganization neglects the creation or destruction of productive specialized jobs, as is the case for sales tax increases. Second, some shocks, like low-wage immigration, would systematically favor more-productive or less-productive firms in a world without reorganization. With reorganization, a broader base of firms can adjust to share in the cost savings of the shock, leading to a different shift in labor across firms and a different aggregate productivity impact.

To compare outcomes with and without internal reorganization, I define the *reallocation equilibrium* as the outcome when firms are allowed to adjust prices (p_j) but organizational structures (B_j) are fixed at the initial equilibrium choices. Because prices control quantities, this equilibrium allows firms to adjust the total labor they hire but not the division of labor within the firm. I define the *reorganization equilibrium* as the outcome when firms are allowed to fully adjust their task assignments. It is the full equilibrium described in Section 4.1.

28. I use this notion rather than revenue per minute of labor because the latter is in general contaminated by markups.

Different economic shocks change the competitive position of firms within an industry, depending on their initial structure and task mix. The reallocation equilibrium quantifies how reallocation of labor across firms shifts outcomes. Because task assignments differ across firms, reallocation can change aggregate productivity and task specialization even when each firm's assignments are held fixed. The spirit of the reallocation equilibrium is not novel; in many models with heterogeneous firms, shocks reallocate labor inputs in ways that change aggregate productivity. However, the reorganization equilibrium is novel, because firms with different organization costs are internally changing in response to an external economic shock.

I solve for counterfactual equilibria for each county in 2021 Q2, as this is the final quarter in the estimation sample and typically has the most salons due to greater adoption of the software over time. I compare counterfactual outcomes to an initial equilibrium where I solve the model taking labor supply as fixed.[29] For additional details, see Appendix Section A.16. I consider four counterfactuals.

- **Sales Tax Increase.** I impose a four-percentage-point increase in the tax on salon services. In Los Angeles County and Cook County, salon services are exempt from sales tax, so in these two counties, this counterfactual changes the sales tax rate from 0% to 4%. In New York County, salon services are already taxed at a rate of 4.5% (New York State Department of Taxation and Finance 2011), so this change brings the tax rate to 8.5%.

- **Management Diffusion.** Each salon adopts the management practices of the next-best salon. I implement this change by ordering the salons by their organization cost parameter and then changing each salon's organization cost to that of the salon one rank above it. I leave the salon with the lowest organization cost unchanged.

- **Low-Wage Immigration.** There is a 5% increase in total labor, and all of this additional labor has the same skill set as the lowest-wage workers in the initial equilibrium. In Los Angeles County, the lowest-wage worker in the initial equilibrium is skill set 1, which Table 10 shows is a medium-skill generalist without an absolute advantage in any task, but that is also not the lowest skill in any task. In Cook County, it is skill set 4, which Table 9 shows has the lowest skill in the nail/misc task and an absolute advantage in the administrative task. In New York County, it is skill set 4, which Table 11 shows has an absolute advantage in the nail/misc task and the lowest skill in the administrative task. I focus on low-wage rather

29. This is necessary, because wages are estimated without fully solving the model.

than low-skill immigration because the latter concept does not have a natural definition when skill is multidimensional.

- **Increase in Market Concentration.** Half of the salons in each market are removed. Because each salon in the data is given a weight corresponding to the number of salons it represents in the initial equilibrium, this counterfactual is implemented by halving the weight and recomputing the equilibrium.

Table 14 presents the results from these counterfactuals. I solve for the reallocation and reorganization equilibria for each counterfactual, and compute the percentage change in the average s-index (weighted by total labor) and labor productivity relative to baseline. For the immigration counterfactual, I assume that in all counterfactual equilibria, firms that employ the immigrant skill set use the same immigrant-worker proportion (that is, there is no sorting across firms), and I report changes in native labor productivity.[30]

Table 14: Counterfactual Productivity and Specialization Changes

		Reallocation		Reorganization	
County	Counterfactual	S-Index Change	Prod. Change	S-Index Change	Prod. Change
Cook	Immigration	-0.012	-0.033	-0.013	-0.028
New York	Immigration	-0.036	0.000	-0.034	-0.001
Los Angeles	Immigration	0.059	-0.017	-0.002	-0.009
Cook	Incr. Concentration	-0.025	-0.015	-0.019	-0.016
New York	Incr. Concentration	0.030	0.016	0.024	0.025
Los Angeles	Incr. Concentration	0.021	0.004	0.037	0.003
Cook	Management Diffusion	0.273	0.189	0.553	0.224
New York	Management Diffusion	0.009	0.003	0.026	0.020
Los Angeles	Management Diffusion	-0.005	0.004	0.136	0.003
Cook	Sales Tax	-0.028	0.000	-0.067	0.009
New York	Sales Tax	-0.030	-0.012	-0.063	0.007
Los Angeles	Sales Tax	0.003	0.006	-0.070	0.007

Note: Effects are proportional changes from the baseline equilibrium (0.01 = 1%).

Consider the impact of low-wage immigration in Los Angeles County: total labor supply rises by 5% with all additional labor concentrated in skill set 1, which has the lowest wage in the initial equilibrium. When salons are prevented from internally reorganizing, aggregate task specialization rises by 5.9% and native labor productivity falls by 1.7% relative to the initial equilibrium. In contrast, when salons are allowed to internally reorganize, aggregate task specialization falls by 0.2% and native labor productivity falls by 0.9%. In this case, neglecting internal reorganization overstates the productivity and specialization impacts of immigration.

30. This excludes the mechanical reduction in productivity due to the introduction of new workers with on average lower skills.

Appendix Figure A2 illustrates how productivity changes by decomposing the total productivity impact into a between-firm component and a within-firm component. This decomposition is coherent because the reallocation equilibrium, where the internal organization of all salons is held fixed, is well defined. In both equilibria, the immigration shock systematically favors some salons, generating a between-firm productivity effect. This effect is negative in both cases, but is more than 4 times larger in the reallocation equilibrium because reorganization allows a wider swathe of salons to share in the labor cost reductions from immigration. In the reorganization equilibrium an additional within-firm productivity force operates. In response to the same labor cost reductions from immigration, some salons increase specialization and productivity while others reduce specialization and productivity. This leads to a net within-firm productivity effect of -0.6 percentage points. The total productivity effect under reorganization is then about half the magnitude of the total reallocation effect.

The results of the immigration counterfactual in Los Angeles County have three practical implications. First, the net productivity impact for native workers depends on how immigrant skills change the competitive landscape among firms and whether immigrant skill sets encourage or discourage task specialization within the firm. The magnitude and sign of these forces are an empirical question. Second, because two workers can be substitutes at one firm and complements at another, the wage impact of immigration on natives is also an empirical question. In Los Angeles County, the 5% increase in total labor composed of skill set 1 leads to a 5.6% reduction in wages for natives with the immigrant skill set, but a 0.6% wage increase for natives in skill set 2 and a 0.3% wage increase for natives in skill set 3 (Appendix Table A10). Thus, although the same pair of skill sets is sometimes a substitute at one salon and complement at another (see Appendix Table A5), the immigration counterfactual reveals that natives with skill set 1 are aggregate substitutes while skill sets 2 and 3 are aggregate complements.

Third, attempting to understand low-wage immigration using attributes of firms in the initial equilibrium, like the initial share of workers with the same skills as the low-wage immigrants or even the initial s-index, can be misleading. In the initial equilibrium, the immigrant skill set is concentrated in salons that are internally task specialized but not particularly productive. A researcher might be tempted to predict that low-wage immigration would favor these salons, increasing aggregate task specialization and reducing productivity. Indeed, this logic is exactly how the reallocation equilibrium operates, and it predicts a 5.9% increase in specialization and a relatively large reduction in native labor productivity. This shift-share or exposure-based logic collapses under internal reorganization because task reassignment causes the labor cost reductions

from immigration to be split across firms in a completely different, and in this case more even, way. Task reassignment looks like specialization at some salons and generalization at others, leading to an aggregate 0.2% reduction in task specialization and an attenuated native productivity impact.

A four-percentage-point increase in the sales tax on services in New York County decreases productivity by 1.2% under the reallocation equilibrium and increases productivity by 0.7% under the reorganization equilibrium. To understand this sign reversal, first recall that when deciding their internal organization, salons minimize quality-adjusted costs:

$$\min_{b_j, E_j} \gamma_j \sum_i E_j(i) D_{KL}(b_j(i,k)||\alpha_j) + \sum_i E_j(i) w_i - \rho^{-1} \sum_i E_j(i) \sum_k \theta_i(k) b_j(i,k).$$

Increasing the sales tax is equivalent to scaling up consumer price sensitivity (ρ), which makes improving quality less important relative to keeping costs low. When wages are similar across workers, the main way salons reduce costs is by reducing organization expenditures, which requires reducing specialization and labor productivity. Thus, a sales tax increase has a theoretically unambiguous impact within the firm: it should reduce specialization and labor productivity. Indeed, in the reorganization equilibrium the s-index falls by 6.3%. Further, Appendix Figure A3 shows that almost all salons reduce their individual s-index and almost all salons see a within-firm productivity reduction, leading to an aggregate within-firm productivity effect of -0.3 percentage points.

The puzzle then becomes how the productivity impact is positive when both the theoretical prediction and the estimated within-firm effects are almost universally negative. It turns out that the Nash equilibrium in prices within the competitive equilibrium in wages generates the conditions necessary for an effect similar to Simpson's paradox. Specifically, almost all salons become less productive, but the sales tax increase changes wages and product market positioning in a systematic way. In the reallocation equilibrium, the sales tax favors initially unproductive salons, generating a between-firm productivity decrease of 1.2 percentage points. In the reorganization equilibrium, the sales tax favors initially productive salons, generating a between-firm productivity increase of 1.0 percentage points. A sales tax increase can therefore simultaneously make almost all firms less productive, while also increasing aggregate productivity. Put another way, the treatment effect of a sales tax increase on individual salon productivity is negative, the selection effect is positive, and the aggregate effect is positive. Importantly, reorganization is responsible for the treatment effect and the selection effect.

A counterfactual increase in market concentration has largely similar impacts between the reallocation and reorganization equilibria, but varies in impact across counties. In New York County

it leads to a 2.5% increase in labor productivity in the reorganization equilibrium while in Cook County it leads to a 1.6% decrease in productivity in the reorganization equilibrium. The diffusion of management practices increases task specialization and productivity in the full reorganization equilibrium in all counties. This makes sense: the counterfactual amounts to reducing the primitive designed to capture the cost of specialization. The productivity effects are largest in Cook County: one step up the management ladder improves aggregate labor productivity by 22.4% and task specialization by 55.3%. This large impact is driven by the sparsity of data in the county, which implies that moving up a rung is a more meaningful percentile increase in efficiency among competitors. As with sales taxes, the reallocation equilibrium shows a muted impact on specialization in all three counties and productivity in tow of three counties because management diffusion directly incentivizes endogenous specialization and this channel is mechanically shut down in the reallocation equilibrium.

8 Conclusion

This paper provides evidence that task specialization within establishments is related to productivity differences across establishments. It also provides a structural model which can be used to understand how endogenous and heterogeneous task assignment reacts to economic shocks. An interesting avenue for future work is to study how dynamics affect internal organization, and how internal organization feeds back into the aggregate economy. On the worker side, this paper illustrates that even among workers who likely work in the same occupation, task assignments can differ substantially. If workers learn by doing, then both aggregate human capital and individual human capital are affected by the sorting of worker types across firms. Making progress on this question requires uniting firm-level heterogeneity in organization costs with dynamic human capital accumulation models such as Adenbaum, Babalievsky, and Jungerman (2024). On the firm side, this paper treats firm-specific organization costs as a primitive. In reality, firms accumulate organizational capital, as in Dessein and Prat (2022), which reduces organization costs in the future. Task specialization of workers may have other costs that are not explicitly explored in this paper. One cost worth highlighting is reduced resilience or adaptability. There is evidence that more specialized firms are more productive in economic booms but more vulnerable during economic busts (Kuhn et al. 2023).

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A Online Appendix

A.1 Interpreting Organization Costs

This section derives the microfoundation for the organization cost functional form and shows that organization costs are proportional to the s-index.

Equivalence to the S-Index. A few algebraic manipulations reveal that the s-index multiplied by the organization cost parameter and the required labor ($\gamma_j \cdot a_j I_j(B_j)$) is equal to the organization cost. First, note that an organization is related to a task assignment in the following way for workers employed at the salon: $\frac{B_j(m,k)}{\sum_{k'} B_j(m,k')} = b_j(m,k)$. The s-index can then be expressed as

$$\begin{aligned}
I(B_j) &= \sum_{m,k} B_j(m,k) \log\left(\frac{B_j(m,k)}{B_j^G(m,k)}\right) \\
&= \sum_{m,k} B_j(m,k) \log\left(\frac{B_j(m,k)}{[\sum_{k'} B_j(m,k')]\alpha_j(k)}\right) \\
&= \sum_{m,k} \left[\sum_{k'} B_j(m,k')\right] b_j(m,k) \log\left(\frac{b_j(m,k)}{\alpha_j(k)}\right) \\
&= \sum_{m} \left[\sum_{k'} B_j(m,k')\right] \sum_{k} b_j(m,k) \log\left(\frac{b_j(m,k)}{\alpha_j(k)}\right) \\
&= \sum_{m} \left[\sum_{k'} B_j(m,k')\right] D_{KL}(b_j(m,k)||\alpha_j) \\
&= \sum_{m} E_j(m) D_{KL}(b_j(m,k)||\alpha_j)
\end{aligned}$$

where the final expression is exactly the organization cost divided by the required labor and the organization cost coefficient.

Microfoundation. Suppose the firm must communicate each worker's task assignment. Prior to communication, each worker knows only their firm's overall task-based production function. Results from information theory imply that the minimum expected amount of information required to communicate a random variable Y given knowledge of a random variable X is exactly the Kullback–Leibler divergence between the joint distribution of the variables and the product of their marginals, also known as the relative entropy (Cover and Thomas 2006). This implies that the amount of information that must be communicated per unit of labor assigned to worker m is given by the Kullback–Leibler divergence between the task assignment of that worker $\{b_j(m,k)\}_{k=1}^K$ and the task mix $\{\alpha_j(k)\}_{k=1}^K$. Thus, the s-index is exactly the minimum amount of information that must be transmitted within the firm to implement the observed task assignment.

A.2 First-Stage Estimates and Alternative Demand Systems

After identifying worker skill sets, the estimation procedure performs a relatively standard demand estimation with instruments for endogenous prices. Importantly, $B_j(i,k)$ are considered data in this step, and under the model they do not depend on additive demand shocks. Therefore, $B_j(i,k)$ interacted with required labor (the average labor per customer at each salon) enter as exogenous variables in the two-stage least squares procedure. This section compares a variety of potential demand estimation specifications.

I consider specifications with different fixed effects and different instruments. The two instruments I consider are a dye instrument, constructed as the share of the color task multiplied by the producer price index of synthetic organic dyes in that quarter, and a Hausman instrument, constructed as the average price of all salons in the two other counties. I use the dye instrument with county×quarter fixed effects because it balances concerns about power and the exclusion restriction. First-stage estimates for my preferred specification along with the alternatives are given in Appendix Table A1. My preferred specification achieves similar first-stage F-statistics in two of the three markets compared to the less saturated quarter-only fixed effects specification. It achieves comparable first-stage F-statistics to the Hausman instrument for two of the three markets.

In terms of the absolute level, the first-stage F-statistic is near the levels commonly used in the instrumental variables literature (Stock and Yogo 2002) for Los Angeles County, and but is below the common threshold in Cook and New York County. The estimated average effect of a one standard deviation increase in the dye instrument on price is between 0.16 and 0.54 standard deviations (between \$28.77 and \$73.17 in levels). Relative to the alternative specifications, the exclusion restriction appears more likely to be satisfied in my preferred specification. The Hausman instruments are more powerful, but the exclusion restriction requires that there must not be national demand shocks in the beauty industry, which seems unrealistic. The quarter-only fixed effects specifications are both less powerful and require demand conditions to have changed similarly across time in the different counties, which is unlikely to be true because the sample spans periods before and after the COVID shutdowns.

In Appendix Table A2, I show that my preferred specification also generates coefficients on price (second-stage coefficients) that are negative in all counties, and therefore consistent with the law of demand. Reassuringly, the coefficient on price is similar across all four specifications in New York County.

It should be noted that my preferred specification omits the 2018 Q1 fixed effect for each county. Because the skill sets are free parameters estimated in the data and there is a unique

skill set matrix in each county, if the required labor of each establishment were the same, a county fixed effect would not be separately identified. The required labor of all salons is not the same, so it is possible to estimate models that do not omit any fixed effects. However, these more saturated specifications have nearly ill-conditioned projection matrices, where the condition of the estimation matrix is measured using the scaled Belsley, Kuh, and Welsch (1980) condition applied to $X'P_ZX$, the relevant index for two-stage least squares (B. E. Hansen 2022). Appendix Table A3 illustrates this: when the 2018 Q1 category is omitted, the scaled $X'P_ZX$ condition is $34,751$, but when it is re-introduced in column 1, the scaled $X'P_ZX$ condition number rises to 1.27 million.

Finally, I consider a slightly more general demand specification where the outside option of no salon services is given its own nest and all salons are put in one nest. Formally, I specify that the utility of consumer z from salon j is given by $u_{z,j} = \xi_j + \nu_j - \rho p_j + \epsilon_{z,j}$, where instead of assuming $\epsilon_{z,j}$ is i.i.d. across consumers and salons, I assume $\epsilon_{z,j} = \zeta_{z,g(j)} + (1-\sigma)\upsilon_{z,j}$ where $g(j) = 1$ for all $j > 0$, $g(0) = 2$, $\epsilon_{z,j}$ is extreme value distributed, and $\upsilon_{z,j}$ is i.i.d. extreme value distributed. Then the market shares satisfy: $\log\left(\frac{s_j}{s_0}\right) = \xi_j + \nu_j - \rho p_j + \sigma \log\left(\frac{s_j}{s_{g(j)}}\right)$.

Estimation requires two instruments to identify ρ and σ, and I use the Hausman and dye instruments. The results are presented in Appendix Table A4. Several specifications generate positive price coefficients (violating the law of demand) and nested logit parameters that are outside of the unit interval, indicating inconsistency with random utility maximization. In the county×quarter specification, New York County and Cook County have positive price coefficients and New York County has a nesting parameter outside the unit interval. In other specifications, the nesting parameter is often negative. These results should be treated with caution because of the high scaled condition number and because they rest on the Hausman instrument satisfying an exclusion restriction.

Table A1: Standard Logit First-Stage Estimates: Instrument and Fixed-Effect Comparison

	County×quarter; omit	Quarter only; omit	County×quarter; omit	Quarter only; omit
Fixed effects	County×quarter; omit	Quarter FE	County×quarter; omit	Quarter FE
Price on Hausman (Cook)	0.1095***	0.0554***		
	(0.0287)	(0.0142)		
Price on Hausman (New York)	0.0515*	0.0524*		
	(0.0231)	(0.0216)		
Price on Hausman (Los Angeles)	0.1089*	0.0996*		
	(0.0505)	(0.0476)		
Price on Dye (Cook)			0.1574	0.0518
			(0.1185)	(0.1084)
Price on Dye (New York)			0.3616	0.2123*
			(0.1976)	(0.1023)
Price on Dye (Los Angeles)			0.5380**	0.4682**
			(0.1989)	(0.1688)
Weak-IV F Price (Cook)	51.45	33.38	2.98	1.28
Weak-IV F Price (New York)	7.95	13.37	5.47	6.59
Weak-IV F Price (Los Angeles)	10.19	10.30	13.90	13.39
Observations	472	472	472	472

Notes: Fully standardized first-stage OLS coefficients on county-specific instruments: each cell reports $\widehat{\beta}_{c,z} \times \widehat{\mathrm{SD}}_c(z)/\widehat{\mathrm{SD}}_c(\text{cust_price})$, the within-county standard-deviation change in `cust_price` (county c) associated with a one-within-county-SD increase in the underlying instrument z. Standard errors are rescaled by the same constant, so t-statistics and stars are unchanged. Within-county SDs used for the rescaling: 17031: SD(Price)=182.805, SD(Hausman)=18.515, SD(Dye)=18.914; 36061: SD(Price)=128.893, SD(Hausman)=10.936, SD(Dye)=18.602; 6037: SD(Price)=136.002, SD(Hausman)=26.892, SD(Dye)=19.958. Location-level clustered standard errors in parentheses, for the same specifications reported in Table A2. Each column regresses the endogenous county-specific `cust_price` column on all exogenous controls plus the excluded instrument(s) for that column. Blank cells indicate that the instrument is not included in the column's specification. *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$. Weak-IV F rows reproduce the diagnostics from `summary(ivreg, diagnostics = TRUE)`.

Table A2: Standard Logit: Instrument and Fixed-Effect Comparison

Instrument	County×quarter; omit Hausman	Quarter only; omit Hausman	County×quarter; omit Dye	Quarter only; omit Dye
Fixed effects	County×quarter; omit	Quarter FE	County×quarter; omit	Quarter FE
Price (Cook)	-0.0212***	-0.0467*	-0.0491*	-0.1146
	(0.0051)	(0.0199)	(0.0220)	(0.2581)
Price (New York)	-0.0324*	-0.0307*	-0.0341	-0.0227
	(0.0146)	(0.0154)	(0.0176)	(0.0302)
Price (Los Angeles)	-0.0423*	-0.0448*	-0.0274*	-0.0234
	(0.0205)	(0.0219)	(0.0111)	(0.0139)
Weak-IV F price (Cook)	51.45	33.38	2.98	1.28
Weak-IV F price (New York)	7.95	13.37	5.47	6.59
Weak-IV F price (Los Angeles)	10.19	10.30	13.90	13.39
Wu-Hausman p-value	$<$0.0001	$<$0.0001	$<$0.0001	$<$0.0001
Scaled $X'P_ZX$ cond.	4,189	3,984	34,751	117,666
Adj. R^2	0.7603	0.7406	0.8595	0.8396
Observations	472	472	472	472

Notes: Standard-logit 2SLS price coefficients with location-level clustered standard errors in parentheses. Columns cross the two instruments (Hausman leave-own-county-out same-quarter price; dye = task_mix_2 × PPI) with two fixed-effect parameterizations. "County×quarter; omit" drops the county-specific 2018.1 cells, and "Quarter only; omit" uses additive quarter FE with an omitted reference quarter. All columns also include county-specific $avg_labor \times B_raw$ cost controls. *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. Weak-IV rows report the diagnostic statistics from `summary(ivreg, diagnostics = TRUE)`.

Table A3: Standard Logit with Dye Instrument: Fixed-Effect Comparison

	County×quarter; full	County×quarter; omit	Quarter only; omit	Quarter only; all	County + quarter
Fixed effects	County×quarter; full	County×quarter; omit	Quarter FE	All quarter FE	County + quarter
Price (Cook)	0.0024	-0.0491*	-0.1146	-0.0068	0.0002
	(0.0023)	(0.0220)	(0.2581)	(0.0702)	(0.0239)
Price (New York)	-0.0709	-0.0341	-0.0227	-0.0040	-0.0918
	(0.5347)	(0.0176)	(0.0302)	(0.0082)	(0.3336)
Price (Los Angeles)	-0.0049	-0.0274*	-0.0234	-0.0064	-0.0011
	(0.0045)	(0.0111)	(0.0139)	(0.0080)	(0.0158)
Weak-IV F price (Cook)	10.11	2.98	1.28	0.80	12.85
Weak-IV F price (New York)	0.03	5.47	6.59	5.14	0.08
Weak-IV F price (Los Angeles)	4.24	13.90	13.39	6.08	3.57
Wu-Hausman p-value	0.2096	<0.0001	<0.0001	0.2808	0.0380
Scaled $X'P_Z X$ cond.	1,268,302	34,751	117,666	806,464	656,358
Adj. R^2	0.8189	0.8595	0.8396	0.9877	0.7152
Observations	472	472	472	472	472

Notes: Standard-logit 2SLS price coefficients with location-level clustered standard errors in parentheses. All columns use the county-specific dye instrument (task_mix_2 × PPI input-cost shifter) and county-specific $avg_labor \times B_raw$ controls. Columns vary only in the fixed-effect parameterization. "County×quarter; omit" drops the county-specific 2018.1 quarter indicators; "Quarter only; omit" and "County + quarter" use an omitted common quarter reference; "Quarter only; all" includes all quarter fixed effects. *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. Weak-IV rows report the diagnostic statistics from `summary(ivreg, diagnostics = TRUE)`.

Table A4: Nested Logit with Hausman + Dye Instruments: Fixed-Effect Comparison

	County×quarter; omit	Quarter only; omit	Quarter only; all	County + quarter
Fixed effects	County×quarter; omit	Quarter FE	All quarter FE	County + quarter
Price (Cook)	0.0002	0.0026	0.0373	-0.0036
	(0.0002)	(0.0070)	(0.2679)	(0.0870)
Sigma (Cook)	0.9125***	0.9283***	-6.1567	-4.5471
	(0.0089)	(0.2605)	(53.3682)	(42.1788)
Price (New York)	0.0060	-0.0531	0.1422	-0.0202
	(0.0460)	(0.0805)	(1.3024)	(0.0418)
Sigma (New York)	1.0851	-0.6715	-3.9637	-0.8887
	(1.2891)	(2.3281)	(31.2156)	(1.2574)
Price (Los Angeles)	-0.0003	0.0028	-0.0552	0.0031
	(0.0003)	(0.0074)	(0.4448)	(0.0285)
Sigma (Los Angeles)	0.9365***	1.0420***	-7.3447	2.1261
	(0.0084)	(0.1960)	(63.1803)	(7.7767)
Weak-IV F price (Cook)	32.72	23.61	22.80	7.09
Weak-IV F price (New York)	3.96	6.69	5.97	0.41
Weak-IV F price (Los Angeles)	7.35	8.00	4.73	2.28
Weak-IV F sigma (Cook)	319.74	278.97	265.64	0.44
Weak-IV F sigma (New York)	111.44	121.50	103.81	3.45
Weak-IV F sigma (Los Angeles)	185.17	167.36	70.14	0.82
Wu-Hausman p-value	<0.0001	<0.0001	<0.0001	<0.0001
Scaled $X'P_Z X$ cond.	1,293,181	100,447	22,333,673	32,665,701
Adj. R^2	0.9987	0.8991	-0.3291	0.9631
Observations	472	472	472	472

Notes: Nested-logit 2SLS coefficients with location-level clustered standard errors in parentheses. Price rows report county-specific coefficients on `cust_price`; sigma rows report county-specific coefficients on `log_within_share`. All columns use the county-specific dye instrument and the county-specific Hausman instrument (leave-own-county-out same-quarter price) together, plus county-specific $avg_labor \times B_raw$ controls. Columns vary only in the fixed-effect parameterization. "County×quarter; omit" drops the county-specific 2018.1 quarter indicators; "Quarter only; omit" and "County + quarter" use an omitted common quarter reference; "Quarter only; all" includes all quarter fixed effects. The fully saturated "County×quarter; full" spec is omitted because it absorbs the instrument's identifying variation in the nested model. *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. Weak-IV rows report the diagnostics from `summary(ivreg, diagnostics = TRUE)`. In a simple nested-logit interpretation, sigma values in $[0, 1)$ are the usual RUM-consistent range.

A.3 Estimated Model Properties

I conduct two partial equilibrium exercises to illustrate how the model changes standard economic intuition. In both, I consider the initial state of the economy to be the initial equilibrium used in the counterfactuals; that is, I solve for wages, task assignments, and prices such that all equilibrium conditions are satisfied.

A.3.1 Labor-Labor Substitution Patterns

First, I explore labor-labor substitution patterns. I do this by increasing the wage of each worker skill set and examining how relative labor demand for each worker skill set responds across the heterogeneous firms in each market. The results are in Table A5.

Table A5: Labor-Labor Substitution Patterns

		Skill Set 1			Skill Set 2			Skill Set 3			Skill Set 4			Skill Set 5		
County	Skill Set	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.
Cook	1	0.000	-0.001	-0.013	0.008	0.000	0.000	0.003	0.000	-0.005	0.006	0.000	0.000	0.007	0.001	0.000
Cook	2	0.001	0.000	0.000	0.000	0.000	-0.008	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Cook	3	0.000	0.000	-0.001	0.000	0.000	0.000	0.000	0.000	-0.005	0.000	0.000	0.000	0.001	0.000	0.000
Cook	4	0.010	0.000	0.000	0.001	0.000	-0.002	0.005	0.000	0.000	0.000	0.000	-0.007	0.002	0.000	0.000
Cook	5	0.006	0.001	0.000	0.000	0.000	-0.002	0.008	0.000	0.000	0.001	0.000	0.000	0.000	-0.001	-0.007
New York	1	0.000	0.000	-0.545	0.007	0.000	-0.005	0.005	0.000	0.000	0.025	0.000	0.000	0.817	0.000	0.000
New York	2	0.041	0.000	-0.006	0.000	-0.002	-0.027	0.043	0.000	0.000	0.018	0.000	0.000	0.001	0.000	-0.042
New York	3	0.006	0.000	0.000	0.027	0.000	0.000	0.000	-0.001	-0.043	0.000	0.000	0.000	0.001	0.000	-0.029
New York	4	0.034	0.000	0.000	0.020	0.000	0.000	0.001	0.000	0.000	0.000	0.000	-0.038	0.004	0.000	0.000
New York	5	0.504	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.003	0.000	0.000	0.000	0.000	-0.772
Los Angeles	1	0.000	0.000	-0.477	0.044	0.000	-0.305	0.045	0.000	-0.334	0.603	0.000	0.000	0.522	0.000	-0.005
Los Angeles	2	0.007	0.000	-0.054	0.000	-0.008	-0.067	0.016	0.000	-0.057	0.128	0.001	0.000	0.000	0.000	-0.032
Los Angeles	3	0.006	0.000	-0.116	0.028	0.000	-0.112	0.000	-0.004	-0.135	0.127	0.002	0.000	0.000	0.000	-0.065
Los Angeles	4	0.599	0.000	0.000	0.539	0.002	0.000	0.582	0.002	0.000	0.000	-0.007	-0.605	0.219	0.000	-0.210
Los Angeles	5	0.156	0.000	-0.001	0.000	0.000	-0.123	0.000	0.000	-0.123	0.264	0.000	-0.305	0.000	0.000	-0.305

Note: This table depicts the labor-labor substitution patterns across different worker skill sets. For each skill set listed in the top row, I increase the wage by 1% holding all other wages fixed. I then measure the change in relative labor demand for the county and skill set listed in the first and second columns. I report the maximum, median and minimum changes across all salons in each county in 2021 Q2 (the quarter for which the initial equilibrium and counterfactual equilibria are solved), for each skill set. Thus, diagonal entries are related to own-wage elasticities while off-diagonal entries are related to cross-wage elasticities.

Although the numbers are not exactly elasticities,[31] the table has an interpretation similar to that of demand-substitution tables common in industrial organization. Three patterns emerge.

First, an increase in own wage weakly reduces relative labor demand across all firms, as expected. Second, in all markets there exist pairs of worker skill sets that are substitutes at one firm and complements at another. Third, substitution effects are small at some firms but large at others. For example, in Los Angeles County and new York County it is common for the median firm to experience a near-zero change in relative labor demand in response to a wage increase, while the most responsive firm sees a change of 20 percentage points or more. These patterns occur not just because of differences in task mixtures across firms but also because of differences in organization costs.

A.3.2 Productivity Spillovers

In this section, I consider the same wage shocks as in the last subsection. However, I now ask how these change the productivity of different worker skill sets at different salons. Theoretically (Theorem 1) and empirically (Table A5), I show that as the wage of a particular skill set rises, the law of demand holds and each salon demands relatively less of that particular worker. However, the tasks that were previously performed by that skill set still need to be completed, and the salon incorporates them into the task assignments of the workers that remain. Intuitively, coworkers must pick up the slack of those who are let go.

Table A6 shows how such task reassignment impacts labor productivity. I only include skill set combinations that are employed together in the initial equilibrium.

31. I do not estimate elasticities because some firms have an initial relative labor demand of 0 for one or more worker skill sets.

Table A6: Effects of Wage Increases on Own and Coworker Productivity

		Skill Set 1			Skill Set 2			Skill Set 3			Skill Set 4			Skill Set 5		
County	Skill Set	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.	Max.	Med.	Min.
Cook	1	0.001	0.001	-0.001	0.004	0.001	0.001	0.001	0.000	-0.001	0.001	0.001	0.000	-0.001	-0.001	-0.001
Cook	2	0.000	-0.001	-0.001	0.010	0.009	0.002	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Cook	3	0.000	0.000	0.000	0.000	0.000	0.000	0.005	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Cook	4	-0.003	-0.003	-0.004	0.001	0.000	0.000	0.005	0.002	0.000	0.002	0.001	0.000	0.000	-0.001	-0.001
Cook	5	0.000	-0.001	-0.003	0.000	0.000	0.000	0.002	0.000	0.000	0.001	0.000	0.000	0.002	0.001	0.000
New York	1	0.003	0.000	-0.009	0.001	-0.001	-0.002	0.000	0.000	-0.002	0.000	0.000	-0.001	NA	NA	NA
New York	2	-0.002	-0.005	-0.045	0.023	0.003	0.000	0.000	-0.002	-0.051	0.000	-0.001	-0.006	0.101	0.011	0.000
New York	3	-0.003	-0.009	-0.059	0.000	-0.023	-0.079	0.165	0.050	0.002	-0.001	-0.004	-0.011	0.126	0.037	-0.002
New York	4	-0.010	-0.013	-0.016	0.001	-0.002	-0.006	0.000	-0.001	-0.001	0.011	0.009	-0.001	0.003	0.000	-0.006
New York	5	NA	NA	NA	0.000	0.000	0.000	0.000	0.000	-0.001	0.000	0.000	0.000	0.002	0.000	0.000
Los Angeles	1	0.003	-0.001	-0.014	0.001	0.000	-0.001	0.000	0.000	-0.003	0.005	0.000	-0.002	0.100	0.019	0.001
Los Angeles	2	0.000	0.000	-0.002	0.003	0.001	0.000	0.000	0.000	0.000	0.000	-0.001	-0.002	0.000	0.000	0.000
Los Angeles	3	0.033	0.000	-0.069	0.017	0.000	-0.009	0.022	0.004	0.000	-0.001	-0.003	-0.043	0.355	0.001	0.001
Los Angeles	4	0.000	-0.001	-0.003	0.002	0.000	0.000	0.001	0.000	-0.001	0.010	0.000	-0.002	0.001	0.001	0.001
Los Angeles	5	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.002	0.001	0.000

Note: This table depicts the proportional change in labor productivity due to increases in own and coworker wages at different salons. For each skill set listed in the first row, I increase the wage by 1% holding all other wages fixed. I then measure the change in quality per unit of labor for each county and skill set listed in the first and second columns. I report the maximum, median and minimum changes across all salons in each county in 2021 Q2 (the quarter for which the initial equilibrium and counterfactual equilibria are solved), for each skill set. Thus, the value in row 1 (Cook County Skill Set 1) and column 3 (Skill Set 1 Max.) represents the maximum change in productivity of workers in skill set 1 after a 1% change in the wage of workers with skill set 1. Salons are only included in these calculations if their relative labor demand for both worker skill sets was greater than 0.01 initially. If a pair of skill sets fails this criterion at every salon in a county, the corresponding cells are filled in with NA; this applies to skill sets 1 and 5 in New York County.

A.4 Data Construction Details

The data throughout the paper are constructed using the following procedure. I limit the data to establishments identified as hair salons, barbershops, or blowout-and-styling salons. The management software company that provided the data for this project sells a subscription to individual businesses. Some of these businesses had existing software, and the management software company appears to have imported some of the records from that software as legacy records. These records are not labeled differently in the data, but 10 salons have what appear to be placeholder records (services with very short, similar durations) that abruptly end when native fields from the software begin to populate. Since these likely represent legacy data recorded by different software, all records rom that salon prior to the apparent transition date are excluded (2.8% of task assignments). The six transition salon-quarters generated by this procedure are kept in the sample. A robustness exercise excluding these salon-quarters found no material impact on the stylized facts.

Two firms appear to record prices in cents rather than dollars, potentially due to legacy data importing, and their prices are adjusted to be in dollars. Two firms appear to have imported some legacy price records in cents rather than dollars. For these firms I rescale to dollars those transactions whose recorded price exceeds a firm-specific threshold. I leave smaller transactions at the same firms unchanged. One transaction has a price of over $1 million and is excluded. Approximately 0.2% of task assignments (26,288 of 12.6 million) have a recorded duration of 0. For these I impute the time spent as the average duration of other tasks in that category and quarter. The data include the ZIP code of each salon, which I use to map to states and counties. One salon does not have a recorded ZIP code and is treated as its own category in regressions with location fixed effects.

The vast majority of service descriptions ($20,560$ of $20,786$) are classified by a licensed cosmetologist. The remaining 226 arose from the incorporation of additional data after the original classification had been completed. For these, I first find any that match the original descriptions up to text encoding differences and assign the cosmetologist's category. Of those that remain, I use edit distance to match them to the original descriptions (the differences are primarily in spelling and case) and assign the cosmetologist's category. Of those that remain, I have an LLM propose close matches from the original set. I then read the proposed pairs and approve the match only if the two descriptions have the same essential meaning. I repeat this process and am able to find an original service description that matches each of the additional descriptions.

Importantly, some service descriptions are classified as multiple task categories. When this occurs, the service descriptions are broken into component tasks. The data record the duration

of appointments that sometimes include multiple services. Thus, for appointments with multiple task categories (either due to classification or a customer purchasing multiple services in one appointment), the duration must be split. To do this, I first compute the average amount of time spent on each task category among only single-task appointments. Second, for appointments with multiple task categories, I compute the fraction of time to assign to each task category as the corresponding task average divided by the sum of the averages of all tasks in that appointment. Third, I distribute the total time spent on the appointment across the tasks using this imputed fraction.

The data.table package (Barrett et al. 2025), fixest package (Bergé 2018), stargazer package (Hlavac 2022), and squarem package (Du and Varadhan 2018) were all used to construct the data and analyze the results.

A.5 Proof of Proposition 1

Proof. First, consider any set of workers at a firm j that differ only in their labor supply. Take any two workers in the set and suppose for the sake of contradiction that in a profit-maximizing task assignment they have different task assignments. Conditional on a task being assigned to this pair of workers, which of the two workers receives it does not affect quality because both have the same skills. There always exists a way to accomplish the same work in which both workers perform identical task assignments. This alternative task assignment is a coarser division of labor, and by the "distraction-free" property of mutual information (Tian 2019) it requires strictly less communication, contradicting optimality of the original task assignment.

Second, consider a set of workers who have the same skill set but different skill levels at a firm j. Take any two workers in the set and suppose for the sake of contradiction that in a profit-maximizing task assignment they have different task assignments. If labor demand is held fixed, wage costs are sunk, and how a firm divides tasks among workers with the same skill set does not impact service quality. However, assigning all workers with the same skill set the same distribution of tasks strictly reduces communication and thus organization costs by the argument from the last paragraph. Therefore, workers with the same skill set but different skill levels are assigned the same distribution of tasks.■

A.6 Proof of Proposition 2

Proof. First, consider two workers that differ only in labor supply. Because they will be assigned the same distribution of time across tasks and they have the same skills, if they have different wages per unit of labor, firms will demand no labor from the one with the higher wage, and the

labor market will not clear.

Second, consider two workers with the same skill set but different skill level $\bar{\theta}_m > \bar{\theta}_{m'}$. Proposition 1 implies that, conditional on being hired, a firm will assign them the same distribution of time across tasks. Therefore the impact of hiring m compared to m' on profit comes only through wage differences and skill level differences. Firm j hires worker m over worker m' if: $w_m - w_{m'} \leq \rho^{-1}(\bar{\theta}_m - \bar{\theta}_{m'})$. Notice that this inequality is the same for all firms. This implies that labor markets do not clear unless wages are such that firms are indifferent: $w_m - w_{m'} = \rho^{-1}(\bar{\theta}_m - \bar{\theta}_{m'})$.■

A.7 Proof of Proposition 3

This result is proven for a general demand system under which demand for product j is D_j. The only restriction is that demand depend on quality and price only through a quality-price index $(\xi_j - \rho p_j)$ and be strictly increasing in that index. Multinomial logit, nested logit, and mixed logit with a nonrandom price coefficient all satisfy this restriction. Mixed logit with consumer price sensitivity heterogeneity would not. Variables with a subscript $-j$ represent the vector of firm-specific objects excluding those of firm j.

For any given task assignment and relative labor demand, the firm will always choose a price that is weakly above the implied marginal cost; otherwise, it receives negative profit. Without loss, I therefore restrict the set of price-structure pairs considered to be those where price exceeds marginal cost. For this proof, I work in the space of organization structures, defined as: $B_j(i,k) := E_j(i)b_j(i,k)$. I also use the notation $\xi(B_j) := \sum_{i,k} \theta_i(k)B_j(i,k)$. The proof is performed without required-labor, quality and marginal cost heterogeneity (a_j, ν_j, ω_j) for expositional convenience only.

First, I prove that if an organization structure B_j^* solves the simpler problem (Equation 2), then it is profit-maximizing ("if" direction). I need to show that for any price-organization structure pair (p_j', B_j') there exists p_j such that profit under (p_j, B_j^*) is weakly higher than profit under (p_j', B_j'). I do this by construction. Denote B_j^* as a structure which solves Equation 2. Such a structure always exists because Equation 2 is a rate-distortion/rational inattention problem, as I prove in the following lemma.

Lemma 1 *Equation 2 is a rate-distortion or rational inattention problem.*

Proof of Lemma. Equation 2 can be rewritten as:

$$\gamma_j \min_{B_j \in \mathbb{B}} \left\{ I(B_j) + \gamma_j^{-1} \left[W(B_j) - \rho^{-1}\xi(B_j) \right] \right\}. \tag{4}$$

I can rewrite (4) as a maximization problem:

$$\max_{B_j \in \mathbb{B}} \left\{ \sum_{i,k} B_j(i,k)(\rho^{-1}\theta_{i,k} - w_i) - \gamma_j I(B_j) \right\}. \tag{5}$$

Comparing (5) to formulations in papers such as Jung et al. (2019) illustrates that this is a rational inattention problem with mutual information attention costs. I rewrite Equation 4 one last time:

$$\gamma_j \min_{B_j \in \mathbb{B}} \left\{ I(B_j) + \gamma_j^{-1} \sum_{i,k} B_j(i,k)(w_i - \rho^{-1}\theta_{i,k}) \right\}. \tag{6}$$

Comparing Equation 6 to formulations such as Equation 6 in Tishby, Pereira, and Bialek (2000) demonstrates that this is a well-understood minimization problem from information theory called a rate-distortion problem. A solution to these problems is known to exist, and a well-known algorithm can constructively recover at least one solution.■

For any price p_j' and any structure B_j', I can construct $p_j = p_j' + \gamma_j I(B_j^*) + W(B_j^*) - \gamma_j I(B_j') - W(B_j')$. The price p_j is positive and therefore feasible. Recall that profit evaluated at (p_j, B_j^*) is

$$D_j(\xi(B_j^*) - \rho p_j, p_{-j}, \xi_{-j}) \Big[p_j - \gamma_j I(B_j^*) - W(B_j^*) \Big].$$

The second multiplicative term of profit is equal under (p_j, B_j^*) and (p_j', B_j'). The first term (demand) is strictly increasing in the quality-price index $\xi(B_j) - \rho p_j$; therefore, it is sufficient to show that this index is weakly higher for (p_j, B_j^*). I show this by rewriting $\xi(B_j^*) - \rho p_j$:

$$= \xi(B_j^*) - \rho[p_j' + \gamma_j I(B_j^*) + W(B_j^*) - \gamma_j I(B_j') - W(B_j')] \tag{7}$$
$$= \xi(B_j^*) - \rho[p_j' + \gamma_j I(B_j^*) + W(B_j^*) - \gamma_j I(B_j') - W(B_j')] + \xi(B_j') - \xi(B_j') \tag{8}$$
$$= \xi(B_j') - \rho[p_j' + \gamma_j I(B_j^*) + W(B_j^*) - \gamma_j I(B_j') - W(B_j') - \rho^{-1}\xi(B_j^*) + \rho^{-1}\xi(B_j')] \tag{9}$$
$$= \xi(B_j') - \rho p_j' - \rho[\gamma_j I(B_j^*) + W(B_j^*) - \rho^{-1}\xi(B_j^*) - \{\gamma_j I(B_j') + W(B_j') - \rho^{-1}\xi(B_j')\}] \tag{10}$$
$$\geq \xi(B_j') - \rho p_j'. \tag{11}$$

The second-to-last line follows because B_j^* is a minimizer. This proves the "if" direction. I now prove that if a structure B_j^* is profit-maximizing, it solves Equation 2 (the "only if" direction). Suppose for the sake of contradiction there exists B_j' which is profit-maximizing but does not solve Equation 2. Then, as in the first part of the proof, there exists B_j^* which does solve Equation 2. Then I can construct p_j as before for any p_j' that is weakly higher than marginal cost under B_j'.

However, because B_j' does not minimize Equation 2, $\xi(B_j^*) - \rho p_j > \xi(B_j') - \rho p_j'$, and thus profit is strictly higher under B_j^*, p_j. This contradicts optimality of B_j' and concludes the proof.

A.8 Proof of Proposition 4

Denote by Q the quality-adjusted wages. Denote by $I^*(Q)$ the optimal s-index as a function of quality-adjusted wages. Rate-distortion equivalence, proven in Lemma 1, implies $I^*(Q)$ is continuous, convex and decreasing. It is also strictly decreasing below some threshold $\bar{Q}$, and equal to zero for $Q \geq \bar{Q}$. (Moser and Chen 2012). The firm's choice of quality-adjusted wages solves $V(\gamma) := \min_Q \gamma I^*(Q) + Q$. The envelope theorem implies: $\frac{dV(\gamma)}{d\gamma} = I^*(Q) \geq 0$. Taking the first-order condition: $\frac{dI^*(Q)+\gamma^{-1}Q}{dQ} = \frac{dI^*(Q)}{dQ} + \gamma^{-1} = 0 \implies \frac{dI^*(Q)}{dQ} = -\gamma^{-1}$. Because I^* is decreasing and convex, its derivative is negative and increasing. Therefore, Q^* is increasing in γ and $I(Q^*)$ is decreasing in γ. This proves the proposition.

For Corollary 1, note that $V(\gamma)$ is exactly the quality-adjusted cost given in Equation 2. Thus, quality-adjusted cost is increasing in γ. The fact that profit and market share are decreasing in γ then follows from the proof of Proposition 3 (provided in Appendix Section A.7).

A.9 Proof of Theorem 1

For this proof, I suppress the firm subscript j. I define $h_{i,k}$ as the fraction of task k performed by worker i. The first-order condition of (2) using this choice variable is given by:

$$
\begin{aligned}
h_{i,k} &= \frac{E_i}{Z(k,\lambda)} exp\Big(-\lambda(\rho w_i - \theta_{i,k})\Big) \\
\sum_i h_{i,k} &= \frac{1}{Z(k,\lambda)} \sum_i E_i exp\Big(-\lambda(\rho w_i - \theta_{i,k})\Big) = 1 && \text{From summing over } i \\
Z(k,\lambda) &= \sum_i E_i exp(-\lambda(\rho w_i - \theta_{i,k})) \\
h_{i,k} &= \frac{E_i exp(\lambda(-\rho w_i + \theta_{i,k}))}{\sum_{i'} E_{i'} exp(\lambda(-\rho w_{i'} + \theta_{i,k}))} && \text{From substitution of } Z(k,\lambda) \\
h_{i,k} &= \frac{E_i exp(-\gamma^{-1} w_i + (\rho\gamma)^{-1}\theta_{i,k})}{\sum_{i'} E_{i'} exp(-\gamma^{-1} w_{i'} + (\rho\gamma)^{-1}\theta_{i',k})} && \text{From substituting for } \lambda \\
b(i,k) &= \alpha(k) h_{i,k}/E(i) && \text{From the definition of } b(i,k) \\
b(i,k) &= \alpha(k) h_{i,k}/E(i) = \frac{\alpha(k) exp(-\gamma^{-1} w_i + (\rho\gamma)^{-1}\theta_{i,k})}{\sum_{i'} E_{i'} exp(-\gamma^{-1} w_{i'} + (\rho\gamma)^{-1}\theta_{i'}(k))} && \text{From substitution of } h_{i,k}
\end{aligned}
$$

This illustrates that optimal jobs take an almost-logit form. I can also derive this result by

applying Theorem 1 from Matějka and McKay (2015).

The fact that all hired worker types spend a positive amount of time on each task is a direct application of Lemma 1 from Jung et al. (2019). An increase in wage corresponds to a decrease in the "payoff" to the firm of using workers of type i in all tasks (i.e., states of the world in the rational inattention literature). This means I can apply Proposition 3 from Matějka and McKay (2015) to say that an increase in w_i leads to a decrease in E_i, all else constant. I can further say that E_i is strictly decreasing in w_i whenever the initial share of worker i is strictly interior, i.e., $0 < E_i < 1$.

The property of maximum coworker diversity follows from noting that the firm's task-assignment problem is a rational inattention problem with a Shannon mutual information cost function; therefore, Corollary 3 of Caplin and Dean (2013) applies (states bound acts): there exists an optimal task assignment strategy that uses weakly fewer worker sill sets than task categories. There may also exist other optimal strategies that use more workers, but they achieve the same profit.

A.10 Proof of Proposition 5

To recover the best responses of the firm's problem, I use the fact that the joint maximization of any function is equivalent to the sequential maximization. Thus I can write the firm's problem as

$$\max_{p_j, b_j, E_j} \frac{exp(\xi(b_j, E_j) - \rho p_j)}{1 + \sum_{j'} exp(\xi(b_{j'}, E_{j'}) - \rho p_{j'})} \left(p_j - a_j \gamma_j \sum_i E_j(i) D_{KL}(b_j(i,\cdot) || \alpha_j) - a_j \sum_i w_i E_j(i) - \alpha_j c - \omega_j \right)$$

s.t.

$$\sum_i E_j(i) b_j(i,k) = \alpha_j(k) \forall k$$

I first study the inner pricing problem. When the organization structure is held fixed, the model reduces to a logit Bertrand game with heterogeneous costs and qualities. Proposition 7 of Caplin and Nalebuff (1991) proves that such a game has a unique pure-strategy Nash equilibrium in prices. Therefore, for any chosen organization structure, there is a single best-response price.

I now move on to the choice of task assignments and relative labor demands (the outer maximization). In Proposition 3, I show that when prices are chosen to maximize profit, the internal organization choice separates from the pricing problem and solves:

$$\min_{b_j, E_j} \gamma_j \sum_i E_j(i) D_{KL}(b_j(i,k) || \alpha_j) + \sum_i E_j(i) w_i - \rho^{-1} \sum_i E_j(i) \sum_k \theta_i(k) b_j(i,k) \tag{12}$$

s.t. $\sum_i E_j(i)b_j(i,k) = \alpha_j(k) \forall k$

The best-response structure will therefore depend on other firms' actions only through wages. Lemma 1 in Appendix Section A.7 shows that this is a rational inattention problem with a mutual information cost function. I can appeal to Matějka and McKay (2015) to say that there exists an organization structure which maximizes profit for each firm. This establishes equilibrium existence.

For uniqueness, the online Appendix of Matějka and McKay (2015) contains a result which implies that whenever the following condition holds, the relative labor demands and task assignments will have a unique solution:

Assumption 1 *(adapted from Matějka and McKay (2015)) Define the wage-quality vector of a worker of type i at firm j as $v_{i,j} = \{exp(\gamma_j^{-1}(\rho^{-1}\theta_i(k) - w_i))\}_{k=1}^K$. The set of wage-quality vectors $\{v_{i,j}\}_{i \in \mathcal{I}}$ is affinely independent, for all firms $j = 1, ..., J$.*

Notice that wage-adjusted skills can be rewritten as:

$$exp(\gamma_j^{-1}(\rho^{-1}\theta_i(k) - w_i)) = exp(\rho^{-1}\gamma_j^{-1}\theta_i(k))exp(-w_i\gamma_j^{-1}) \tag{13}$$

I assume that $\det(exp[c\Theta]) \neq 0$ for all $c > 0$, which implies that the vectors $\{exp(\rho^{-1}\gamma_j^{-1}\theta_i(k))\}_{k=1}^K$ are linearly independent. Note that $exp(-w_i\gamma_j^{-1}) > 0$; therefore, the vectors $v_{i,j}$ are formed by multiplying N linearly independent vectors by positive constants. Multiplication by a non-zero constant is an elementary row operation; therefore, this transformation preserves the rank of the matrix. Thus, the vectors $\{v_{i,j}\}_{i=1}^N$ are linearly independent, and linear independence implies affine independence at all firms j, and thus uniqueness of task assignment strategies at all firms.

Note that $\det(exp[c\Theta]) \neq 0$ for all $c > 0$ is one of several sufficient conditions. Another is that there exists a positive semi-definite symmetric matrix which has all the skill set vectors as rows. To use this condition, note that the vectors given by (13) are then the function $f(x) = exp(x)^{(\gamma\rho)^{-1}}exp(-w_i\gamma_j^{-1})$ applied element-wise to each skill set vector θ_i. An application of the Schur Product Theorem, which is Theorem 7.5.9 in Horn and Johnson (2012), yields that the element-wise exponential of a positive semi-definite matrix is positive definite, provided no two rows of the initial matrix are identical (trivially true in this setting). Thus the matrix derived from applying $exp(x)^{(\gamma\rho)^{-1}}$ point-wise to this matrix is positive definite, and therefore the skill set vectors are linearly independent and by implication affinely independent. While this condition is somewhat easier to check and state, it requires symmetry, which is not empirically appealing.

Notice that this result is true for any wage vector w. Thus there is a unique task assignment and relative labor demand for all wage vectors w. Earlier in the proof, it was demonstrated that

for each task assignment, pricing strategies are unique. Combining the two results, I now have that there is a unique Nash equilibrium for any fixed wage vector w.

A.11 Proof of Theorem 2

Proof. I first set aside all firms where one of the K task types is not performed at all ($\exists k$ s.t. $\alpha_j(k) = 0$) or where the s-index is 0 ($I_j = 0$). Until stated otherwise, I work with only the subsample of firms where all tasks are performed and the s-index is strictly positive. Propositions 1 and 2 establish that all workers at the same firm are assigned the same distribution of time across tasks if they have the same skill set. As a result, time-use data allow workers to be grouped into mutually exclusive types within each firm. That is, I can aggregate the workers at each firm into representative workers, one for each skill set present at the firm. However, these representative workers are not comparable across firms. If I observe two groups of workers at firm A and two at firm B, I do not know which, if any, of the groups are the same skill set. I know only that group 1 and group 2 at firm A have different skill sets from each other, and group 1 and group 2 at firm B have different skill sets from each other. Note that the firms currently being analyzed all have strictly positive s-index, which implies they employ at least two worker skill sets.

In order to partition all workers in the market into skill sets, I need to group workers across firms. The key challenge is that the task content of a worker's job depends on both the worker's skill and the firm at which the worker works. Given a worker l at firm j with unknown skill set t_l, I can capture this issue using the characterization from Theorem 1:

$$b_j(t_l, k) = \alpha_j(k) \frac{exp(-\gamma_j^{-1} w(t_l) + (\rho\gamma_j)^{-1}\theta_{t_l}(k))}{\sum_{i'} \frac{E_j(i')}{\sum_{i''} E_j(i'')} exp(-\gamma_j^{-1} w(i') + (\rho\gamma_j)^{-1}\theta_{i'}(k))}$$

To identify t_l, recall that at this stage workers with different types are grouped into mutually exclusive groups within the firm. At any firm with at least two worker skill sets, I can divide one worker's distribution of time across tasks by that of a coworker known to have a different skill set, and take the logarithm to obtain:

$$\log\left(\frac{b_j(t_l, k)}{b_j(t_{l'}, k)}\right) = (\rho\gamma_j)^{-1}\Big([\theta_{t_l}(k) - \rho w(t_l)] - [\theta_{t_{l'}}(k) - \rho w(t_{l'})]\Big)$$

This expression does not depend on firm-specific factors except for the fact that it is scaled by the reciprocal of organization costs γ_j. It is also defined for all K tasks and for every coworker who has a different skill set. Thus every worker has several such log-ratio vectors, representing how

similar the task content of their job is to all of their coworkers. The following lemma, which uses the fact that wage-adjusted skills are linearly independent, establishes that the normalized log-ratio vectors of two worker pairs, possibly at different firms, coincide if and only if the workers' skill sets coincide pair by pair.

Lemma 2 *Suppose workers 1 and 2 are at firm j and workers 3 and 4 are at firm j'. Suppose firms j and j' are in the same market. The log ratio of relative time-use between workers 1 and 2 divided by its Euclidean norm is equal to the log ratio of relative time-use between workers 3 and 4 divided by its Euclidean norm if and only if worker 1 is the same type as worker 3 and worker 2 is the same type as worker 4.*

Proof. Normalizing the log ratio removes the firm-specific object γ_j:

$$\frac{\log\left(\frac{\frac{\tilde{B}_j(t_l,k)}{\tilde{E}_{t_l}}}{\frac{\tilde{B}_j(t_{l'},k)}{\tilde{E}_{t_{l'}}}}\right)}{\left|\left\{\log\left(\frac{\frac{\tilde{B}_j(t_l,k')}{\tilde{E}_{t_l}}}{\frac{\tilde{B}_j(t_{l'},k')}{\tilde{E}_{t_{l'}}}}\right)\right\}_{k'=1}^{K}\right|} = \frac{(\rho\gamma_j)^{-1}\left[\tilde{\theta}_{t_l}(k)-\tilde{\theta}_{t_{l'}}(k)\right]}{\left(\sum_{k'}\left((\rho\gamma_j)^{-1}\left[\tilde{\theta}_{t_l}(k')-\tilde{\theta}_{t_{l'}}(k')\right]\right)^2\right)^{1/2}}$$

$$\frac{\log\left(\frac{\frac{\tilde{B}_j(t_l,k)}{\tilde{E}_{t_l}}}{\frac{\tilde{B}_j(t_{l'},k)}{\tilde{E}_{t_{l'}}}}\right)}{\left|\left\{\log\left(\frac{\frac{\tilde{B}_j(t_l,k')}{\tilde{E}_{t_l}}}{\frac{\tilde{B}_j(t_{l'},k')}{\tilde{E}_{t_{l'}}}}\right)\right\}_{k'=1}^{K}\right|} = \frac{\left[\tilde{\theta}_{t_l}(k)-\tilde{\theta}_{t_{l'}}(k)\right]}{\left(\sum_{k'}\left[\tilde{\theta}_{t_l}(k')-\tilde{\theta}_{t_{l'}}(k')\right]^2\right)^{1/2}}$$

From this expression it is clear that for any fixed pair of types $t_l, t_{l'}$ the normalized log ratios will be the same at all firms. This proves one direction. I now prove that if the normalized log ratios are the same, the types correspond. Suppose for the sake of contradiction that the normalized log ratios correspond but the types are different. Then:

$$\frac{\left[\tilde{\theta}_{t_1}(k)-\tilde{\theta}_{t_2}(k)\right]}{\left(\sum_{k'}\left[\tilde{\theta}_{t_1}(k')-\tilde{\theta}_{t_2}(k')\right]^2\right)^{1/2}} = \frac{\left[\tilde{\theta}_{t_3}(k)-\tilde{\theta}_{t_4}(k)\right]}{\left(\sum_{k'}\left[\tilde{\theta}_{t_3}(k')-\tilde{\theta}_{t_4}(k')\right]^2\right)^{1/2}}$$

Denoting the denominators as $\frac{1}{A}$ and $\frac{1}{B}$: $A\left[\tilde{\theta}_{t_1}-\tilde{\theta}_{t_2}\right] = B\left[\tilde{\theta}_{t_3}-\tilde{\theta}_{t_4}\right]$ which can be rewritten as $A\tilde{\theta}_{t_1} - A\tilde{\theta}_{t_2} - B\tilde{\theta}_{t_3} + B\tilde{\theta}_{t_4} = 0$. This can be rearranged as saying that one wage-adjusted skill can be written as a combination of other wage-adjusted skills: $\tilde{\theta}_{t_1} = \frac{B}{A}\tilde{\theta}_{t_3} - \frac{B}{A}\tilde{\theta}_{t_4} + \tilde{\theta}_{t_2}$ $(*)$. Recall that $t_1 \neq t_2, t_3 \neq t_4$ are assumed for sake of contradiction. Also observe that the left-hand side of

$(*)$ is a valid linear combination and therefore $(*)$ is a contradiction of linear independence unless $t_1 = t_3$ and $t_2 = t_4$, in which case $A = B$ and we have that all coefficients are 0. Even in the case when $t_1 = t_4, t_2 = t_3$ this is a contradiction because then $B = A$ but the coefficients are $2, -2$. Thus the ratios coincide if and only if the types of the workers coincide.■

With this result in hand, I can classify two workers as having the same skill set if one of their log-ratio vectors matches. By repeating this process, I can partition all workers within the connected set of firms into the N mutually exclusive skill set groups. Note that by assumption, every skill set is employed by at least one firm among the connected set of firms.

At this point in the proof, workers are classified into skill set groups. Although the skill set parameters and wages are not yet identified, the labeling itself is arbitrary. As a result, I can treat the index of each worker's skill set as identified, and jobs $b_j(i,k)$ as data for the rest of the proof. I now identify the market parameters Ω. It is convenient to define $B_j(i,k) := \frac{E_j(i)b_j(i,k)}{\sum_{i'} E_j(i')}$, the share of total labor assigned to worker i and task k. This object is identified and can be considered data in what follows because $b_j(i,k), E_j(i)$ are identified. Under the demand system, market shares can be written as:

$$\log\left(\frac{s_j}{s_0}\right) = \sum_{i,k} \theta_i(k) a_j B_j(i,k) - \rho p_j + \nu_j \tag{14}$$

The theoretical results imply the firm's task-assignment strategy ($\{b_j(i,k)\}_{i,k}$) and relative labor demands ($\frac{E_j(i)}{\sum_{i'} E_j(i')}$) do not depend on ν_j or prices p_j. They do depend on the task mix (α_j) and organization costs (γ_j), but I assumed that these objects are independent of ν_j. Therefore $B_j(i,k)$ is independent of ν_j and I have $N \times K$ moment conditions $E[\nu_j B_j(i,k)] = 0, \forall i,k$. Prices are set strategically, and they depend directly on ν_j. However, the model provides a natural instrument: relative organization costs ($\frac{\gamma_j}{\gamma_1} I_j$). This object has already been identified and can therefore be considered data. It is independent of ν_j because organization cost parameters are independent of ν_j and the amount of communication is proven to not depend on ν_j. It also has a positive relationship with prices because it increases marginal costs. Thus I obtain the moment condition $E[\nu_j \frac{\gamma_j}{\gamma_1} I_j] = 0$. Collecting these I obtain the following set of moment conditions:

$$E\left[\left(\log\left(\frac{s_j}{s_0}\right) - \sum_{i,k} a_j \theta_i(k) B_j(i,k) + \rho p_j\right)\begin{pmatrix} \{B_j(i,k)\}_{i,k}^{N,K} \\ \frac{\gamma_j}{\gamma_1} I_j \end{pmatrix}\right] = 0 \tag{15}$$

This is a linear system of $N \times K + 1$ equations and $N \times K + 1$ unknowns, so there exists a single solution under the assumed full rank conditions. Thus the skill set parameters (Θ) and price sensitivity (ρ) are identified. To identify the remaining parameters, consider the firm's first-order

condition for price, which takes the standard Bertrand-logit form of a markup plus marginal cost:

$$p_j = \frac{1}{\rho(1-s_j)} + \gamma_1 \frac{\gamma_j}{\gamma_1} a_j I_j + \sum_i w_i a_j \frac{E_j(i)}{\sum_{i'} E_j(i')} + \alpha_j' c + \omega_j$$

Price sensitivity has already been identified, so the markup $\frac{1}{\rho(1-s_j)}$ is identified. I can therefore adjust prices by the markup to obtain marginal costs:

$$p_j - \frac{1}{\rho(1-s_j)} = \gamma_1 \frac{\gamma_j}{\gamma_1} a_j I_j + \sum_i w_i a_j \frac{E_j(i)}{\sum_{i'} E_j(i')} + \alpha_j' c + \omega_j \tag{16}$$

The theoretical results imply the firm's relative labor demands ($\frac{E_j(i)}{\sum_{i'} E_j(i')}$) and relative organization cost ($\frac{\gamma_j}{\gamma_1} I_j$) do not depend on ω_j or prices p_j. They do depend on the task mix (α_j) and organization costs (γ_j), but I assumed that these objects are independent of ω_j. Therefore both are independent of ω_j and I have the moment conditions $E[\omega_j \frac{E_j(i)}{\sum_{i'} E_j(i')}] = 0$, $E[\omega_j \frac{\gamma_j}{\gamma_1} I_j] = 0$ and $E[\omega_j \alpha_j] = 0$. Collecting these I obtain the following set of moment conditions:

$$E\left[\left(p_j - \frac{1}{\rho(1-s_j)} - \gamma_1 \frac{\gamma_j}{\gamma_1} a_j I_j - \sum_i w_i a_j \frac{E_j(i)}{\sum_{i'} E_j(i')} - \alpha_j' c\right) \begin{pmatrix} \{a_j \frac{E_j(i)}{\sum_{i'} E_j(i')}\}_i^N \\ a_j \frac{\gamma_j}{\gamma_1} I_j \\ \alpha_j \end{pmatrix}'\right] = 0 \tag{17}$$

This is a linear system of $N+K+1$ equations and $N+K+1$ unknowns, so there exists a single solution under the assumed full rank conditions. Thus the skill-set-specific wages ($\{w_i\}_{i=1}^N$), the material-cost parameters (c), and the organization cost of the reference firm (γ_1) are all identified. With this, all of the parameters in Ω are now identified.

The only objects that remain to be identified are the organization cost parameters and skill sets of workers at firms that either (1) do not use all task types but have a strictly positive s-index or (2) have an s-index of 0.

I begin with the firms in group (1). By Proposition 4, the s-index I_j is strictly decreasing in γ_j until it reaches 0. The s-index chosen by the firm for each γ_j is known because all market parameters are now identified. Therefore, the organization cost parameter γ_j is identified for these firms as the value where the model-predicted level of communication matches the observed s-index. Task-assignment strategies are unique by Proposition 5; therefore, identification of γ_j implies identification of the skill sets of the workers employed at the firm.

Now I address (2). By Proposition 4, the organization cost parameter of these firms is only set-identified: γ_j can be any number above some threshold $\bar{\gamma}_j$. Because the s-index is strictly

decreasing in γ_j below this threshold, the threshold is identified. The firm's task-assignment strategy (and therefore the composition of its workforce) is the same for all $\gamma_j > \bar{\gamma}_j$; therefore, since task-assignment strategies are unique by Proposition 5, identification of the threshold $\bar{\gamma}_j$ implies identification of the skill sets of the workers employed at the firm.■

A.12 Estimation Procedure

The estimation procedure consists of the following steps:

1. Classify workers within each establishment based on their jobs' task content. Set aside salon-quarters where workers cannot be grouped across establishments.[32]

2. Classify workers across establishments using their normalized coworker log-ratio vectors.

3. Using recovered task assignments and relative labor demands, recover price sensitivity, skill set parameters, and county-quarter demand levels via two-stage least squares applied to (14). As an instrument for price, use the producer price index for synthetic organic dyes multiplied by the task mix for the color task.

4. Guess relative wages. Recover the organization cost parameter of each salon by inverting the s-index. Solve each salon-quarter's internal organization problem. Compare the average relative labor demands to those implied via the classification procedure. Repeat this step until convergence.

5. Since relative wages are now known, recover the organization costs and relative labor demands of the set-aside establishments by inverting the s-index.

6. Estimate material costs, county-quarter wage levels, and cost levels via linear GMM applied to (16) with all variables instrumenting for themselves. Material costs are allowed to vary by quarter, are are relative to the omitted haircut/shave task category, whose level is absorbed by the county-quarter cost level. Constrain wage levels such that all wages are above county-quarter-specific minimum wages.

I use inversion, as opposed to the connected-set procedure from the identification proof, to recover salon-quarter organization costs for two reasons. First, this imposes that each establishment's s-index in the data match its model-generated s-index at the market parameters. Second, it

32. This includes establishments with an s-index of 0, establishments with only one worker skill set, and establishments that do not perform at least one of the task categories.

leads to more reasonable organization costs in practice. The main drawback of inversion is that it greatly increases computational time, as it requires solving each establishment's internal organization problem for each guess of the wage vector. I mitigate this cost by accelerating the contraction mapping proposed in Appendix Section A.14.

I classify workers within a salon using hierarchical clustering with complete linkage where the distance is Euclidean. Formally, I fix a stopping threshold, call it h^*. Even though clustering is within each salon, the stopping threshold is common across salons within a county (subject to the salon-specific override described below). For each salon, I start with all workers in the salon in separate skill sets. I then measure the Euclidean distance between each pair of task-assignment vectors $\{b_j(m,k)\}_{k=1}^K$. If none are less than h^*, I stop. Otherwise, I group the two workers with the smallest distance into one skill set. Then, I compute the Euclidean distance between every pair again, but for the two-worker skill set, I define its distance to any worker outside the group as the largest of the distances between that worker's task-assignment vector and each group member's vector (this is the complete linkage method). Again, if none are less than h^*, I stop. Otherwise, I group the pair with the smallest distance and repeat until no pairwise distance is below h^*.

I set the within-establishment classification threshold h^* in the following way. For each salon-quarter I find the smallest threshold $h_{j,t}$ such that there are at most $N = K = 5$ clusters of workers at that salon. I then set h^* to be the 90th percentile of $h_{j,t}$ in that county (where salons are weighted in the percentile determination based on their number of employees). For salons with $h_{j,t} < h^*$ (salons with low task-assignment diversity), this generates five or fewer clusters, which is consistent with theory. For salons with $h_{j,t} > h^*$ (salons with high task assignment diversity), using h^* can result in more than five clusters, which is inconsistent with theory. For these salons only, I allow the clustering procedure to use $h_{j,t}$. This procedure ensures that there exists a salon with N skill sets (a sufficient condition for identification). It should be noted that by construction a salon cannot have more skill sets than actual workers. However, if a salon has many workers, that does not guarantee that it has all five skill sets. The procedure will group these workers into representative workers based on their task assignments.

The practical implication of this rule for h^* is that salons in the top 10 percent in terms of unique task assignment vectors are assumed to employ all five worker skill sets. Another natural rule would be to set h^* to be such that there are no more than 5 skill sets observed at any salon (a maximum rule). The maximum rule is more sensitive to outliers, which is why it was not used as the main specification. After grouping workers into skill sets within a salon, I treat each group as a representative worker, with a single task assignment vector which is the weighted average of

their individual task assignments, with weights given by their relative labor demand.

Classifying workers across establishments requires accounting for the fact that task assignments across salons are not directly comparable due to salon-specific confounders like γ_j and α_j. The proof of identification implies a clustering procedure, where workers are grouped across establishments into skill sets based on the ratio of their time use with coworkers from other skill sets. To translate the theoretical procedure into a practical estimation procedure, I must choose an algorithm to implement it. Several factors constrain the procedures I can use. First, the ratio comparisons are not proper distances. That is, I can make statements of the form: workers 1 and 2 are close. But I cannot say that 1 and 2 are closer than 3 and 4, and I cannot say that 1 and 2 are close to some fixed point in Euclidean space. I also cannot say that just because 1 and 2 are far, they are not the same type. This is because two workers may have different log ratios simply because they have different coworker types.

These issues imply I cannot use any clustering algorithms that rely on centroids or that aggregate over multiple coworker pairs when forming clusters. This rules out popular non-hierarchical algorithms like k-means and also many hierarchical algorithms that use linkage methods which involve some sort of aggregation (average, complete, Ward, median, etc.). To address this, I develop a single-linkage procedure which uses the fact that there exists a salon with all $N = K$ skill sets. This firm is guaranteed to exist computationally and assumed to exist for estimation (its existence is a sufficient condition for the weaker connected set assumption used for identification).

1. Start with any establishment that has all worker skill sets (N distinct worker groups). Call this the reference establishment. Fix the labels of the worker skill sets to be the labels of the N distinct worker groups at the reference establishment.

2. Consider any establishment that is not the reference establishment. Take all the distinct worker groups at this establishment and label them every possible way. For example, if there are three worker skill sets at an establishment, label these groups all 60 ways (1-2-3,2-3-1,1-3-2, etc.).

3. Under any potential labeling, denote worker l's type t_l and treat it as known. Compute the log ratio of relative time use between l and every $l' \neq l$ and the log ratio of relative time use between t_l and $t_{l'}$ at the reference salon:

$$\frac{\log\left(\frac{\frac{\tilde{B}_j(l,k)}{\tilde{E}_j(l)}}{\frac{\tilde{B}_j(l',k)}{\tilde{E}_j(l')}}\right)}{\left|\left\{\log\left(\frac{\frac{\tilde{B}_j(l,k')}{\tilde{E}_j(l)}}{\frac{\tilde{B}_j(l',k')}{\tilde{E}_j(l')}}\right)\right\}_{k'=1}^{K}\right|} \qquad \frac{\log\left(\frac{\frac{\tilde{B}_0(t_l,k)}{\tilde{E}_0(t_l)}}{\frac{\tilde{B}_0(t_{l'},k)}{\tilde{E}_0(t_{l'})}}\right)}{\left|\left\{\log\left(\frac{\frac{\tilde{B}_0(t_l,k')}{\tilde{E}_0(t_l)}}{\frac{\tilde{B}_0(t_{l'},k')}{\tilde{E}_0(t_{l'})}}\right)\right\}_{k'=1}^{K}\right|}$$

and compute the Manhattan distance between the two vectors. Sum this distance for all pairs $l, l' \neq l$ to compute a measure of fit for each potential labeling. Select the potential labeling with the smallest measure of fit. Repeat for all establishments in the market.

This procedure is both internally consistent and computationally feasible because, under the assumption that some firm employs all worker skill sets, every other firm can be linked to that firm both theoretically (under optimal task assignment) and numerically (one labeling of workers at each firm will minimize the distance of normalized coworker log ratios from the reference firm's normalized coworker log ratios). Additionally, the procedure never contradicts itself. For example, if workers A and C are at the same firm, and the first clustering procedure grouped them into different skill sets within the firm, the algorithm will never generate a grouping that places workers A and C in the same skill set.

The proof of identification uses variation in relative organization costs across firms to identify price sensitivity. In practice, relative organization costs are estimated using small samples and are therefore noisy. Instead, I use the share of the color task multiplied by the producer price index of synthetic organic dyes in that quarter. In practice this provides better power. Because the task mix is taken to be exogenous, this instrument satisfies an exclusion restriction under the theory. It is also relevant to firm pricing decisions both in theory and empirically.[33] Intuitively, variation across time in the price of dye and variation across firms in the share of color tasks generates variation in costs, and observing how this variation in costs is passed on to consumers via prices identifies the price sensitivity parameter (ρ).

For the classification step of estimation only, I apply Lidstone smoothing to the task assignments within a firm. Specifically, I add 50.22 minutes (equivalent to adding a single task realization with an average duration) to the time spent by each employee on each task in a quarter. The idea is to capture the fact that because the sample is finite, there is a positive probability that even if a worker is supposed to be assigned a task, they are not observed being assigned that task. The practical reason for doing this is to avoid undefined coworker log ratios.

Within the estimation routine, I implement two guards to ensure that the equilibrium implied

33. It is strongly positively correlated with prices.

by the model parameters remains interior. First, I penalize the labor demand GMM objective whenever the implied share of a worker skill set is close to 0. Second, I require that the observed s-index of all salon-quarters be achievable by some finite γ. In practice, I penalize the GMM objective whenever the parameters are such that a single skill set has the highest wage-adjusted skills for more than one task, and this results in a maximum s-index that is below the observed s-index (which would make inversion of the s-index impossible).

A.13 Computation of Standard Errors

Estimation can be viewed as three sequential generalized method of moments (GMM) procedures on a panel with establishments indexed by j and time indexed by t:

1. the demand step that recovers skills and price sensitivity:

$$E\left[\left(\log\left(\frac{s_{j,t}}{s_{0,t}}\right) - \sum_{i,k} a_{j,t}\theta_i(k)B_{j,t}(i,k) + \rho p_{j,t}\right)\begin{pmatrix}\{B_{j,t}(i,k)\}_{i,k}^{N,K}\\ Z_{j,t}\end{pmatrix}\right] = 0 \tag{18}$$

2. the labor market demand step that recovers wages (where $\tilde{E}_{j,t}(i)(w, I_{j,t}, \theta, \rho)$ denotes model-implied labor demand):

$$E\left[\begin{pmatrix}E_{j,t}(2) - \tilde{E}_{j,t}(2)(w, I_{j,t}, \theta, \rho)\\ \dots \\ E_{j,t}(5) - \tilde{E}_{j,t}(5)(w, I_{j,t}, \theta, \rho)\end{pmatrix}\right] = 0 \tag{19}$$

3. and then the pricing step, which recovers cost parameters:

$$E\left[\left(\tilde{p}_j - \gamma_1\frac{\gamma_j}{\gamma_1}a_j I_j - \sum_i w_i a_j \frac{E_j(i)}{\sum_{i'} E_j(i')} - \alpha_j' c\right)\begin{pmatrix}\{a_j\frac{E_j(i)}{\sum_{i'} E_j(i')}\}_i^N\\ \frac{\gamma_j}{\gamma_1}I_j\\ \alpha_j\end{pmatrix}'\right] = 0. \tag{20}$$

Note that the expressions omit time subscripts and fixed effects for brevity, but these are included in estimation. The labor market demand step also imposes a penalty whenever there is close to 0 labor market demand for a skill set, but at the estimated parameters labor demand is interior and the penalty does not apply. Note that each step in the sequence uses the estimates from the earlier steps as inputs. As a result, standard errors are computed so that the estimation error from earlier steps propagates through to later steps (Murphy and Topel 2002, Newey

and McFadden 1994). Following L. P. Hansen (1982), I construct a "sandwich"-style estimator of the variance-covariance matrix clustered at the establishment level. Denote the Jacobian of the moment conditions at the parameter estimates as J. The derivatives of all demand moments are given analytically by the standard 2SLS form and the derivatives of the price step with respect to the parameters that appear only in that step are given by the standard OLS form. All other parts of the Jacobian are computed using numerical derivatives. The "meat" of the variance-covariance estimator is then the variance-covariance matrix of the moments, computed within establishment across time and then summed across establishments, which I denote S. I form a small-sample correction denoted $c := G/(G-1) \cdot (N-1)/(N-K_{total})$ where G is the total number of establishments, N is the total number of establishment-quarters used in estimation, and K_{total} is the number of parameters. The variance-covariance matrix is then $V_{joint} := cJ^{-1}S(J^{-1})'$.

Computing the variance-covariance estimate this way involves forming and inverting a large Jacobian. However, because estimation is sequential, parameters estimated earlier affect later moments but parameters estimated later do not impact earlier moments, so the Jacobian is block lower-triangular. I exploit this fact to compute the variance-covariance matrix by summing influence functions within establishments, block by block of moments. For the parameters recovered from the demand moments (price sensitivity and skill set parameters), the standard errors generated by this procedure are equivalent to two-stage least squares standard errors reported in Appendix Section A.2, up to the small-sample adjustment. Parameters that sit on their lower bound (the wage level in four Cook County quarters) are considered fixed and excluded from standard error calculation.

The classification procedure of workers into skill sets depends only on the support of observed task assignments, not on the distribution of observations across the support.[34] Thus, uncertainty from this initial step is not reflected in the standard errors. However, because the finite number of worker skill sets is small relative to the number of establishments and workers, it is possible that an asymptotic super-consistency argument similar to that given in Bonhomme and Manresa (2015) holds. If that is the case, not accounting for classification in asymptotic inference is justified.

A.14 Full Solution Contraction Mapping

Recovering the organization costs of firms that employ only one skill set (the final step of estimation) and performing counterfactuals both require solving for the firm's optimal task-assignment

34. To make this more concrete: if the Bayesian bootstrap were used to obtain standard errors, the classification step would be identical across all bootstrap replications.

strategy. Although the theoretical results imply that the firm's task-assignment strategy is fully characterized by the division of time across worker skill sets, this is still an $N \times K$ matrix that must be found for all J firms. It turns out that there exists a globally convergent contraction mapping which delivers the firm's profit-maximizing organization structure given values for the market parameters (Ω).

Proposition 6 *Given market parameters (Ω) and firm j's task mix and organization cost parameter, the Blahut–Arimoto algorithm delivers relative labor demands and jobs for each skill set that jointly maximize firm j's profit.*

The proposition follows directly from the fact that the firm's strategy solves an equivalent rate-distortion problem, which can be solved using the Blahut–Arimoto algorithm.[35] This equivalence is proven in the course of the proof of Proposition 3 in Appendix Section A.7. The Blahut–Arimoto algorithm (Blahut 1972) is a fixed-point algorithm which iterates on two optimality conditions and can be described as follows, where the firm subscript j is suppressed:

0. Guess some relative labor demands E^0. Create matrix V: $V_{i,k} = \exp[\gamma^{-1}(\rho^{-1}\Theta(i,k) - w_i)]$.
1. Compute interim organization structure $B(i,k)^t = \alpha(k)\frac{V_{i,k}\frac{E^t(i)}{\sum_{i'} E^t(i')}}{\sum_{i'}\frac{E^t(i')}{\sum_{i''} E^t(i'')}V_{i',k}}$.
2. Compute interim relative labor demands $\frac{E^{t+1}(i)}{\sum_{i'} E^{t+1}(i')} = \sum_k B(i,k)^t$.
3. If converged, exit; else return to Step 1 and advance t.

This algorithm converges to a global optimum from any feasible starting point (Tishby, Pereira, and Bialek 2000). For fixed wages, there is one global optimum, so the algorithm converges to the unique profit-maximizing strategy for each firm. Using the algorithm allows the researcher to estimate the model and perform counterfactuals without numerically searching for the firm's profit-maximizing strategy. Solving for a counterfactual equilibrium then consists of only two additional steps: solving for equilibrium product prices (a standard problem in industrial organization) and solving for the wages which clear the labor market.

A.15 Closed-Form Logit Price Expression

Demand for a product j is given by $s_j(p_j) = \frac{\exp(-\rho p_j + \xi_j)}{\sum_{j'=0}^{J}\exp(-\rho p_{j'} + \xi_{j'})}$. Optimal pricing in a Bertrand Nash equilibrium with single-product firms is then given by $p_j = MC_j + \frac{1}{\rho(1-s_j(p_j))}$, where c_j

35. See Tishby, Pereira, and Bialek 2000 or Blahut 1972.

denotes marginal cost. I now follow the arguments laid out in Aravindakshan and Ratchford (2011). I rewrite this expression as

$$p_j = c_j + \frac{1}{\rho(1 - \frac{\exp(-\rho p_j + \xi_j)}{exp(-\rho p_j + \xi_j) + \sum_{j' \neq j} \exp(-\rho p_{j'} + \xi_{j'})})}.$$

I rewrite it again as

$$p_j = c_j + \frac{1}{\rho} + \frac{\exp(-\rho p_j + \xi_j)}{\rho \sum_{j' \neq j} \exp(-\rho p_{j'} + \xi_{j'})}.$$

Multiplying by ρ and subtracting ξ_j yields

$$\rho p_j - \xi_j = \rho c_j + 1 + \frac{\exp(-\rho p_j + \xi_j)}{\sum_{j' \neq j} \exp(-\rho p_{j'} + \xi_{j'})} - \xi_j.$$

Now denote $E_j = \sum_{j' \neq j} \exp(-\rho p_{j'} + \xi_{j'})$. Rearranging the previous equation yields

$$\frac{exp(-\rho p_j + \xi_j)}{E_j} + \xi_j - \rho p_j = -1 - \rho c_j + \xi_j.$$

Exponentiating both sides and multiplying by E_j^{-1} gives

$$exp\Big(\frac{exp(\xi_j - \rho p_j)}{E_j}\Big) exp\Big(\xi_j - \rho p_j\Big) E_j^{-1} = exp\Big(-1 + \xi_j - \rho c_j\Big) E_j^{-1}.$$

Finally, define$\tilde{W} = exp\Big(\xi_j - \rho p_j\Big) E_j^{-1}$ where $\tilde{W}$ is a constant, unrelated to the wage, whose notation will be clear in a moment. Then the expression becomes

$$\tilde{W} e^{\tilde{W}} = \exp\Big(-1 + \xi_j - \rho c_j\Big) E_j^{-1}.$$

The left-hand side is of the form required by the Lambert W function, so, by definition, the $\tilde{W}$ that solves the equation is given by the Lambert W function evaluated at the right-hand side. I denote the Lambert W function by $L(\cdot)$, and the optimal price solves

$$L\Big(\exp\Big(-1 + \xi_j - \rho c_j\Big) E_j^{-1}\Big) = \exp\Big(\xi_j - \rho p_j\Big) E_j^{-1}.$$

A property of this function is that $\log(L(x)) = \log(x) - L(x)$. Using this fact yields

$$-1 + \xi_j - \rho c_j - \log(E_j) - L\Big(\exp\Big(-1 + \xi_j - \rho c_j\Big) E_j^{-1}\Big) = \xi_j - \rho p_j - \log(E_j),$$

which can be solved for the optimal price:

$$\frac{1}{\rho} + c_j + \rho^{-1} L\left(\exp\left(-1 + \xi_j - \rho c_j \right) E_j^{-1} \right) = p_j^*. \tag{21}$$

A.16 Counterfactual Procedure

In order to perform counterfactuals, it is necessary to impose additional assumptions, which I outline in four steps. First, the data contain only a small fraction of the total set of salons, but to solve for a new equilibrium I must understand how all firms respond. I do this by assuming there are n copies of each salon in the data, where n is set to be the number such that the sum of market shares in the data multiplied by n equals 1 minus the share of consumers choosing the outside option. Recall that the outside option is set using the fraction of people with no salon transaction in the Consumer Expenditure Survey. Combined with the multinomial logit demand system, this implies all salons in the market (including those outside the sample) have symmetric substitution patterns with each other and the outside option. This has clear limitations, most notably that it overestimates the level of competition faced by the salons in the data, which are positively selected relative to their competitors.

Second, I assume inelastic labor supply. To implement this, I recover the total amount of labor of a skill set i as the sum of labor demand for skill set i across all firms in the data, multiplied by n defined earlier. I then hold the total labor of each skill set fixed across all counterfactuals except the immigration counterfactual where it is explicitly changed.

Third, I fix exogenous quality and marginal cost. Each salon-quarter has exogenous quality heterogeneity (ν_j) and marginal cost heterogeneity (ω_j). I estimate ν_j as the residual of the log relative market share after accounting for endogenous quality from task assignments and the price effect. I estimate ω_j as the observed price minus organization costs (evaluated at estimated wages) and minus wage costs (evaluated at the relative labor demands that are optimal at estimated wages).[36] I fix these exogenous components across counterfactuals, with the caveat that I truncate the estimate if it would cause a negative predicted price. By fixing quality and cost heterogeneity, I am implicitly fixing the skill level of workers employed at each salon.

Fourth, I assume the initial state of the economy in all counterfactuals is the model solved to clear the total labor supplies recovered in the second step. That is, I solve for all endogenous objects, including wages, task assignments, and prices, such that the model-implied labor demand of each skill set equals the data-based labor supplies. This approach has the advantage that it fully

36. Optimal task assignments at estimated wages are distinct from the task assignments observed in the data because the estimation procedure does not impose all theoretical restrictions.

imposes the structure of the model at the initial point, and as a result I can be sure that changes from the initial point are due entirely to the counterfactual rather than misalignment between the data and theory.

I exclude one salon from the counterfactuals which appears to be a booth-rental salon with one employee and an outlier required labor (61.33 hours relative to the median required hours of 2.36). This salon has an s-index of 0, and as a result does not contribute to the sequential GMM estimation procedure.

For each counterfactual, I begin by guessing an initial wage for each skill set. For the reorganization equilibrium, I first solve each salon's task-assignment problem via the contraction mapping given in Appendix Section A.14. For the reallocation equilibrium, I fix each salon's task assignments at their initial values. I then iterate on each firm's Bertrand pricing best response function until I achieve convergence. I use the formulation derived in Appendix Section A.15. Using the implied labor demand from each firm derived from optimal prices and task assignments, I check whether the labor market clears. If it does not, I repeat with a new wage guess until it does. I consider a labor market cleared if the maximum difference between labor supply and demand is within 1%, or the labor-weighted mean absolute difference is within 1%. In practice, I am able to clear the market for each skill set in each county in each counterfactual equilibrium (including the initial equilibrium) to 4.05×10^{-6} in the loosest case, and several orders of magnitude tighter in most cases.

Solving for wages in this model is complicated by the rich set of possible labor-labor substitution patterns, and the two nested loops involved. I address these challenges using the BB package developed in Varadhan and Gilbert (2010). The package provides a root-finding algorithm that works well for this model.

A.17 The Empirical and Theoretical Limits of Worker Skill Heterogeneity

The model in the main text primarily speaks to skill sets, which are meant to capture differences in comparative advantage. The results in the main text also assume all tasks enter consumer utility with the same weight, meaning that utility is then a weighted average of the skills of the worker doing the tasks, with weights equal to how much of each task is done. A natural question is how differences in absolute advantage interact with heterogeneous utility weights. This section explores this question, and shows that the theory and data can accommodate these additional features, provided they can be represented as a redefined set of skill sets.

Suppose that consumers have a utility weight attached to each task. Specifically, the utility

from product j for consumer z is given by $U_{z,j} = \sum_{m,k} \kappa(k)(\bar{\theta}_m + \theta_m(k))B_j(m,k) - \rho p_j + \epsilon_{j,z}$. Note that this extended model is outcome-equivalent to a model where skill sets are redefined to absorb skill levels and task utility weights, and indexed by $m = 1, ..., M$: $\theta'_m(k) := \kappa(k)(\bar{\theta}_m + \theta_m(k))$. The results of the paper can be applied directly to these redefined worker skill sets. For example, Theorem 2 guarantees identification of worker skill sets provided that $\{\kappa(k)(\bar{\theta}_m + \theta_m(k))\}_{m=1}^M$ are linearly independent, and Proposition 5 implies uniqueness of firm task assignment strategies if there is sufficient variation in comparative advantage; formally, $\det(\exp[c\Theta^M]) \neq 0$ for all $c > 0$, where Θ^M is the skill set matrix constructed from the redefined skill sets.

Thus both skill level heterogeneity and task utility weights can be incorporated into the model by recasting them as skill set differences. However, this construction also shows there are limits imposed by both the theory and data. The maximum number of linearly independent vectors of length K is K, so if M is larger than K, worker skill heterogeneity is generally too rich for the theory or the data to accommodate. In this way, both the theory and the data constrain the ability of the model to handle more skill sets than tasks. A researcher who wants to incorporate skill heterogeneity that requires more than K skill sets should expect both multiple equilibria in task assignments and identification failure from task assignment data alone.

It should be noted that limitless amounts of some types of heterogeneity can be absorbed essentially without loss by the model because of Proposition 2. For example, if two redefined skill sets have a constant gap in skill across all tasks ($\theta_m(k) = d + \theta_{m'}(k)$), Proposition 2 implies that a no-arbitrage condition on wages holds in every equilibrium, so that all firms are indifferent between the two skill sets and the task assignments of m and m' are the same at any firm at which they are both employed. Then the estimation procedure will treat m, m' as one skill set, m, m' will not be separately identified, and the quality and wage differences stemming from salons hiring more or less of the higher skill type will load as quality and cost residuals during estimation. To the extent the researcher is primarily concerned about comparative advantage rather than absolute advantage, this may or may not be appropriate.

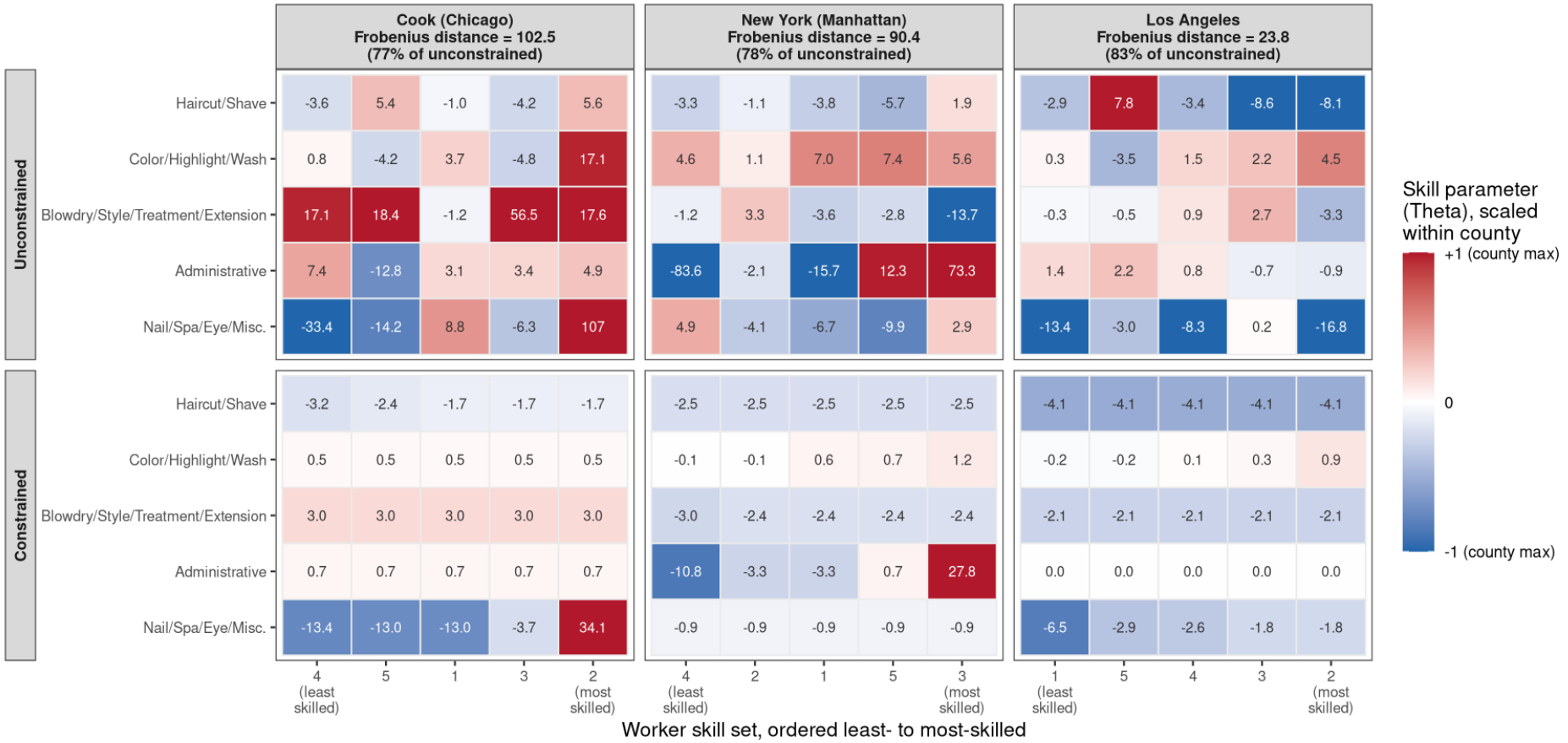
A.18 Monotone Skill Restriction

In some models with multiple dimensions of worker skill, workers can be ordered in their absolute skill across all tasks. The model estimated in this paper does not impose this assumption. A natural question is how imposing this monotonicity assumption impacts results. To answer this question, I estimate the model with the monotonicity restriction: I require that it be possible to order skill sets so that the first skill set has lower skill at all tasks than the second skill set, and

the second skill set has lower skill at all tasks than the third, etc. To implement the restriction, I consider all 120 possible monotonicity orderings of the recovered worker skill set groups, minimize the two-stage least squares GMM objective for each ordering, and choose the ordering which achieves the lowest objective.

Appendix Figure A1 compares the unrestricted skill set parameter estimates to those under the monotonicity restriction. The Frobenius distance is a notion of Euclidean distance between the two skill set matrices, defined as the square root of the sum of squared differences between each element of the constrained and unconstrained matrices. The percentage underneath the distance is the distance divided by the square root of the sum of squared elements of the unconstrained skill set matrix. According to the raw Frobenius distance, the unconstrained Cook County skill set matrix is the farthest from its monotone counterpart, while Los Angeles is closest. Using the percentage Frobenius distance metric, all counties have skill set estimates that are similarly far from monotone.

Figure A1: Monotone vs. Unconstrained Skill Sets



Note: Each cell is an estimated skill parameter Theta: the skill of a worker skill set (column) at a task (row), in the same units as the paper's skill-parameter tables. Unconstrained estimates are the baseline; constrained estimates add the requirement that skill sets can be ranked by absolute advantage, i.e. that one ordering exists in which each skill set is at least as skilled as the previous one at every task. Columns are ordered least- to most-skilled under that ranking, so under the constraint values never fall from left to right; skill sets are numbered as in the paper and the ordering differs by county. Printed numbers are the estimates themselves; colour is rescaled within each county and saturates at that county's 90th percentile of |Theta|, so that a single extreme cell does not wash out the rest. Colours are therefore comparable within a county but not across counties. Each heading reports the Frobenius distance between that county's unconstrained and constrained matrices (the square root of the summed squared cell-by-cell changes), with that distance as a percentage of the unconstrained matrix's own Frobenius norm.

How much does the monotonicity restriction impact the ability of the model to fit the data used for estimation? In Appendix Table A7, I show how well the restricted model matches the estimation moments compared to the main unrestricted model. Because monotonicity is a restriction, it must weakly decrease model fit, so the question is one of magnitude. Estimation of the unrestricted model is essentially a series of just-identified GMM estimations, and as a result unrestricted model fit is near perfect, with an R-squared from regressing model-generated moments on data moments near 1 for all sets of moments. The monotone estimates have worse fit across all sets of moments except the log market share moments. The reduction in model fit is most notable in the labor demand moments and the price county-quarter moments, where R-squared becomes negative. The failure to fit the labor demand moments is evidence that the observed shares of each skill set employed in the data are difficult to rationalize at any wages when skill sets are restricted to be monotone. The failure to fit some price moments is evidence that the price variation across counties and time is difficult to rationalize under the relative wages, markups, and minimum wage constraints.

Table A7: Model Fit Under the Monotonicity Restriction

Equation	Instrument	Count	R2 (Main)	R2 (Monotone)	Δ R2
Labor Demand	County-Skill Set	12	1.000	-0.290	-1.290
Log Market Share	County-Dye Instrument	3	1.000	1.000	0.000
Log Market Share	County-Quarter	33	1.000	1.000	0.000
Log Market Share	County-Task Assignments	75	1.000	1.000	0.000
Price	County-Quarter	36	0.977	-1.700	-2.677
Price	County-Quarter-Labor	36	0.999	0.978	-0.021
Price	Quarter-Task Mix	48	0.971	0.452	-0.519

When restrictions similar to monotonicity are invoked, there are often theoretical justifications. If these theoretical justifications reflect underlying economic truth, then the monotonicity restriction can prevent overfitting and potentially improve performance on data not used in estimation. I test this in Appendix Table A8. For reference, the same validation exercise is performed for the unrestricted main model in Table 13. The results are mixed. Monotonicity improves two variance moments and three correlations, but worsens three variance moments and six correlations, three of them substantially. For example, the monotonicity restriction results in a model-implied correlation between task 3 and task 1 of -0.016, the data imply a correlation of -0.229, and the unrestricted model achieves a correlation of -0.166.

Table A8: Model Validation Under the Monotonicity Restriction

	Task	Variance	Cor. Task 1	Cor. Task 2	Cor. Task 3	Cor. Task 4	Cor. Task 5
Model	1	0.096	1.000	-0.780	-0.016	-0.267	-0.120
Data	1	0.105	1.000	-0.760	-0.229	-0.265	-0.153
Model	2	0.109		1.000	-0.325	-0.124	-0.255
Data	2	0.100		1.000	-0.113	-0.159	-0.247
Model	3	0.013			1.000	0.092	-0.097
Data	3	0.014			1.000	0.028	-0.089
Model	4	0.015				1.000	-0.079
Data	4	0.018				1.000	-0.018
Model	5	0.020					1.000
Data	5	0.020					1.000

A.19 A Quantity-Based Model

In some contexts, such as manufacturing, one may wish to model organizational efficiency as allowing firms to produce greater quantity rather than greater quality. Indeed, this is the default definition of productivity in economics. The model can also be extended to accommodate this: one can simply interpret the skill sets as denoting the amount of time required by the worker to complete task k (therefore smaller $\theta_i(k)$ are better). Then the production function becomes a function of organization structure:

$$F_{\alpha,B}(a_j) = \min\left\{\frac{a_1}{\alpha(1)\sum_i \theta_{i,1}B_j(i,1)}, ...\frac{a_k}{\alpha(k)\sum_i \theta_{i,k}B_j(i,k)}, ..., \frac{a_K}{\alpha(K)\sum_i \theta_{i,K}B_j(i,K)}\right\}.$$

Given any fixed organizational structure, the efficient way to produce a single unit of output is to set $a_k = \alpha(k)\sum_i \theta_{i,k}B_j(i,k)$. Thus the per-unit wage bill is given by

$$\sum_i w_i \sum_k \alpha(k)\theta_i(k)B_j(i,k).$$

Marginal costs are constant and consist of the per-unit wage bill and organization costs:

$$MC_j = \sum_i w_i \sum_k \alpha(k)\theta_i(k)B_j(i,k) + \gamma_j I(B_j).$$

All of the benefits of a more complex organization come through a reduction in the per-unit wage bill. In this way, the intuition from the original model extends directly to the quantity case: firms with greater organizational efficiency (lower γ_j) can produce more of the good with the same workforce. I did not use this as the main model because the following property is not compatible with the empirical application to hair salons:

Proposition 7 *Under a quantity model with multinomial logit demand, prices are increasing in the organization cost parameter γ.*

The proof of this proposition is given in the next paragraph. Intuitively, under the quantity model with logit demand, all the benefits of greater organization come from greater output rather than from greater revenue per unit. The reduction in marginal cost outpaces the increase in the markup, resulting in lower prices. This implies a negative correlation between prices and the s-index, which is shown not to be true for hair salons. However, for manufacturing firms, it appears to be true. Caliendo et al. (2020) find that prices (revenue-based productivity) decline when manufacturing firms reorganize.

Proof. Equation 21 from Appendix Section A.15 provides a closed-form expression for price in any Nash equilibrium under logit demand:

$$\frac{1}{\rho} + c_j + \rho^{-1} L\bigg(\exp\bigg(-1 + \xi_j - \rho c_j \bigg) E_j^{-1} \bigg) = p_j^*.$$

where L denotes the Lambert W function. Taking the derivative w.r.t. c_j yields

$$\frac{\partial p_j^*}{\partial c_j} = 1 - \exp\bigg(-1 + \xi_j - \rho c_j \bigg) E_j^{-1} L'\bigg(\exp\bigg(-1 + \xi_j - \rho c_j \bigg) E_j^{-1} \bigg).$$

A property of the Lambert W function is that $L'(x) = \frac{L(x)}{(1+L(x))x}$. Thus, I can simplify the expression to $\frac{\partial p_j^*}{\partial c_j} = 1 - \frac{L(\exp[-1+\xi_j - \rho c_j] E_j^{-1})}{1+L(\exp[-1+\xi_j - \rho c_j] E_j^{-1})}$. The Lambert W function is weakly positive for values which are weakly positive; therefore, the derivative is positive, and price is increasing in marginal cost. The firm minimizes cost: $\min_{B_j \in \mathbb{B}} \gamma I(B_j) + W(B_j)$. This is again a rate-distortion problem. Denoting the optimal wage bill as $D = W(B_j^*)$, I can reformulate the problem as before, with the firm choosing D given some optimal organization cost and wage bill: $\min_D \gamma I(D) + D$, where I and is expressed as a function of D instead of B_j. Then, as before, there is a negative cross-partial derivative: $\frac{\partial^2 \gamma I(D) + W(D)}{\partial D \partial \gamma} = I'(D) < 0$ with strict inequality whenever $I(D)$ is strictly positive. This establishes strict decreasing differences of D in γ; thus D is strictly increasing in γ, and since $I(D)$ is a strictly decreasing function, it is strictly decreasing in γ. Therefore, prices should be decreasing as γ decreases, while the s-index should be increasing.

A.20 Measurement Error in the S-Index

Task assignments are treated as measured without error when computing the s-index and other quantities. One justification is that many assignments are observed per salon per quarter, so es-

timation error should be small. If estimation error at the quarter level is small, the correlation between monthly s-index measures within a quarter should be large. This section illustrates that this is indeed the case. To do this, I recompute the s-index for each month within a quarter so that I have three measurements of the s-index per salon-quarter observation. Correlations are computed on the 3,950 of 4,495 salon-quarters for which all three months are observed. In the full sample, the pairwise correlation between the first and second month is 0.947, the first and third is 0.938, and the second and third is 0.957. When 2020 (the onset of the coronavirus pandemic) is excluded, the pairwise correlations are 0.968, 0.959 and 0.978, respectively. The high correlation between the s-index measurements within quarters suggests that the s-index at the quarter level has little measurement error.

A.21 Supplementary Tables and Figures

Table A9: Alternative Units for Skill Set Estimates

	Willingness-to-pay (USD)					Pct. Share (pooled)				
Worker Skill Set	Haircut	Color	Style	Admin	Nail	Haircut	Color	Style	Admin	Nail
Cook										
1	0.70	2.19	0.00	1.69	1.27	3.44	10.77	0.00	8.31	6.22
2	2.11	5.63	1.11	1.88	4.22	10.39	27.67	5.46	9.24	20.73
3	0.00	0.00	3.41	1.72	0.81	0.00	0.00	16.76	8.46	3.99
4	0.14	1.43	1.09	2.15	0.00	0.68	7.02	5.33	10.56	0.00
5	2.08	0.15	1.16	0.00	0.58	10.22	0.75	5.69	0.00	2.83
New York										
1	0.42	1.79	0.67	0.82	0.09	1.44	6.09	2.28	2.79	0.30
2	1.01	0.00	1.12	0.99	0.16	3.44	0.00	3.82	3.35	0.55
3	1.68	1.38	0.00	1.90	0.35	5.72	4.70	0.00	6.45	1.21
4	0.52	1.07	0.83	0.00	0.41	1.77	3.64	2.82	0.00	1.40
5	0.00	1.93	0.72	1.16	0.00	0.00	6.56	2.46	3.95	0.00
Los Angeles										
1	1.59	1.77	0.36	0.10	0.07	4.36	4.84	0.98	0.27	0.20
2	0.14	3.73	0.00	0.00	0.00	0.38	10.22	0.00	0.00	0.00
3	0.00	2.66	0.73	0.01	0.37	0.00	7.29	1.99	0.02	1.02
4	1.46	2.31	0.51	0.07	0.19	3.98	6.31	1.41	0.20	0.51
5	4.57	0.00	0.34	0.14	0.30	12.51	0.00	0.93	0.37	0.83

Notes: Each cell is the parameter for worker w at task t minus the parameter for the worst worker at that same task *within the county*, so the worst worker reads 0 and all others are gains relative to that county-specific baseline. Both blocks price the same experiment — reassigning 1% of task t's own mass from the worst worker to worker w at the mean firm — and differ only in units. *Willingness-to-pay (USD):* cell $= (\theta[w,t] - \theta_{\min,t}) \,/\, |\rho_c| \cdot \bar{L}_c \cdot 0.01\, \bar{\alpha}_{c,t}$, where $\theta[w,t]$ is the estimated skill parameter, $\theta_{\min,t}$ is the worst worker's value in column t, $|\rho_c|$ is the negative-utility coefficient on price, $\bar{L}_c$ is the unweighted mean of hours of labor per customer over firm-quarters in county c, and $\bar{\alpha}_{c,t}$ is the mean share of task t in the firm's appointment mix. Reads as the consumer WTP per customer (USD) at a mean-labor firm in county c. *Pct. Share (pooled):* the same recentered θ gain multiplied by $\bar{L}_c\,(1-\bar{s}_c)\,\bar{\alpha}_{c,t}$, where each $\bar{\cdot}$ is the unweighted mean over firm-quarters in county c ($\bar{L}_c$ in hours/customer, $\bar{s}_c$ the within-subdivision market share, $\bar{\alpha}_{c,t}$ the share of task t in the firm's appointment mix). A cell value $\approx$ the percent change in the mean firm's market share (i.e., $\Delta s_j / s_j \cdot 100$, not $\Delta s_j \cdot 100$); the 0.01 of the reassignment cancels against the $\times 100$ that converts to percent.

Figure A2: Productivity Effects of Immigration in Los Angeles County

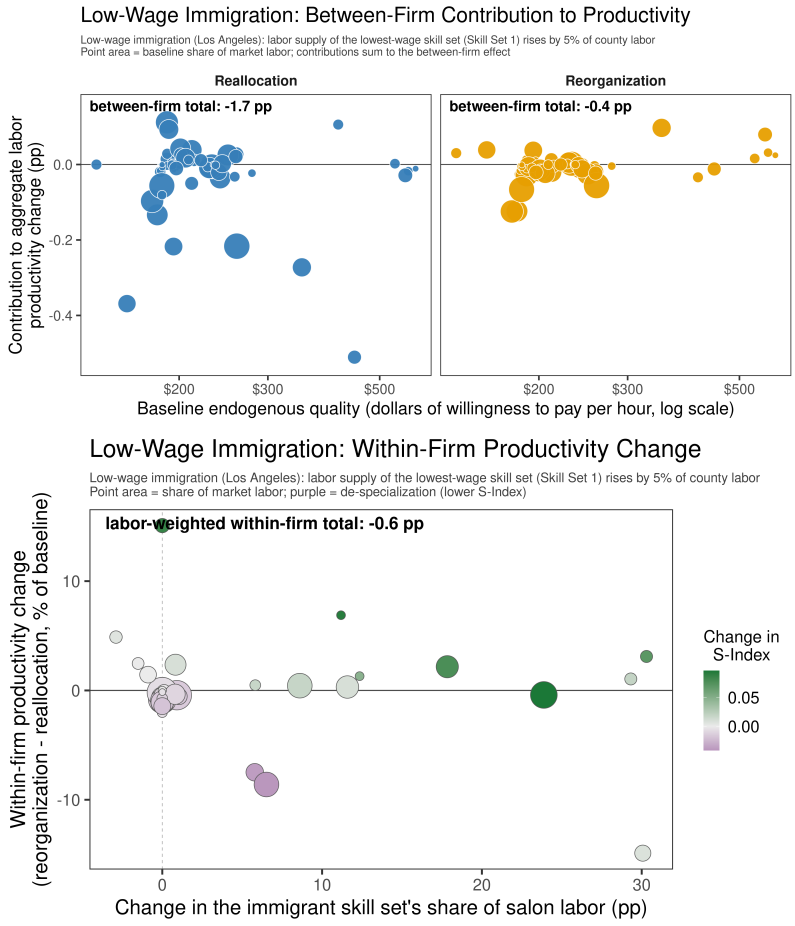


Table A10: Counterfactual Productivity and Wage Effects by Worker Skill Set

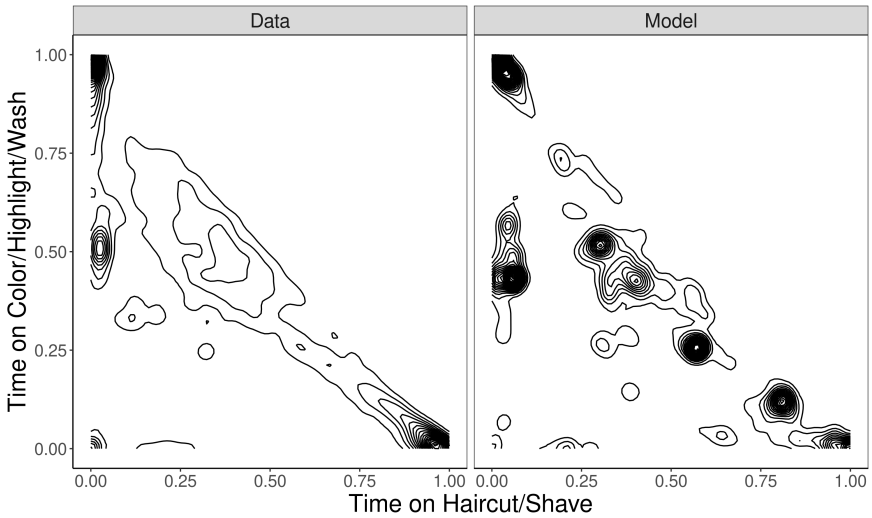
		Skill Set 1		Skill Set 2		Skill Set 3		Skill Set 4		Skill Set 5	
Counterfactual	County	Prod.	Wage	Prod.	Wage	Prod.	Wage	Prod.	Wage	Prod.	Wage
Immigration	Cook	-0.002	-0.019	0.015	0.014	0.003	0.002	-0.028	-0.100	0.002	-0.013
Immigration	Los Angeles	0.001	-0.056	0.003	0.006	0.008	0.003	0.001	-0.034	0.000	-0.011
Immigration	New York	0.001	-0.006	0.009	-0.010	0.010	-0.004	-0.048	-0.059	0.000	-0.005
Incr. Concentration	Cook	0.000	-0.134	-0.027	-0.017	0.002	-0.014	-0.045	-0.226	-0.004	-0.153
Incr. Concentration	Los Angeles	-0.014	-0.163	0.001	-0.073	0.018	-0.120	0.002	-0.149	0.000	-0.026
Incr. Concentration	New York	0.005	-0.059	0.014	-0.122	0.050	-0.053	0.007	-0.078	0.001	-0.052
Management Diffusion	Cook	0.045	0.025	0.296	0.401	0.107	0.032	0.526	0.088	0.066	0.104
Management Diffusion	Los Angeles	-0.004	0.000	0.007	0.036	0.006	0.013	0.001	0.006	0.001	0.011
Management Diffusion	New York	0.000	0.007	0.013	0.023	0.042	0.006	-0.004	0.003	0.000	0.008
Sales Tax	Cook	0.016	-0.079	0.017	-0.011	-0.009	-0.017	0.018	-0.099	-0.012	-0.111
Sales Tax	Los Angeles	0.018	-0.028	-0.005	-0.067	0.036	-0.076	0.001	-0.034	-0.001	-0.037
Sales Tax	New York	-0.004	-0.047	-0.016	-0.050	0.035	-0.034	-0.017	-0.039	-0.002	-0.051

Note: Effects are proportional changes in the reorganization equilibrium relative to the baseline equilibrium (0.01 = 1%). The table presents effects for each worker skill set.

Figure A3: Productivity Effects of a Sales Tax Increase in New York County

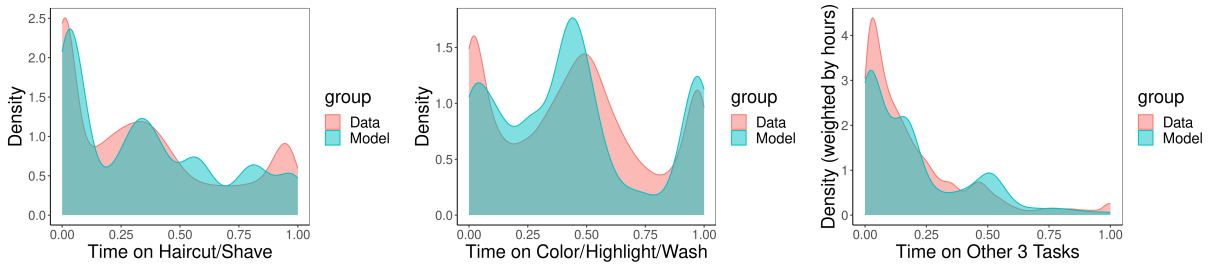


Figure A4: Joint Distribution of Job Task Content in the Model and Data



Note: This model validation exercise compares the task content of model-generated jobs with that of jobs in the data. Jobs are weighted by the amount of labor assigned. The figure is a contour density plot, where more closely spaced contour lines indicate greater probability mass. The model replicates many of the important peaks of the distribution.

Figure A5: Model Validation: Marginal Task Distributions



(a) Haircut/Shave **(b)** Color/Wash **(c)** Other Three Categories

Note: The figure shows the distribution of time spent on each task; density is based on the amount of time assigned to each worker-quarter in the data and to each firm-by-skill-set-by-quarter cell in the model.

Table A11: Auxiliary Parameter Estimates

County	Parameter	2018Q1	2018Q2	2018Q3	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4	2020Q1	2020Q4	2021Q1	2021Q2
All	Material Cost Administrative	-15.971	-6.817	-102.365	-15.252	-40.732	-63.587	-33.231	-82.203	-25.835	-5.186	-15.236	-9.767
All	Material Cost Administrative	(997.113)	(836.194)	(2275.543)	(1131.610)	(2003.452)	(778.703)	(410.787)	(1128.442)	(332.321)	(364.871)	(918.695)	(432.966)
All	Material Cost Blowdry/Style/Etc.	-8.000	-2.062	-49.544	-5.197	-18.066	-28.965	-11.079	-38.307	-12.860	1.352	-5.162	-7.295
All	Material Cost Blowdry/Style/Etc.	(1602.102)	(668.998)	(1086.715)	(859.842)	(657.075)	(993.046)	(1311.158)	(488.269)	(271.102)	(424.857)	(748.199)	(552.645)
All	Material Cost Color/Highlight/Wash	-12.128	0.627	-135.744	-7.236	-45.675	-83.277	-29.222	-99.139	-32.707	1.226	-40.150	-20.863
All	Material Cost Color/Highlight/Wash	(1455.792)	(515.000)	(2298.561)	(1036.412)	(782.345)	(1080.817)	(358.923)	(961.744)	(304.202)	(348.829)	(935.331)	(427.441)
All	Material Cost Nail/Misc.	-1.868	-0.543	-12.823	-0.994	-4.270	-9.995	-4.608	-10.537	-10.156	-1.589	-5.167	-3.784
All	Material Cost Nail/Misc.	(7530.800)	(1572.643)	(2729.630)	(1596.190)	(982.863)	(1080.555)	(1096.624)	(2873.179)	(995.582)	(702.251)	(1047.954)	(1696.478)
Cook	Cost Level	-81.750	-40.049	-431.519	-69.381	-166.608	-266.808	-132.363	-339.689	-164.853	-42.577	-128.094	-90.620
Cook	Cost Level	(679.425)	(222.352)	(1242.892)	(537.393)	(563.733)	(592.799)	(401.553)	(655.045)	(235.117)	(361.299)	(1347.152)	(1290.680)
Cook	Demand Level	-	0.247	-0.083	-2.841	-2.351	-3.805	-1.348	-3.204	-2.238	-1.153	-1.990	-0.750
Cook	Demand Level	-	(1.472)	(1.827)	(2.633)	(2.426)	(1.867)	(1.808)	(2.083)	(1.525)	(1.526)	(1.817)	(1.693)
Cook	Wage Level	18.308	18.484	18.458	18.458	18.458	27.650	24.041	41.572	45.216	25.734	41.480	37.632
Cook	Wage Level	-	(68.107)	-	-	-	(62.718)	(60.731)	(45.583)	(57.272)	(167.773)	(426.582)	(570.104)
Los Angeles	Cost Level	14.884	14.437	20.834	8.500	15.166	16.287	15.954	19.760	23.222	18.825	15.912	18.079
Los Angeles	Cost Level	(677.264)	(459.919)	(1357.860)	(636.544)	(587.196)	(535.589)	(347.243)	(273.824)	(247.779)	(727.501)	(518.837)	(295.282)
Los Angeles	Demand Level	-	-1.210	-0.916	-1.867	-0.908	-0.460	0.024	-0.268	-1.174	-1.333	-0.503	0.524
Los Angeles	Demand Level	-	(.672)	(.721)	(1.003)	(.771)	(.888)	(.942)	(1.163)	(1.155)	(1.316)	(1.549)	(1.463)
Los Angeles	Wage Level	40.827	48.502	70.382	44.673	56.045	61.642	64.755	65.970	81.701	71.820	61.865	69.106
Los Angeles	Wage Level	(158.239)	(191.940)	(196.504)	(165.506)	(204.674)	(152.196)	(241.813)	(292.808)	(195.601)	(374.302)	(304.530)	(203.161)
New York	Cost Level	27.246	18.251	43.688	32.049	25.362	30.090	27.937	43.661	30.484	19.404	18.954	19.109
New York	Cost Level	(1455.960)	(299.994)	(574.328)	(423.588)	(265.251)	(620.742)	(380.499)	(347.293)	(385.223)	(358.725)	(531.905)	(327.040)
New York	Demand Level	-	-0.020	0.890	1.453	0.066	0.871	0.183	1.241	-0.437	0.058	-0.153	1.260
New York	Demand Level	-	(.903)	(1.890)	(1.216)	(1.207)	(1.283)	(1.167)	(1.598)	(1.407)	(1.040)	(.802)	(1.539)
New York	Wage Level	109.056	93.554	142.110	123.169	115.531	122.407	115.930	159.534	121.144	106.759	103.626	108.313
New York	Wage Level	(1303.657)	(250.291)	(516.993)	(350.191)	(168.768)	(363.836)	(253.691)	(270.088)	(184.713)	(137.776)	(203.227)	(107.334)

Note: This table reports the parameters associated with material costs, demand levels, wage levels and cost levels. Standard errors are not reported for Cook County wage level parameters in quarters where the minimum wage bound binds.